

SCHEM, ANGEL_ISLAND, MLB, K17

Rev.A 02/23/10

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD
			2009-05-19

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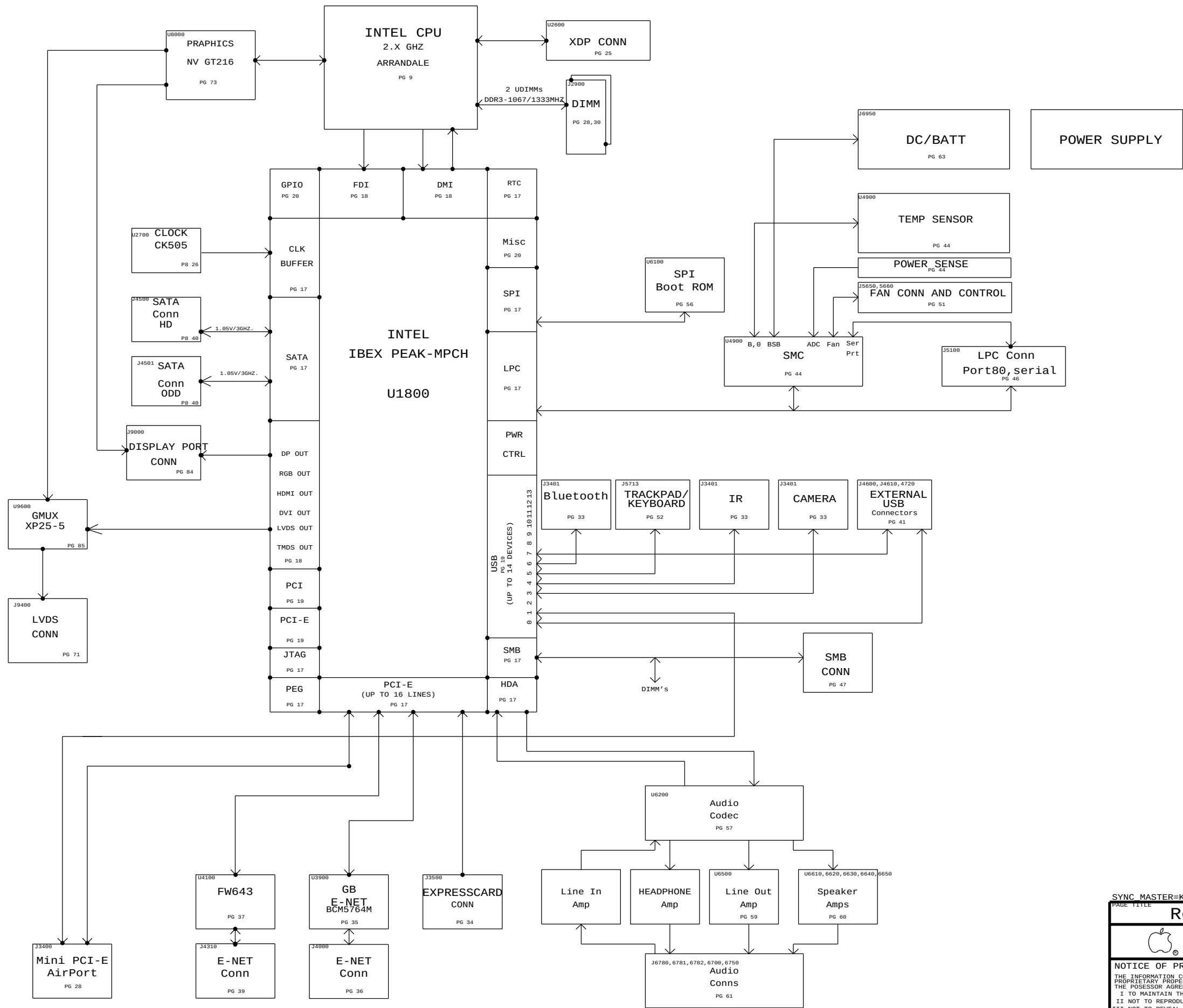
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Schematic / PCB #'s

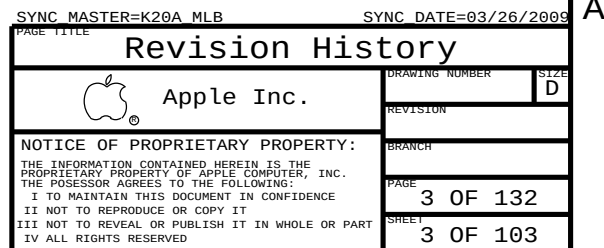
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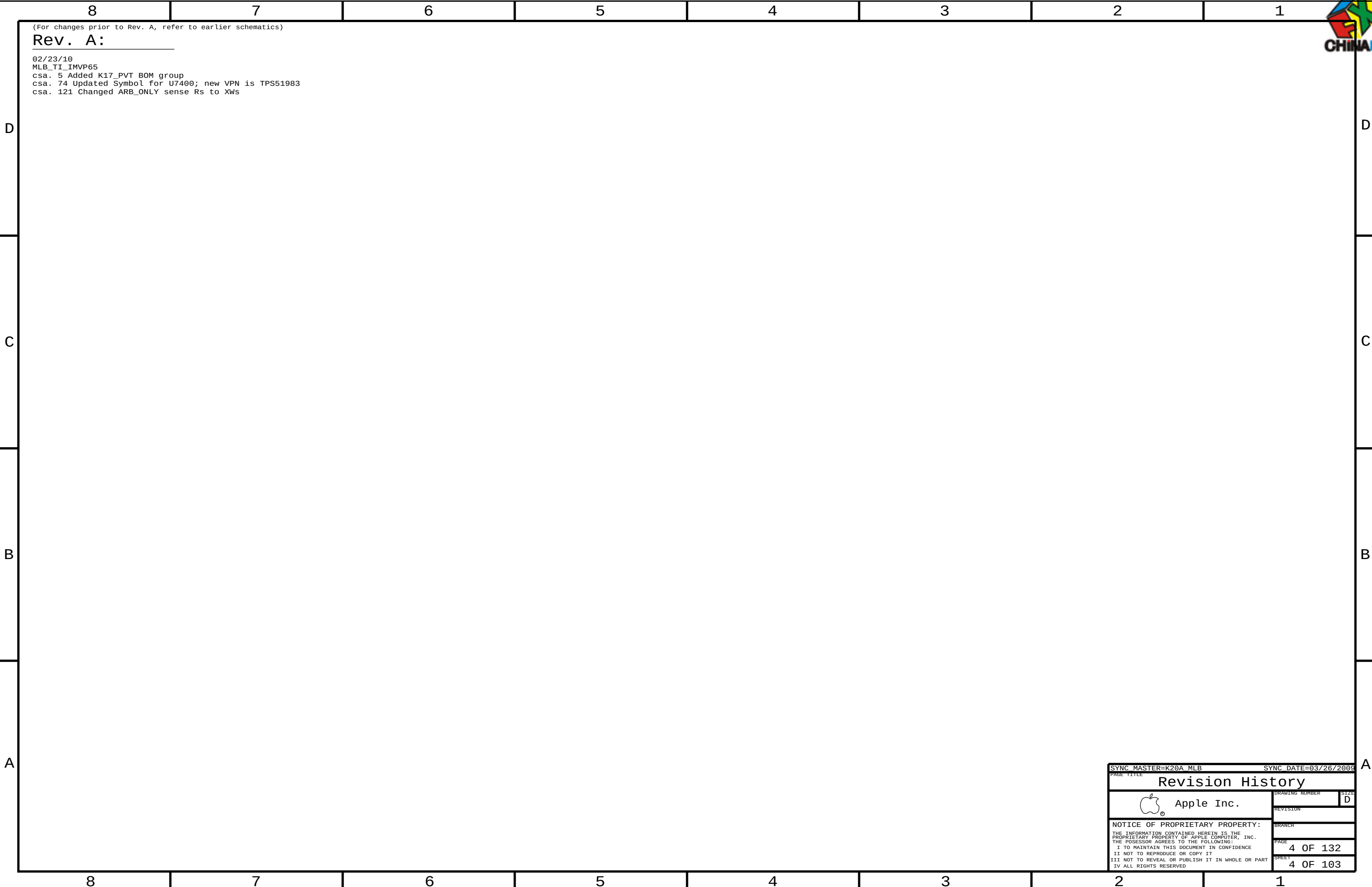
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(For changes prior to Rev. A, refer to earlier schematics)

Rev. A:

02/23/10
MLB_TI_IMVP65
csa. 5 Added K17_PVT BOM group
csa. 74 Updated Symbol for U7400; new VPN is TPS51983
csa. 121 Changed ARB_ONLY sense Rs to Xws



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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-0973	PCBA, 2.53GHZ, 512SAM_VRAM, K17	K17_COMMON, CPU_2_53GHZ, FB_512_SAMSUNG, EEEE_DCMV, K17_PVT
639-0971	PCBA, 2.53GHZ, 512HYN_VRAM, K17	K17_COMMON, CPU_2_53GHZ, FB_512_HYNIX, EEEE_DCMR, K17_PVT
639-0972	PCBA, 2.66GHZ, 512SAM_VRAM, K17	K17_COMMON, CPU_2_66GHZ, FB_512_SAMSUNG, EEEE_DCMT, K17_PVT
639-0970	PCBA, 2.66GHZ, 512HYN_VRAM, K17	K17_COMMON, CPU_2_66GHZ, FB_512_HYNIX, EEEE_DCMQ, K17_PVT
085-1425	K17_MLB_DEVELOPMENT	K17_DEVEL_ENG

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0603	138S0602		ALL	Murata alt to Samsung
157S0058	157S0055		ALL	Delta alt to TDK Magnetics
152S0896	152S0518		ALL	MAG LAYERS ALT TO CYNTEC
152S0915	152S0796		ALL	MAG LAYERS ALT TO CYNTEC
155S0457	155S0329		ALL	MAG LAYERS ALT TO MURATA
516S0806	516S0805		ALL	FOXCONN ALT TO MOLEX
138S0612	138S0602		ALL	Taiyo Yuden alt to Samsung
353S2805	353S2603		ALL	Fairchild 8 in alt to 6 in wafer
127S0111	127S0060		ALL	Rohm alt to Kemet
128S0299	128S0218		ALL	NEC/TOKIN alt to Sanyo
337S3808	337S3839		ALL	GT216 A02 alt to A03 part
376S0887	376S0749		ALL	Fairchild alt to Vishay

K17 BOM GROUPS

BOM GROUP	BOM OPTIONS
K17_COMMON	ALTERNATE, COMMON, K17_COMMON1, K17_COMMON2, K17_PROGPARTS
K17_COMMON1	BCM5764M, DCI, GMUX_VSYNC, CPUPOC_IMAX_40_50, PCH_NAND_3V3, CPUMEM_S0, EXT_HP_AMP, VFRQ_SLPS3, SMC_DEBUG_YES, DPMUX_EN_PLD, FB1V35, USBHUB_2061
K17_COMMON2	GPUVID_0P90V, BKLT_PWR_PBUS, DP_ESD, DP_CA_DET_EG_PLD, SMC_EXCARD_NOT, GPU_SS_INT, RDRV_8515_A2, GMUXPLL_3V3, HUB1_2NONREM, HUB2_2NONREM, RAIL_MON
K17_DEVEL_ENG	ARB_ONLY, CALPELLA_XDP, DEBUG_ADC, LPCPLUS, VREFMRGN, GMUX_JTAG_CONN, EFI_DEBUG, BMON_ENG, SMC_OSC_YES
K17_PROGPARTS	GMUX_PROG, BOOTROM_PROG, SMC_PROG, TPAD_PROG
K17_PVT	BMON_PROD, VREFMRGN_NOT, XDP, XDP_NORMAL, XDP_CPU_BPM

BOM GROUP	BOM OPTIONS
FB_512_SAMSUNG	VRAM4, VRAM_512_SAMSUNG
FB_512_HYNIX	VRAM4, VRAM_512_HYNIX
CALPELLA_XDP	XDP, XDP_CONN, XDP_CPU_BPM, XDP_NORMAL, XDP_PCH

Bar Code Labels / EEEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCMQ]	CRITICAL	EEEE_DCMQ
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCMR]	CRITICAL	EEEE_DCMR
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCMT]	CRITICAL	EEEE_DCMT
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DCMV]	CRITICAL	EEEE_DCMV

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3847	1	ARD, SLBPF, PRQ, 2.53, 35W, C2, 3M, BGA	U1000	CRITICAL	CPU_2_53GHZ
337S3848	1	ARD, SLBPE, PRQ, 2.66, 35W, C2, 4M, BGA	U1000	CRITICAL	CPU_2_66GHZ
337S3846	1	ARD, SLBNA, PRQ, 2.40, 35W, C2, 3M, BGA	U1000	CRITICAL	CPU_2_4GHZ
337S3849	1	IBEX (HM55), SLGZS, PRQ, B3	U1800	CRITICAL	
337S3839	1	IC, GPU, NV GT216 LP++, 969BGA, 40NM, A03	U8000	CRITICAL	
343S0493	1	IC, ASIC, BCM5764M, ENET CONTROLLER, 8x8, 64 QFN	U3900	CRITICAL	BCM5764M
338S0753	1	IC, FW643-E, 1394B PHY/DHCI LINK/PCI-E, 12	U4100	CRITICAL	
338S0563	1	IC, SMC, HS8/2117, 9MMX9MM, TLP	U4900	CRITICAL	SMC_BLANK
341T0229	1	IC, SMC, K17	U4900	CRITICAL	SMC_PROG
335S0610	1	IC, FLASH, SPI, 32MBIT, 3.3V, 86MHZ, 8-SOP	U6100	CRITICAL	BOOTROM_BLANK
341T0244	1	IC, EFI ROM, K17	U6100	CRITICAL	BOOTROM_PROG
341S2384	1	IR, ENCORE II, CY7C63833-LFXC	U4800	CRITICAL	
341S2616	1	IC, TP PSOC, K17, K18	U5701	CRITICAL	TPAD_PROG
336S0025	1	IC, XP2-5, HF, CPLD, BLANK	U9600	CRITICAL	GMUX_5K_BLANK
341S2568	1	IC, CPLD, LATTICE, 132CSBGA, K17MLB	U9600	CRITICAL	GMUX_PROG
333S0533	4	IC, SDRAM, GDDR3, 32MX32, 1GHZ, D-DIE, 136 FBGA	U8400, U8450, U8500, U8550	CRITICAL	VRAM_512_SAMSUNG
333S0535	4	IC, SDRAM, GDDR3, 32MX32, 900MHZ, TIVA, HF	U8400, U8450, U8500, U8550	CRITICAL	VRAM_512_HYNIX
341S2731	1	IC, 1MBIT, SPI FLASH K17/K18	U3990	CRITICAL	
516S0805	1	CONN, 204P, SOD1MM, SOCKET, DDR3, RAM, NON/SC	J3100	CRITICAL	
197S0350	1	OSC, XTAL, 32.768KHZ, 9-3.6V, 12P S01C, HF	U5010	CRITICAL	SMC_OSC_YES

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BOM Configuration

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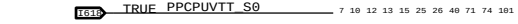
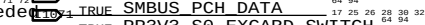
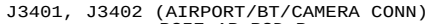
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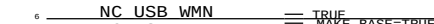
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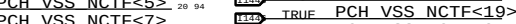
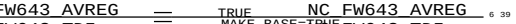
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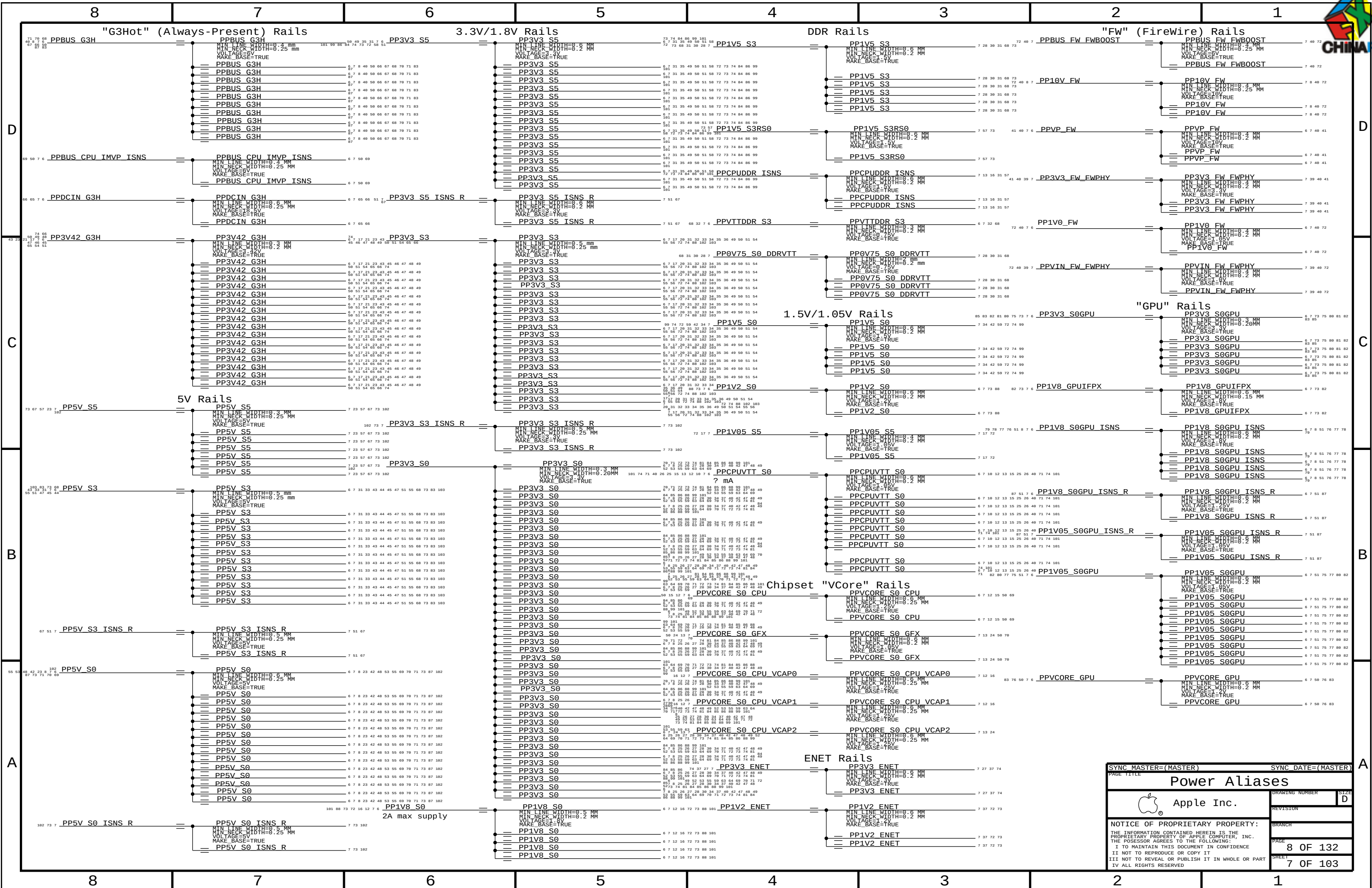


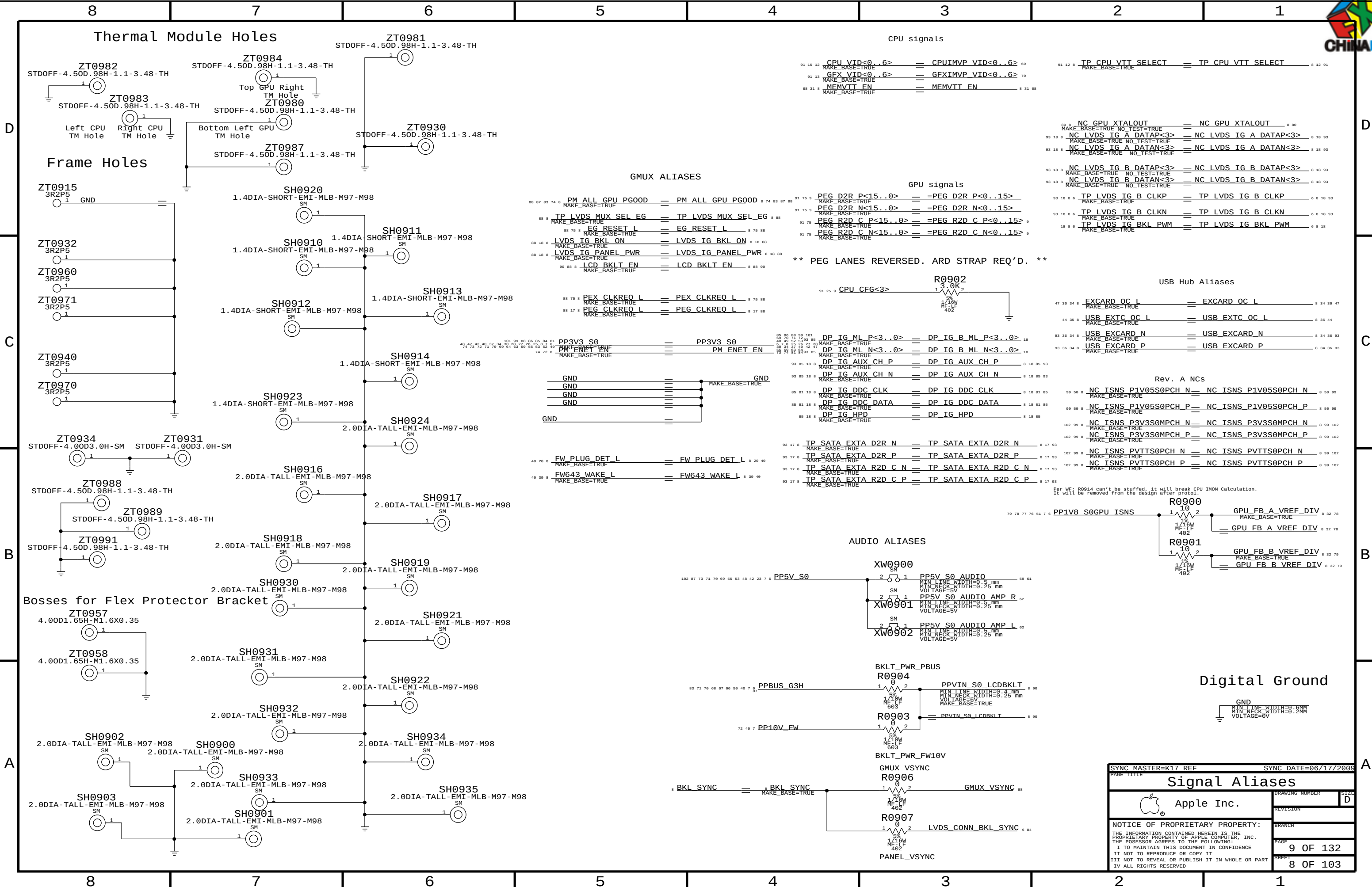
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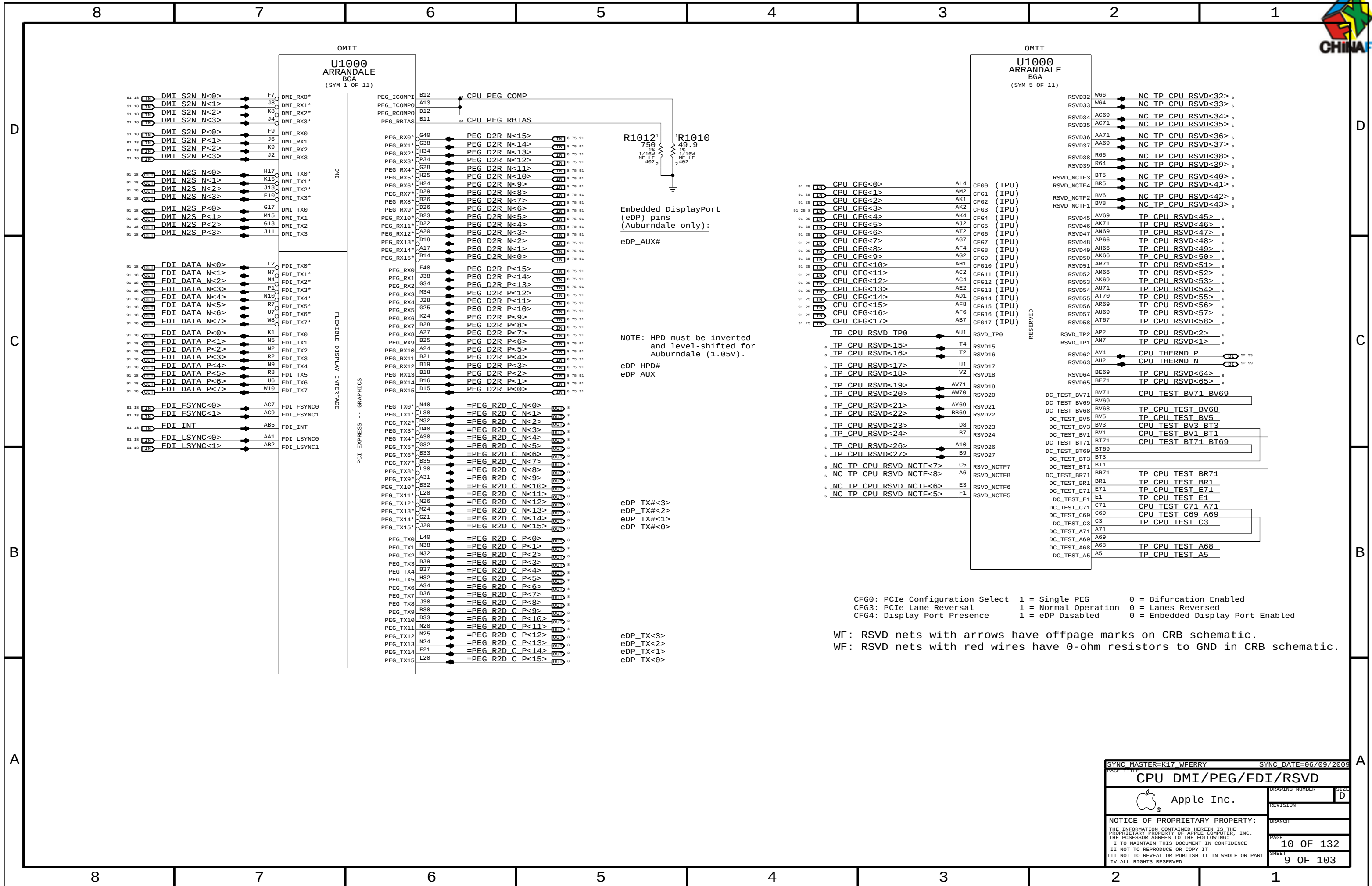
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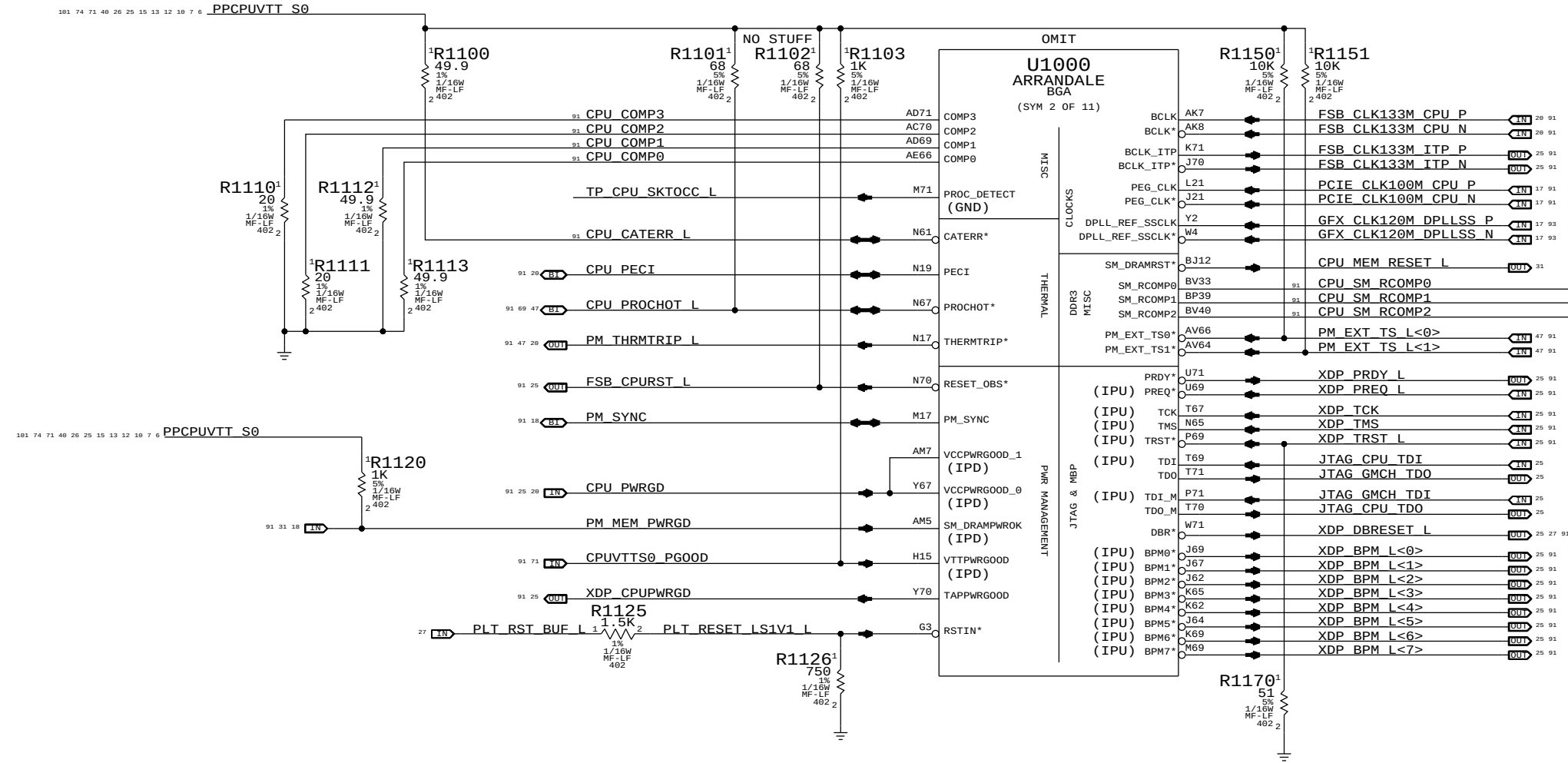
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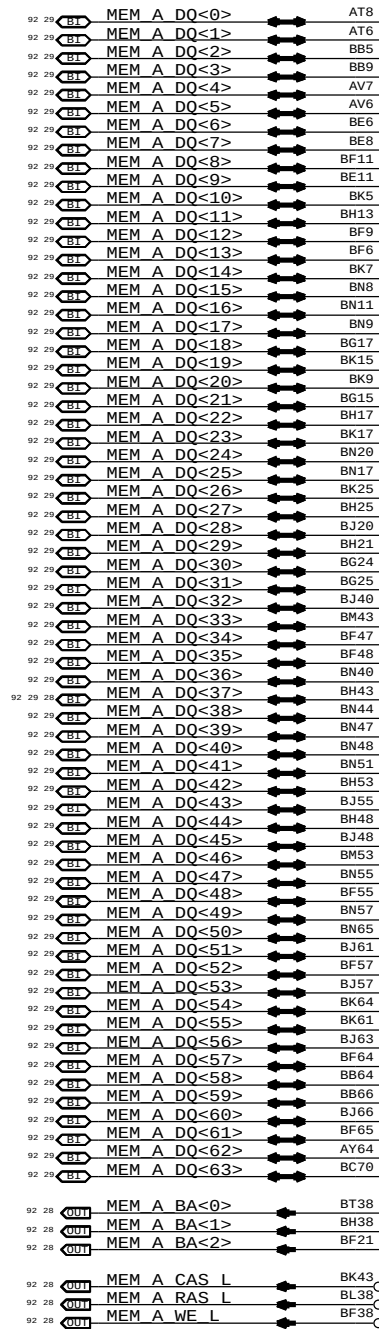




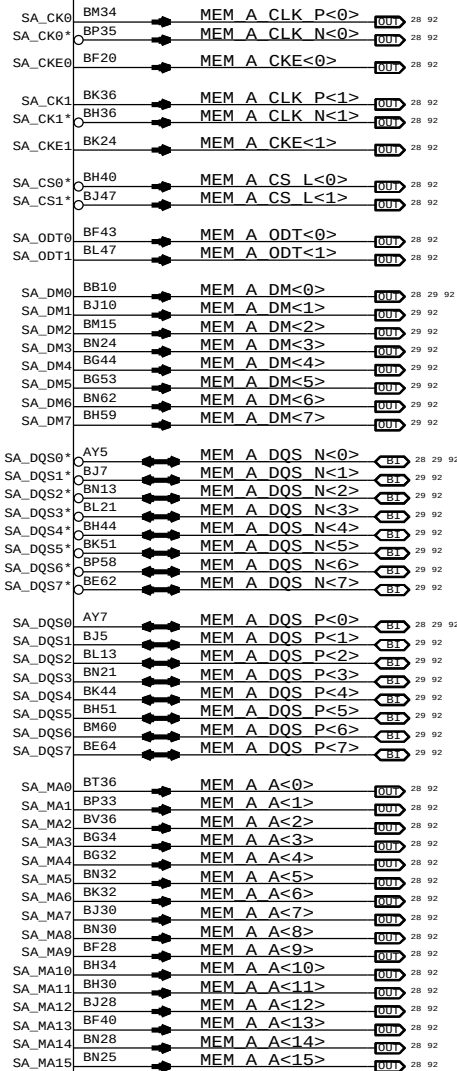
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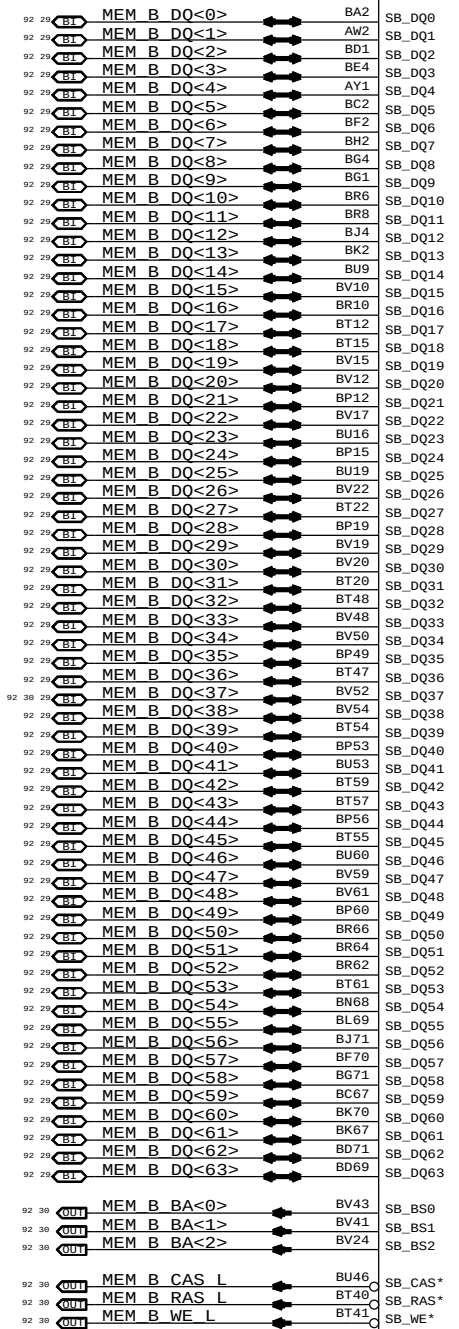


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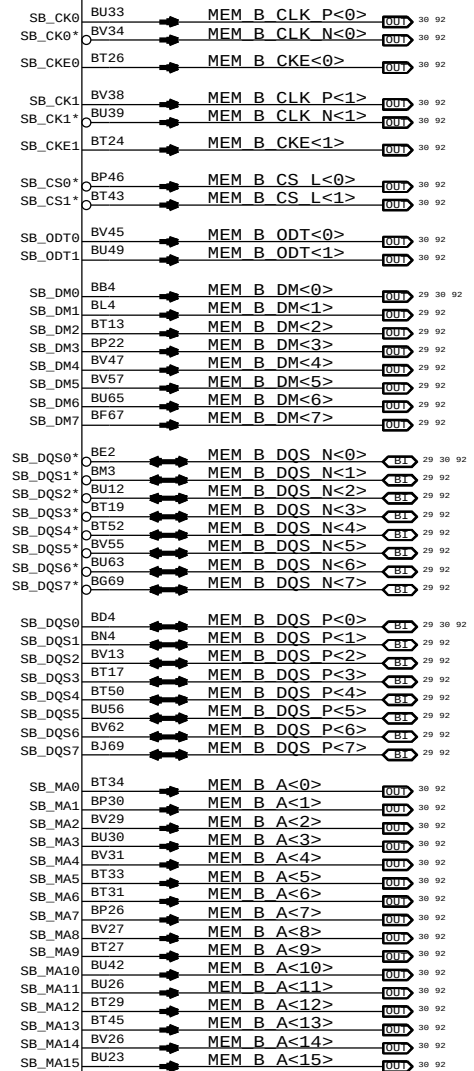


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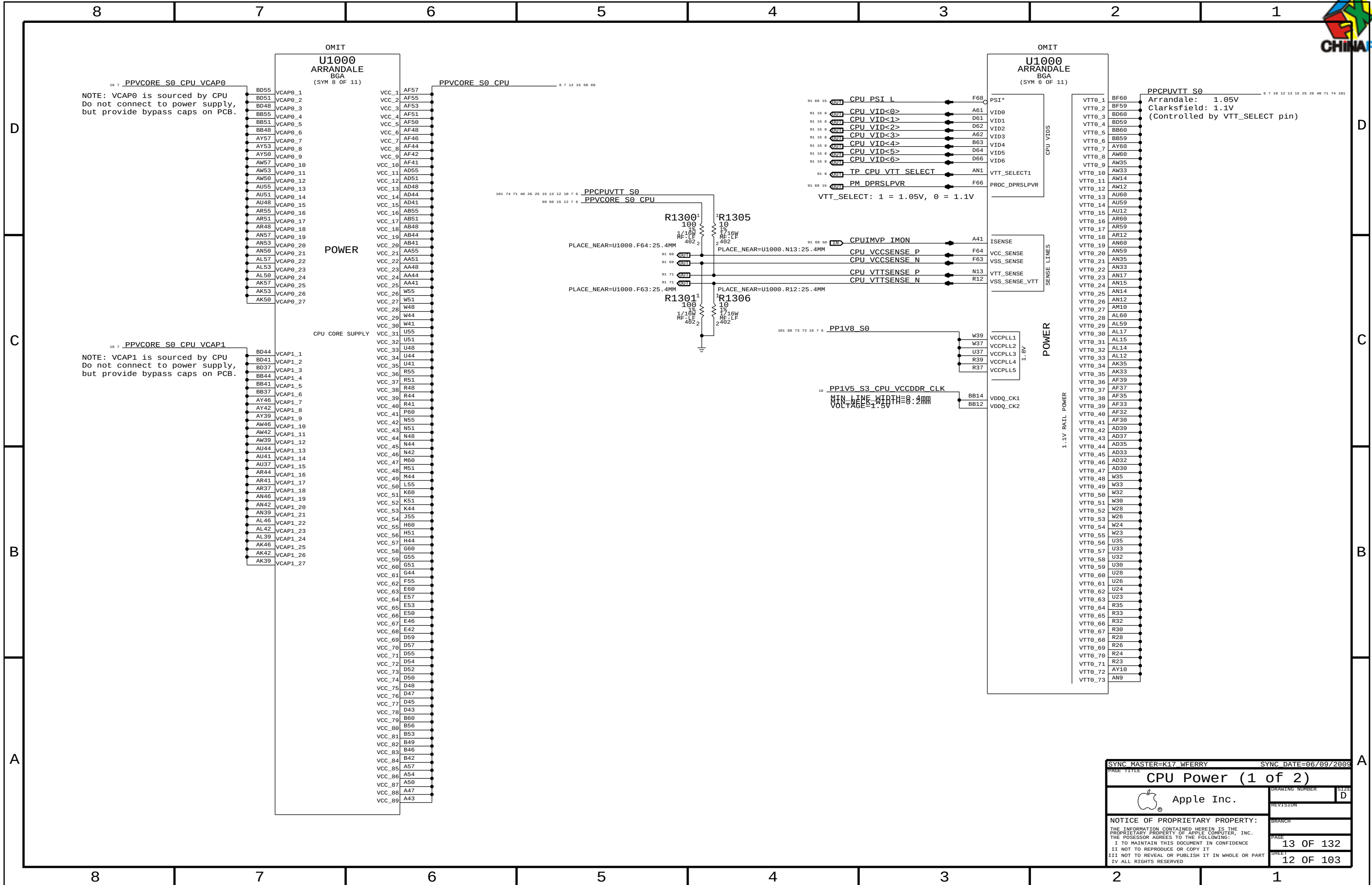
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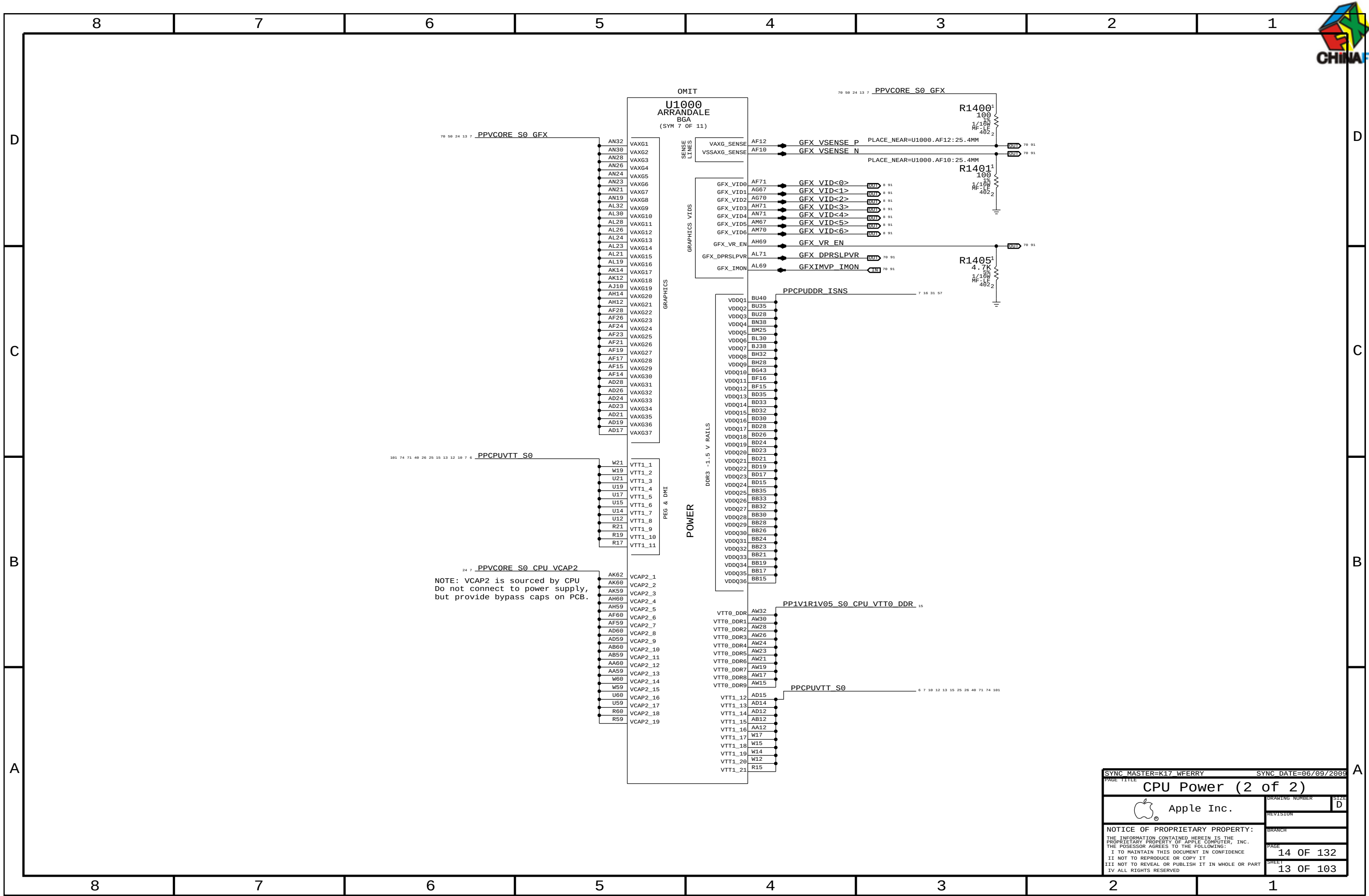


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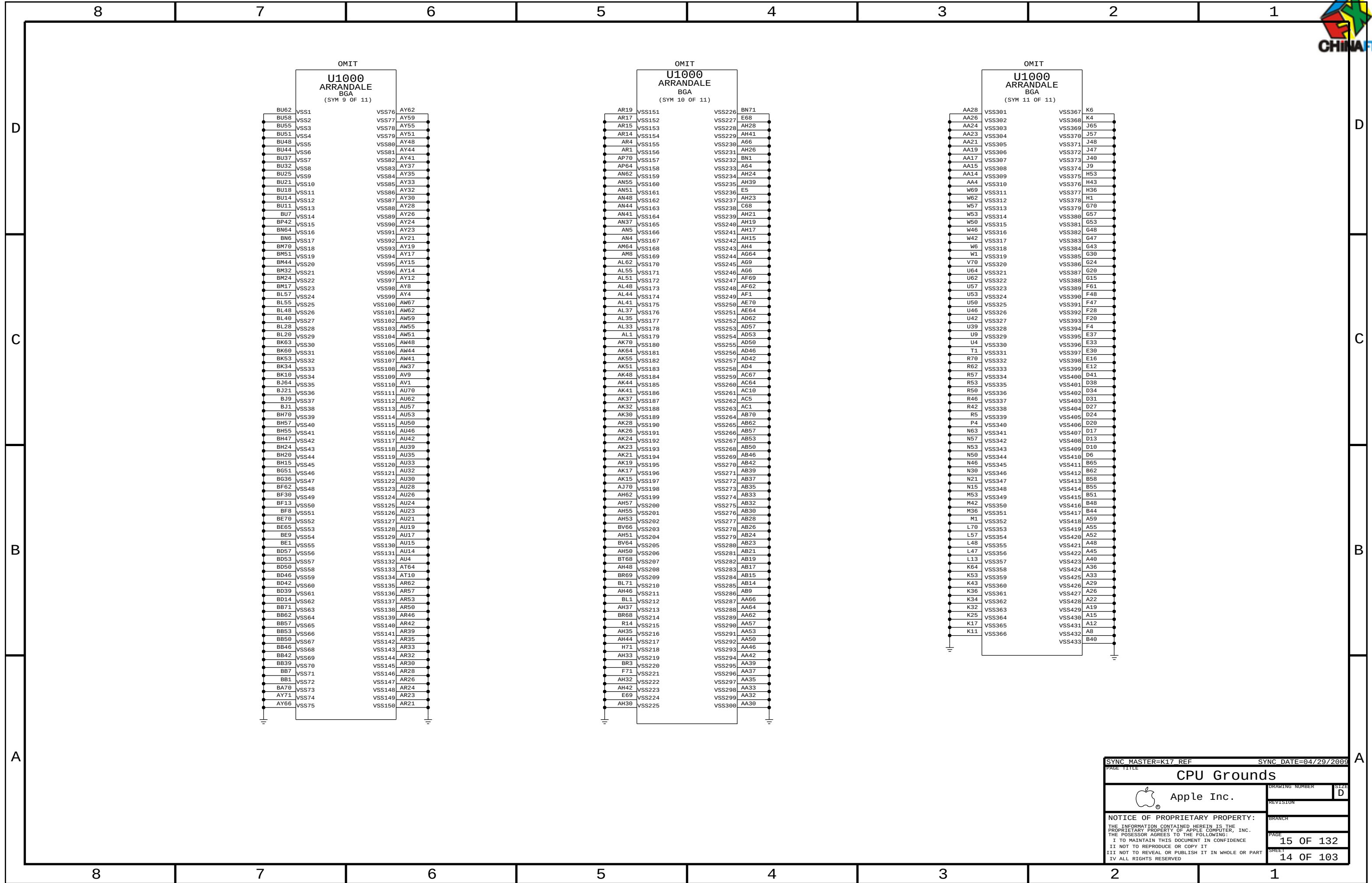


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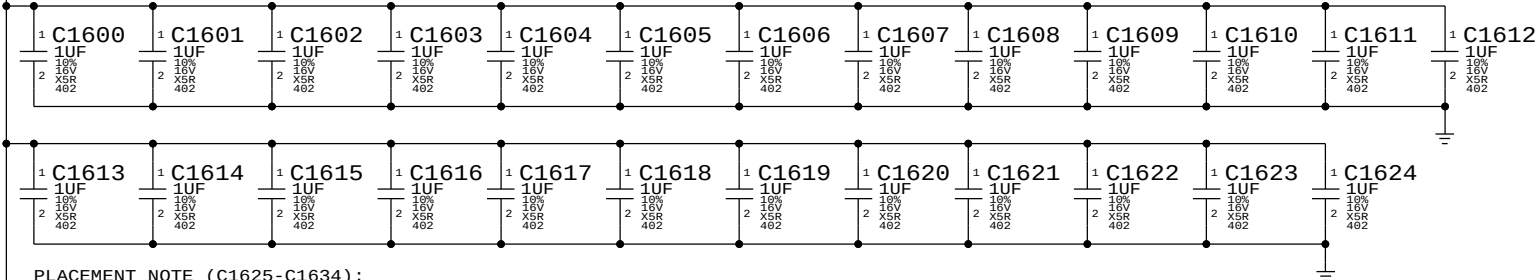


CPU VCore HF and Bulk Decoupling

3x 470uF 4.5mOhm, 1x 330uF, 15x 22uF 0603, 25x 1uF 0402

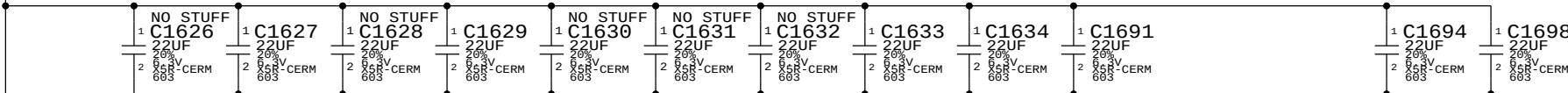
PLACEMENT_NOTE (C1600-C1624):

Place on bottom side of U1000..



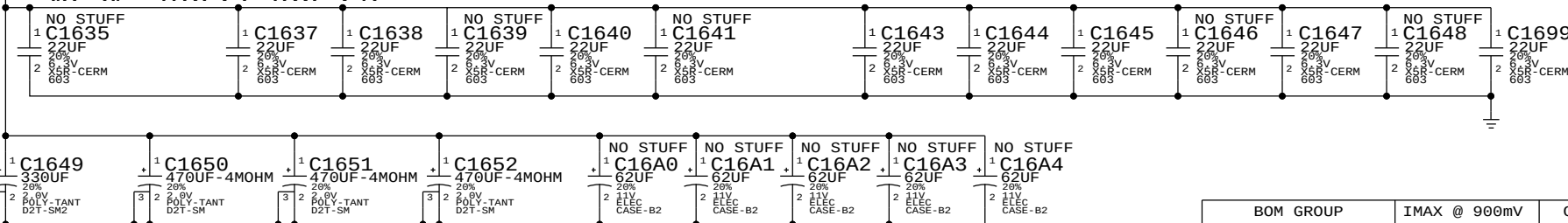
PLACEMENT_NOTE (C1625-C1634):

Place near U1000 on bottom side.



PLACEMENT_NOTE (C1635-C1648):

Place near inductors on bottom side.

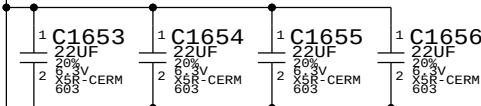


VTT (CPU Uncore) DECOUPLING

3x 330uF 6 mOhm, 4x 22uF 0805, 7x 10uF 0603, 24x 1uF 0402

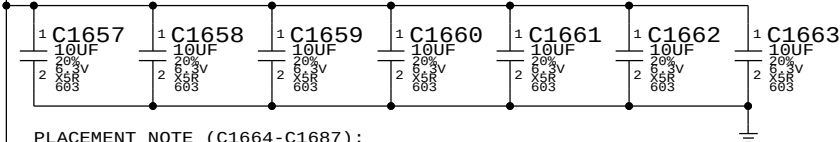
PLACEMENT_NOTE (C1653-C1656):

Place on bottom side of U1000..



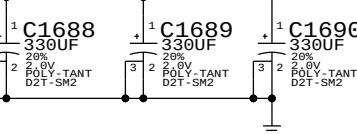
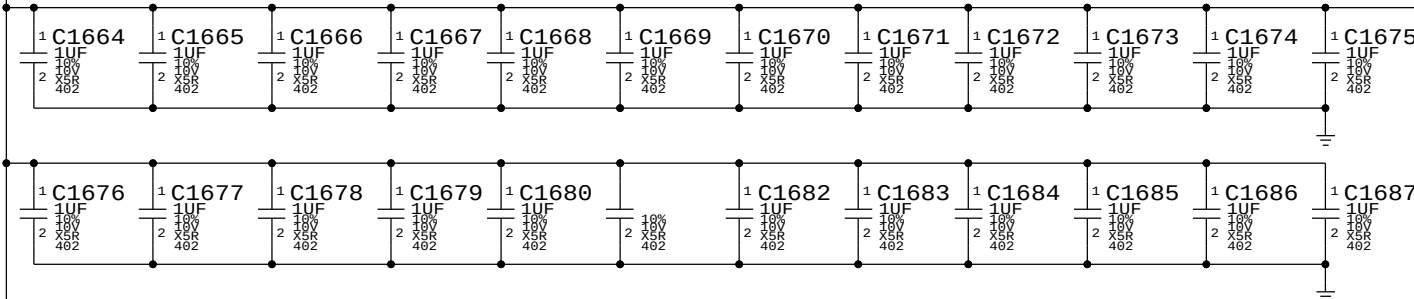
PLACEMENT_NOTE (C1657-C1663):

Place on bottom side of U1000..



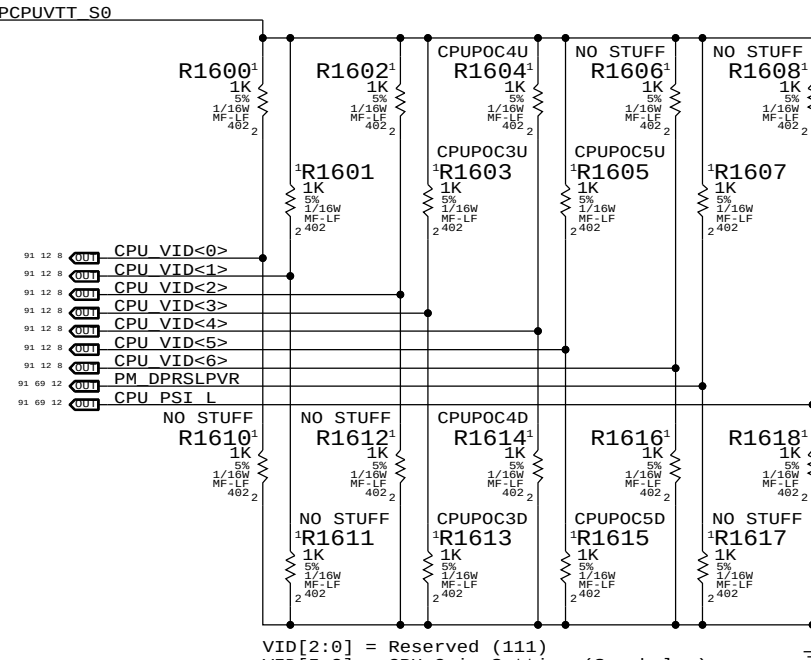
PLACEMENT_NOTE (C1664-C1687):

Place on bottom side of U1000.



CPU Power On Configuration (POC) Straps

Intel recommends all option straps should be provided in layout



VID[2:0] = Reserved (111)
VID[5:3] = GPU Gain Setting (See below)
VID[6] = Reserved (0)
DPRSLPVR = 1 - IMVP-6.5 compliant controller
PSI# = Reserved (0)

BOM GROUP	IMAX @ 900mV	CPU Gain Setting	BOM OPTIONS	Equivalent Gain
CPUPOC_IMAX_DIS		000	CPUPOC3D, CPUPOC4D, CPUPOC5D	
CPUPOC_IMAX_0_20	20A	001	CPUPOC3D, CPUPOC4D, CPUPOC5U	45
CPUPOC_IMAX_20_30	30A	010	CPUPOC3D, CPUPOC4U, CPUPOC5D	30
CPUPOC_IMAX_30_40	40A	011	CPUPOC3D, CPUPOC4U, CPUPOC5U	22.5
CPUPOC_IMAX_40_50	50A	100	CPUPOC3U, CPUPOC4D, CPUPOC5D	18
CPUPOC_IMAX_50_60	60A	101	CPUPOC3U, CPUPOC4D, CPUPOC5U	15
CPUPOC_IMAX_60_70	70A	110	CPUPOC3U, CPUPOC4U, CPUPOC5D	12.857
CPUPOC_IMAX_70_90	90A	111	CPUPOC3U, CPUPOC4U, CPUPOC5U	10

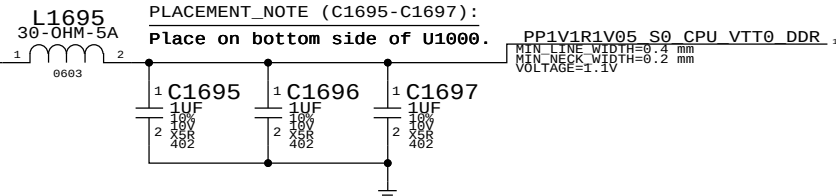
NOTE: BOM Configurations should not call out CPUPOCnU/D BOMOPTIONS directly.
Instead call out appropriate BOM GROUP defined in tables above.

VTT0_DDR DECOUPLING

3x 1uF 0402

PLACEMENT_NOTE (C1695-C1697):

Place on bottom side of U1000.



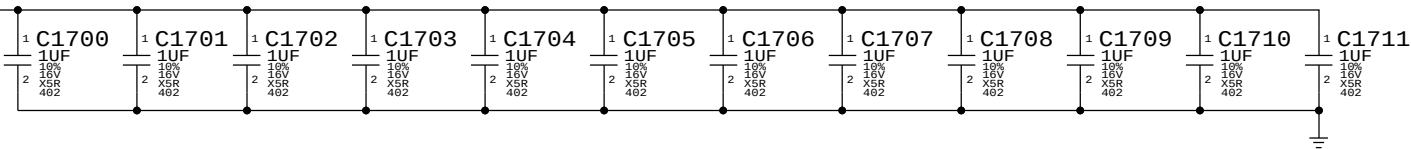


VCAP0 (CPU BSC Package) DECOUPLING

12x 1uF 0402

PLACEMENT_NOTE (C1700-C1711):

Place on bottom side of U1000.

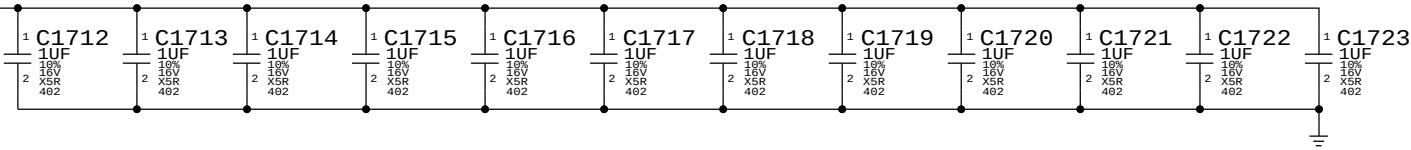


VCAP1 (CPU BSC Package) DECOUPLING

12x 1uF 0402

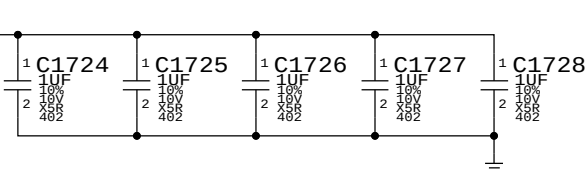
PLACEMENT_NOTE (C1712-C1723):

Place on bottom side of U1000.

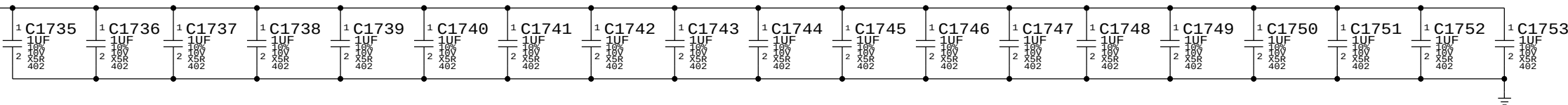


Memory (CPU VCCDDR) DECOUPLING

5x 1uF 0402



NOTE: 19x 1uF 0402 caps per Apple SI for CMD and CNTRL lines.

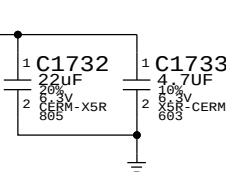


NOTE: 3x 330uF 6 mOhm caps to be shared between CPU and S0-DIMMs. 2x330uF on CSA73. DG recommends 2x 22uF at S0_DIMM not provided. Decoupling caps at S0-DIMMs on CSA 29 and CSA 31.



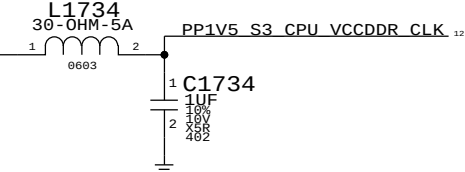
PLL (CPU VCCSFR) DECOUPLING

1x 22uF 0805, 1x 4.7uF 0603

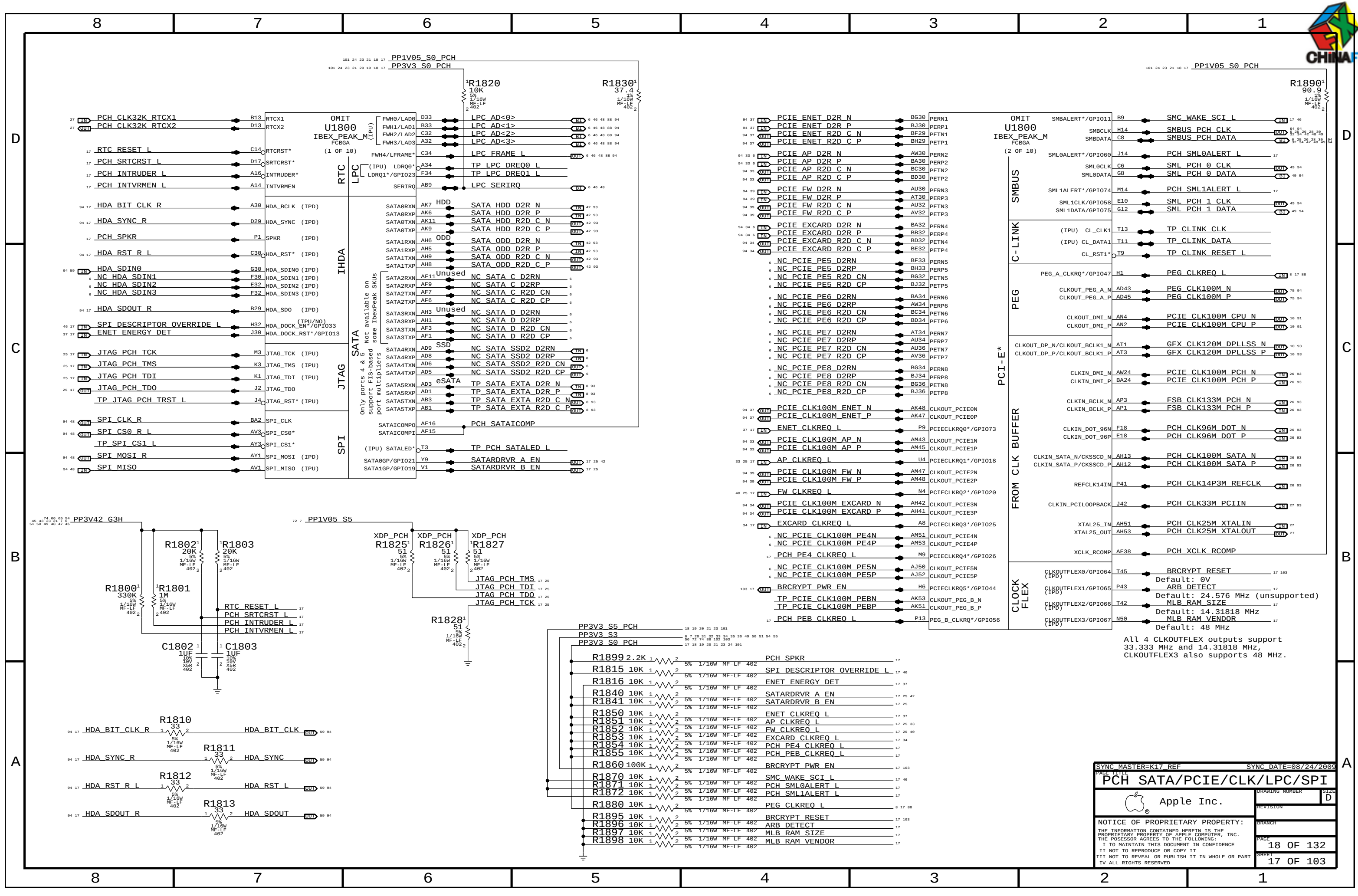


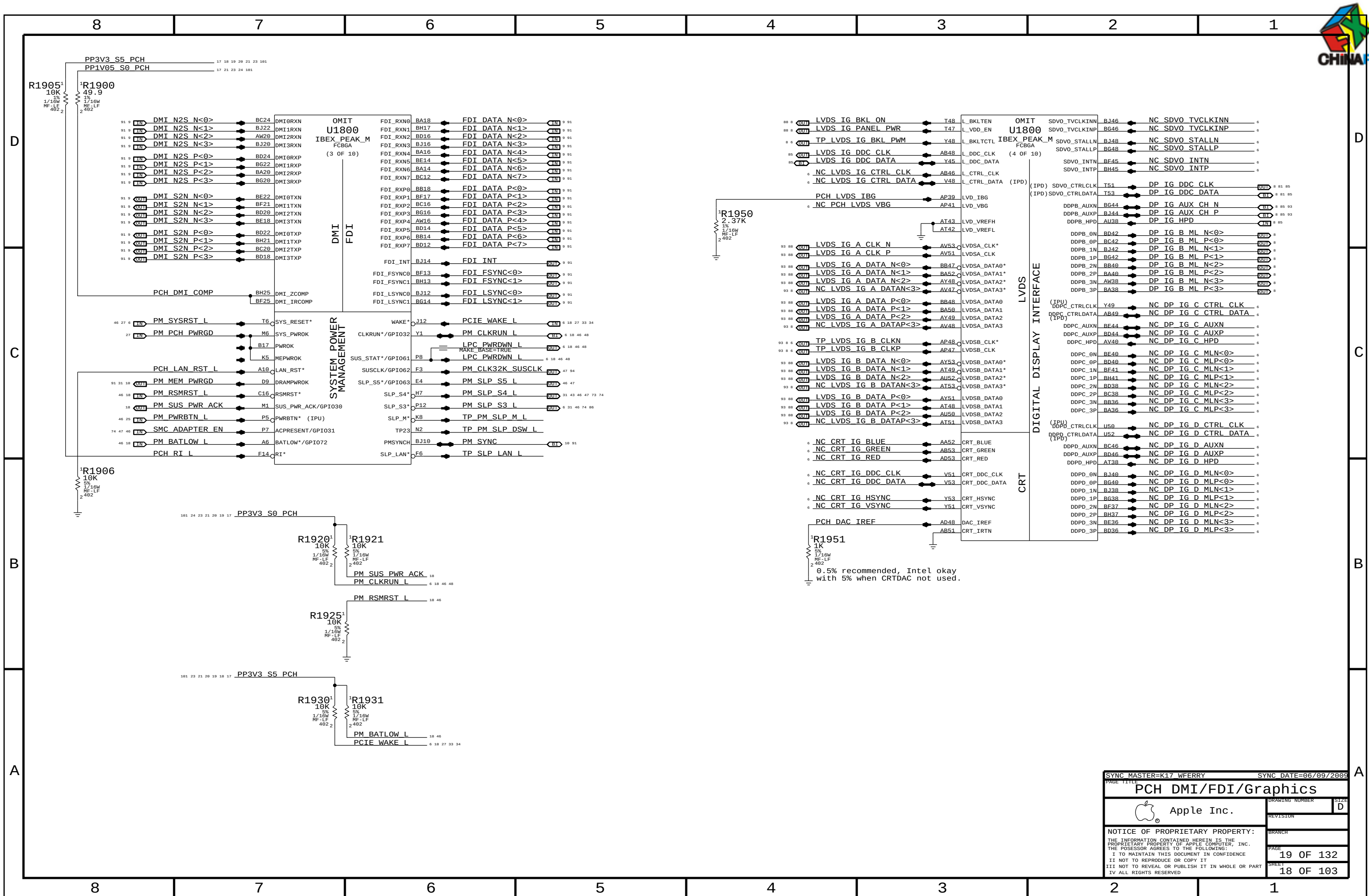
DDR Clock (CPU VDDQ_CK) DECOUPLING

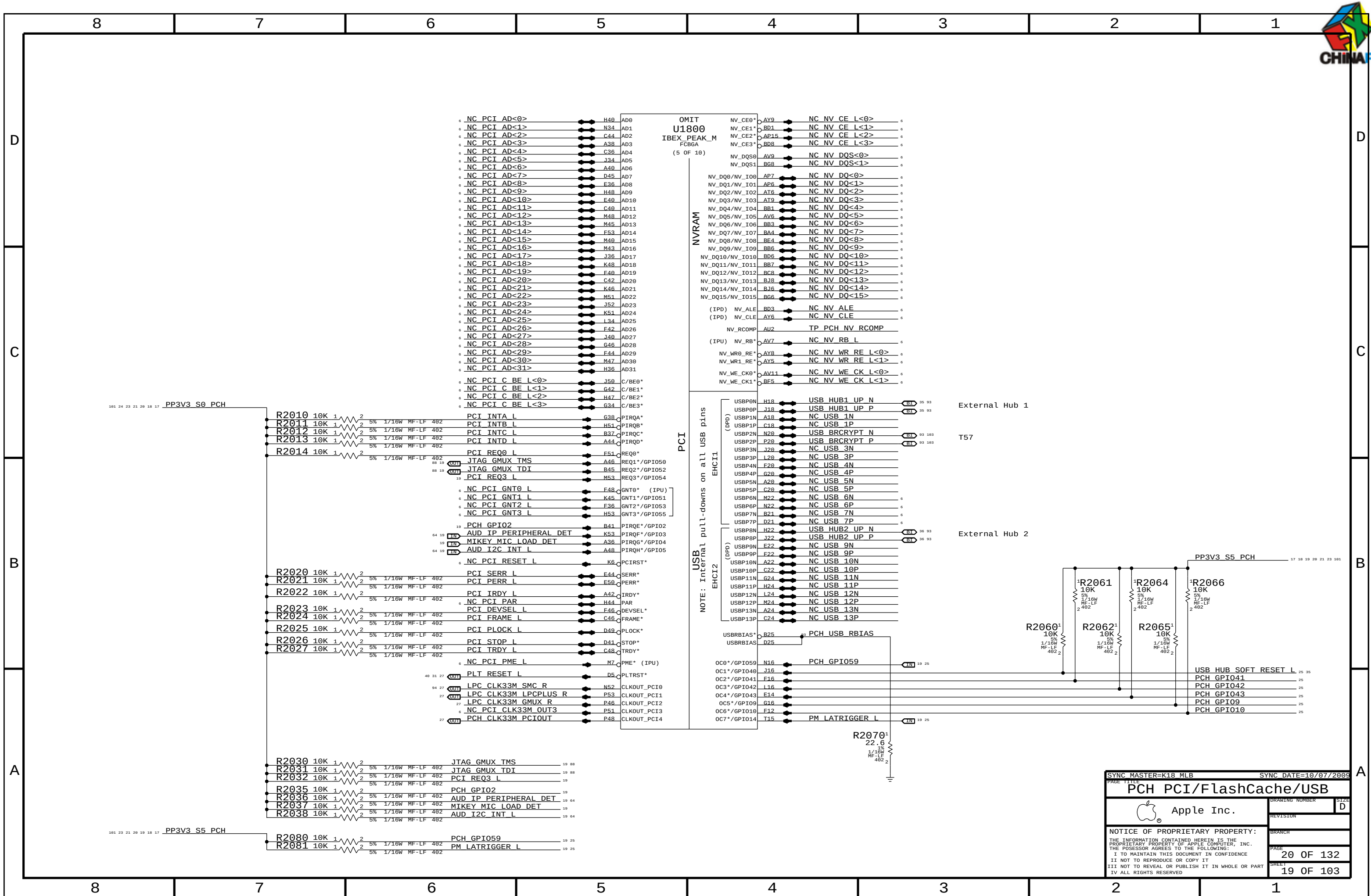
1x 1uF 0402

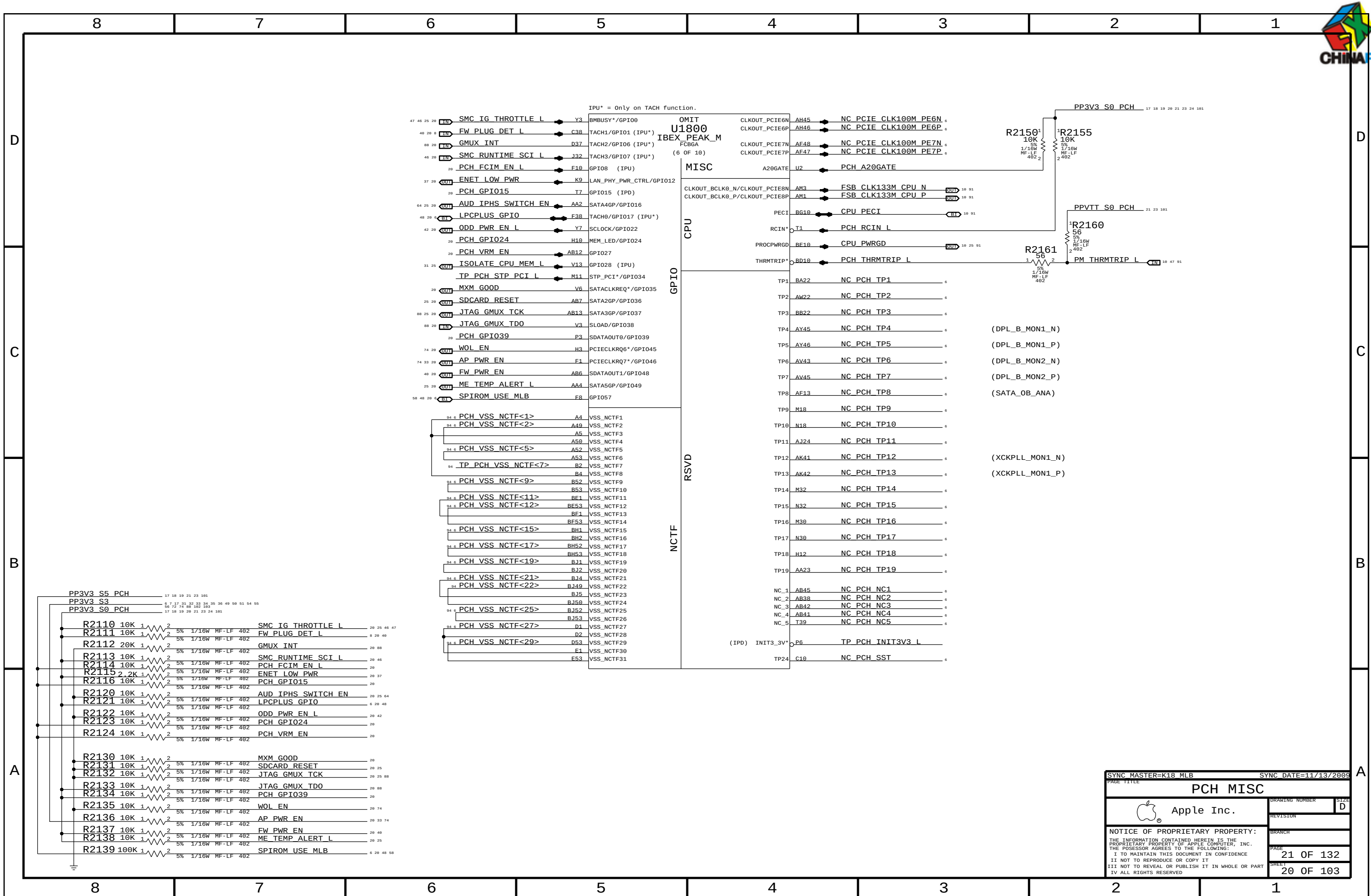


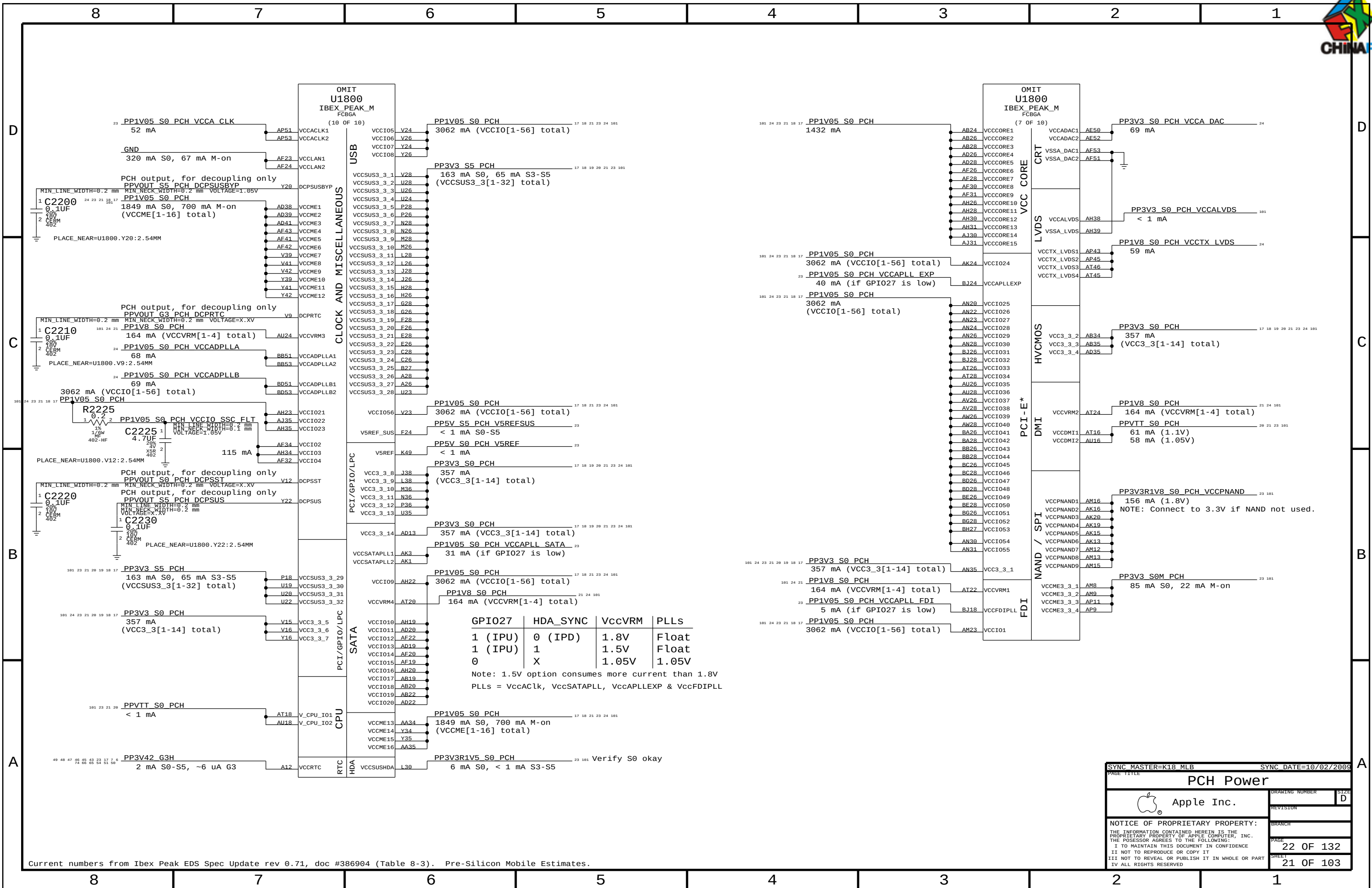
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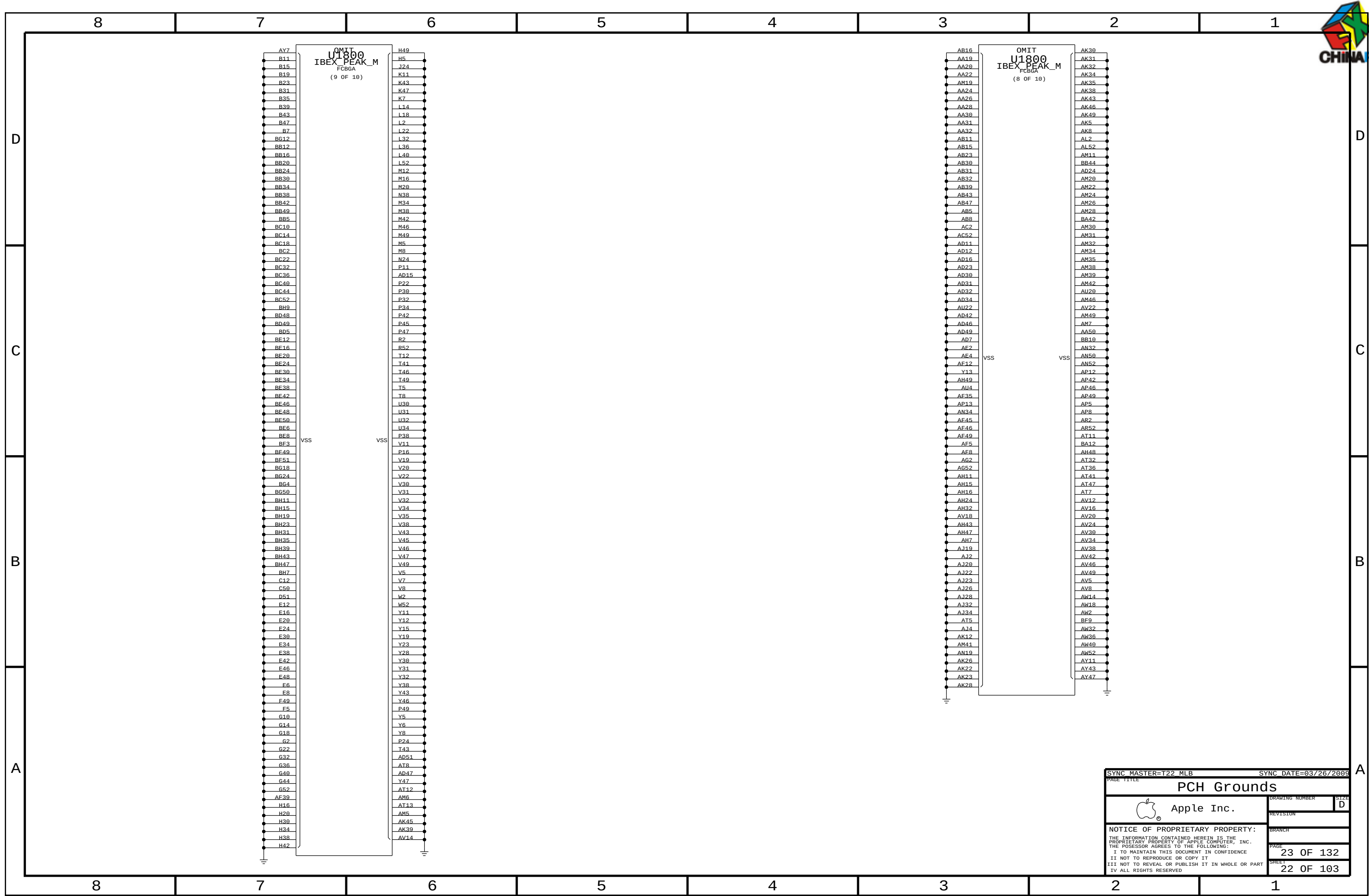


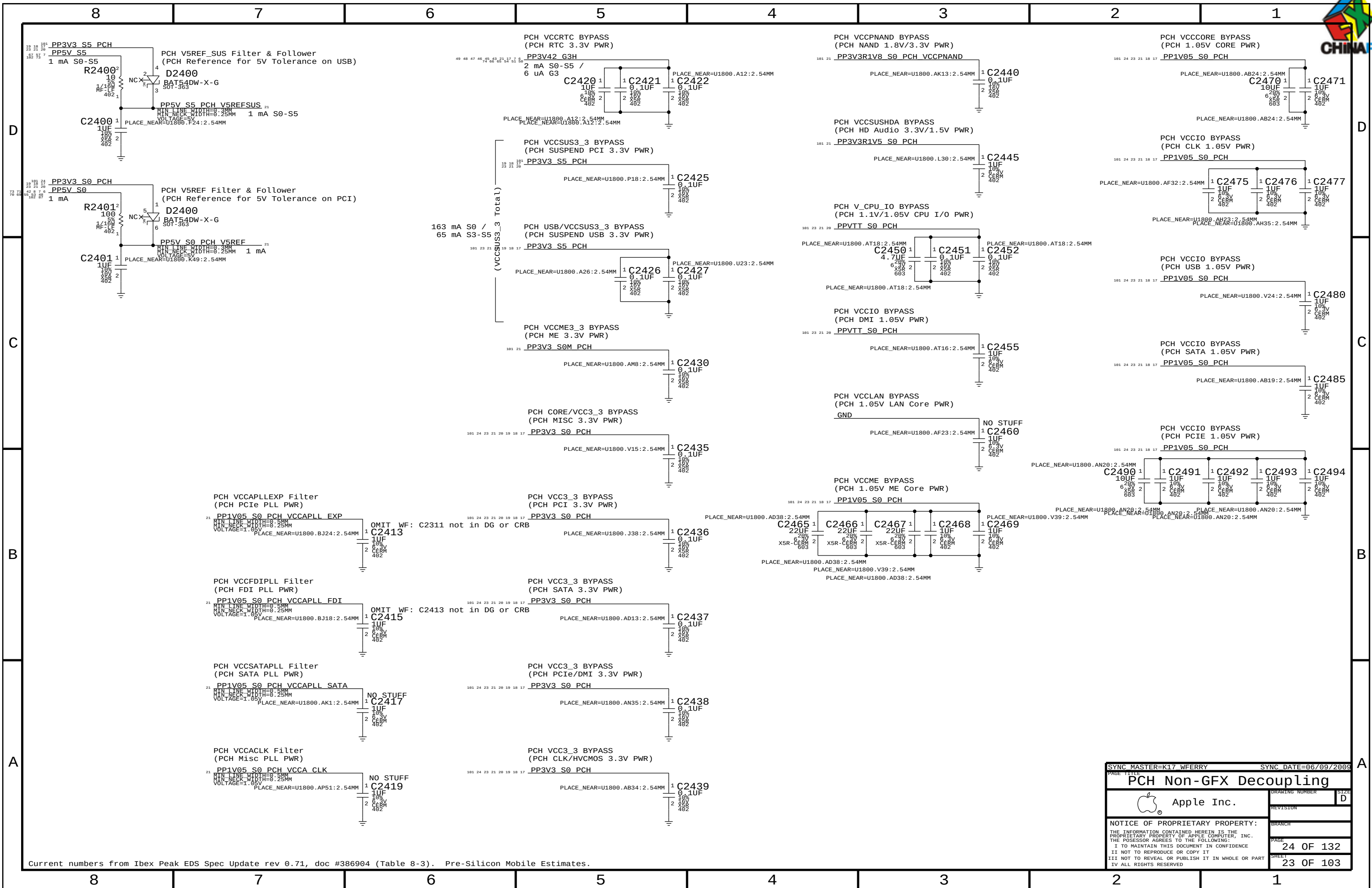












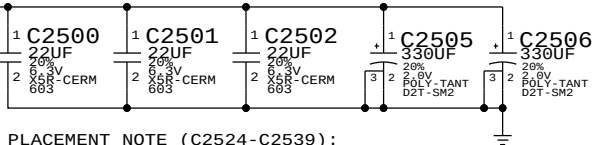


GFX (CPU VCCAXG) DECOUPLING

3x 330uF 6 mOhm (2 stuffed), 3x 22uF 0603, 16x 1uF 0402

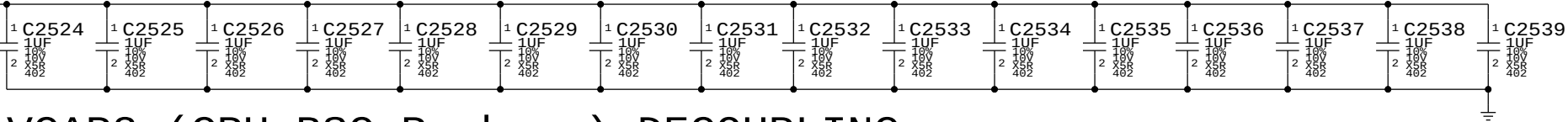
PLACEMENT_NOTE (C2500-C2506):

Place on bottom side of U1000.



PLACEMENT_NOTE (C2524-C2539):

Place on bottom side of U1000.

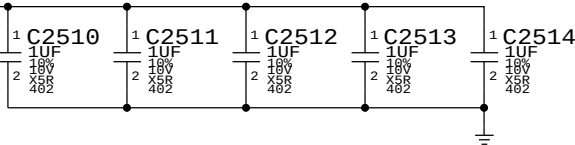


VCAP2 (CPU BSC Package) DECOUPLING

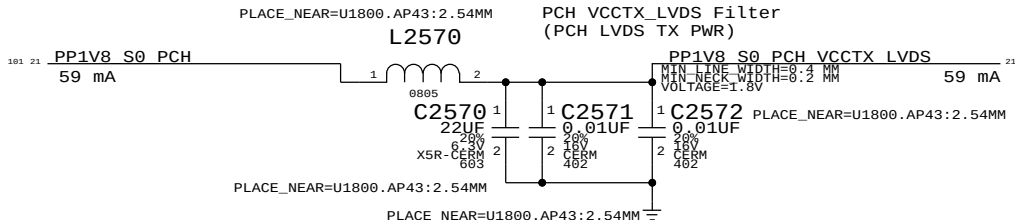
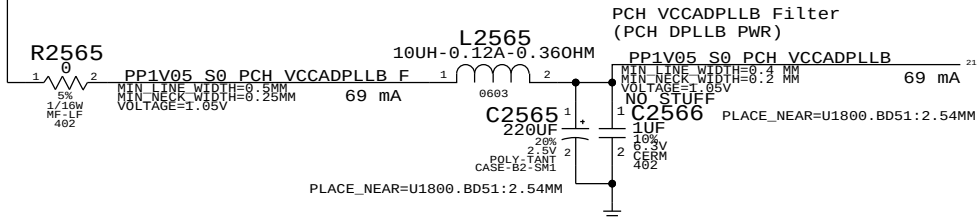
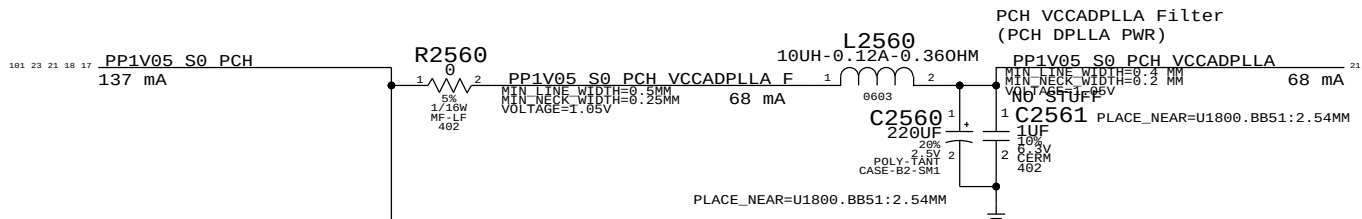
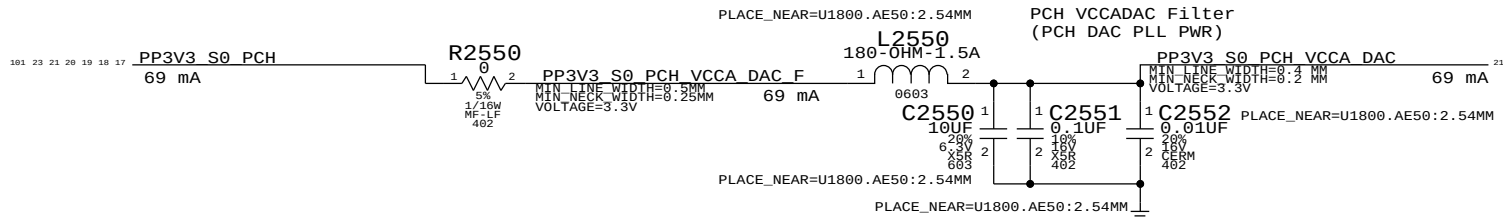
5x 1uF 0402

PLACEMENT_NOTE (C2510-C2514):

Place on bottom side of U1000.



Design recommendations from Calpella Small Form Factor Design Guide Rev 1.5 (doc #407364) table 2-34 and Calpella Small Form F actor Schematic Check List Rev 1.1 (doc #395914) table 3.26.



Design recommendations from Calpella Design Guide Rev 1.5 (doc #398905) Section 3.25.3 tables 161 and 162.

Current numbers from Ibox Peak EDS Spec Update rev 0.71, doc #386904 (Table 8-3). Pre-Silicon Mobile Estimates.

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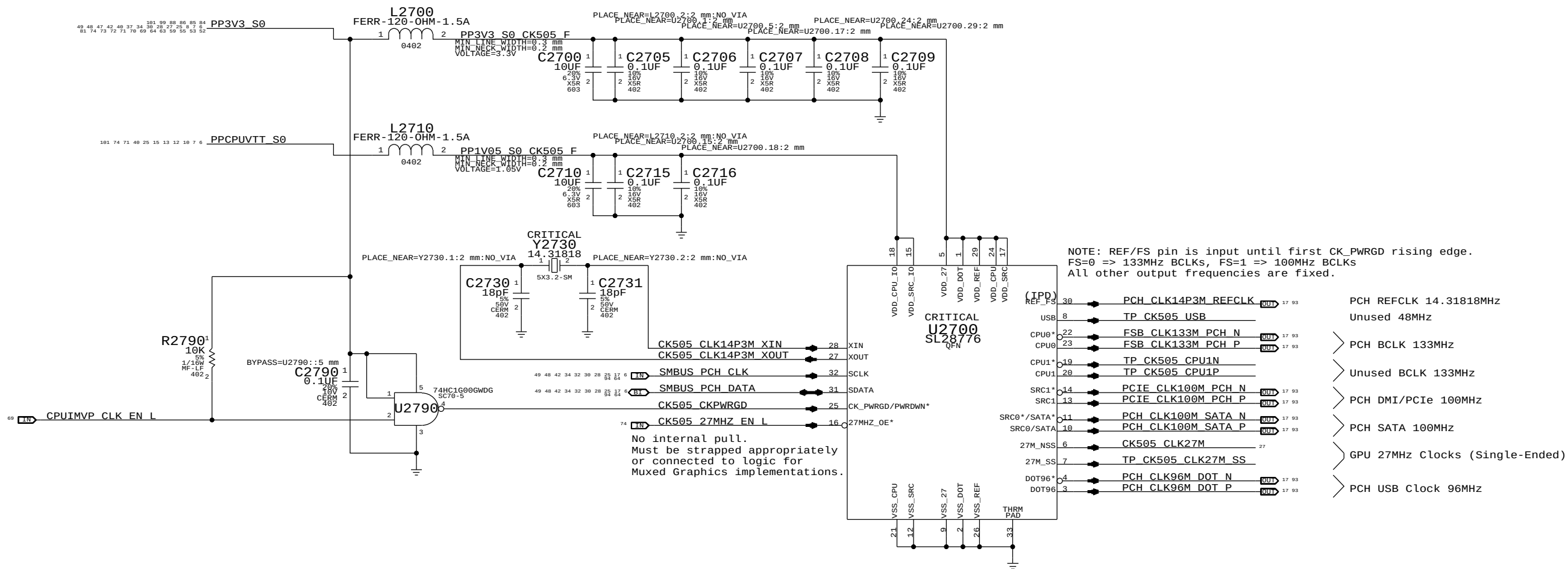
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SYNC_MASTER=K17_REF		SYNC_DATE=05/19/2009	
PAGE TITLE		Clock (CK505)	
Apple Inc.		DRAWING NUMBER	SIZE
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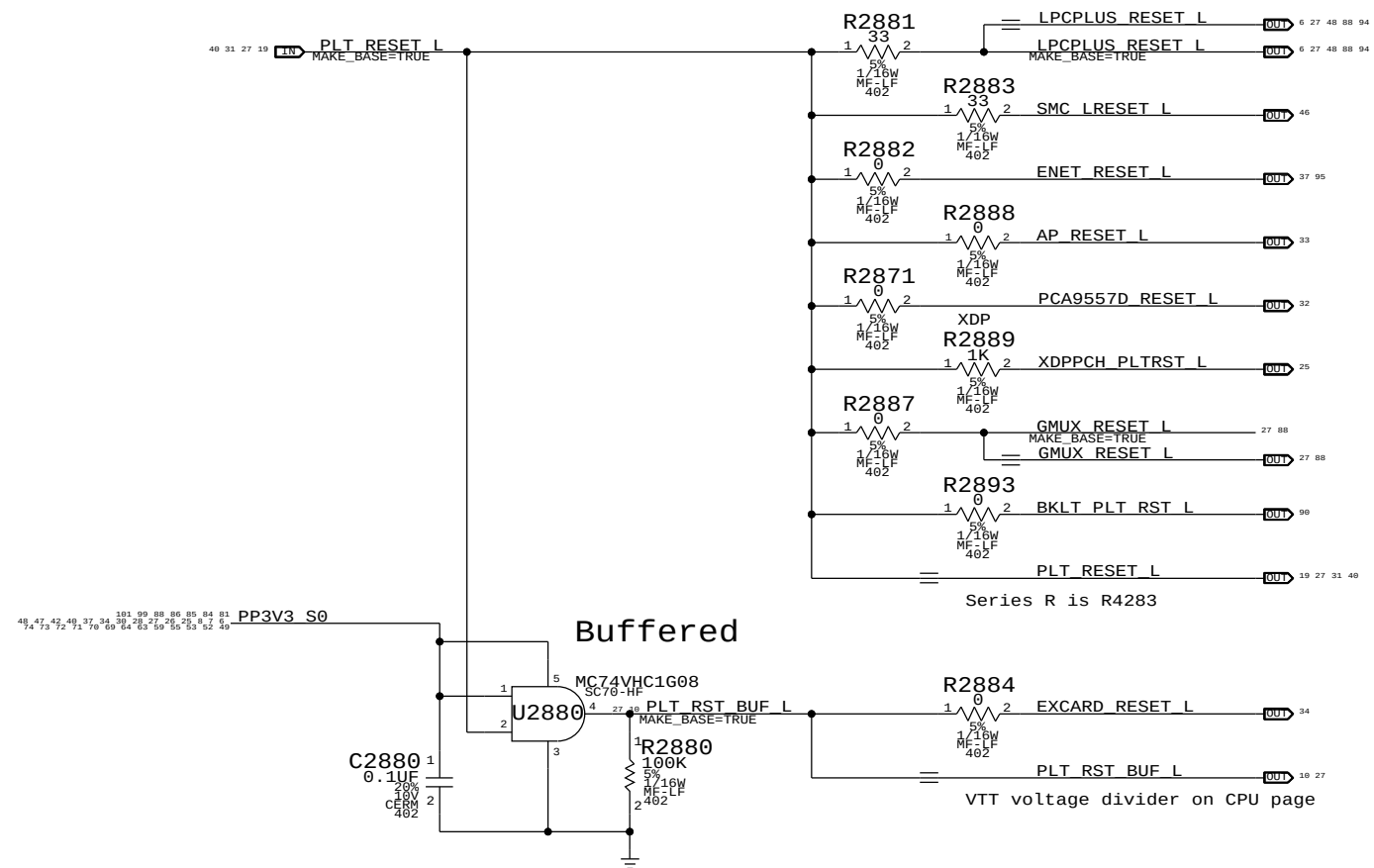
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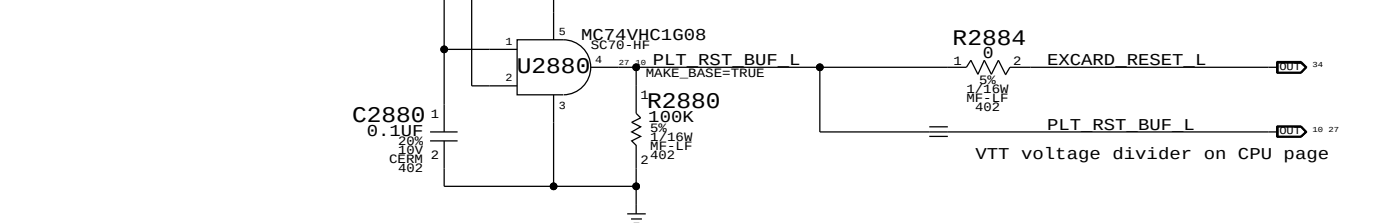
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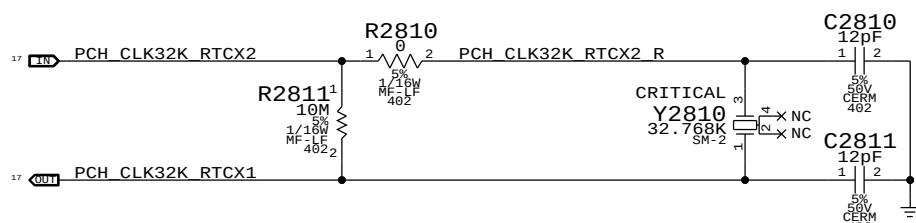
Platform Reset Connections
Unbuffered



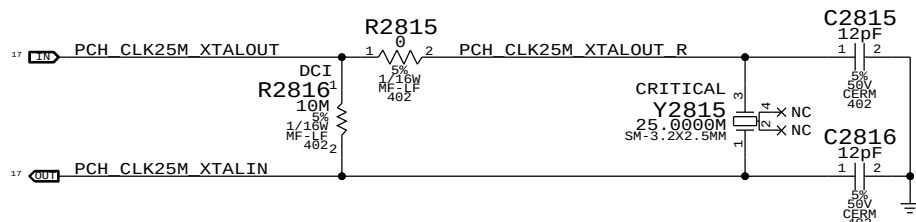
Buffered



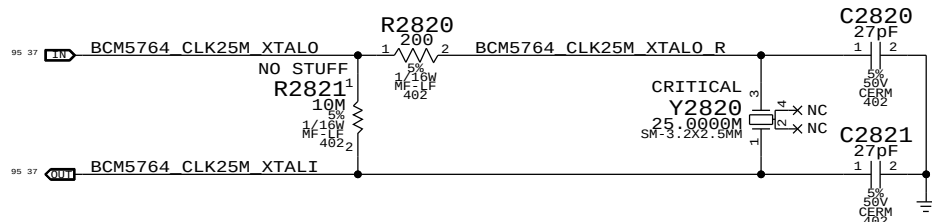
PCH RTC Crystal



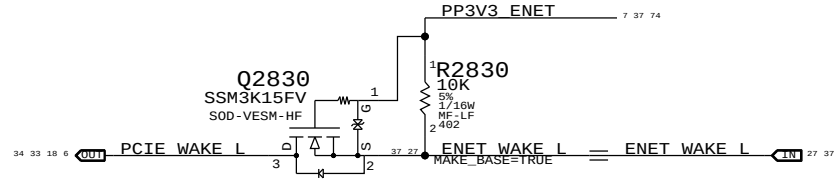
PCH 25MHz Crystal



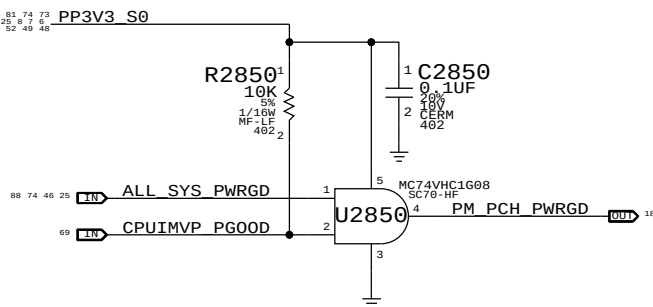
Caesar II (ENET) 25MHz Crystal



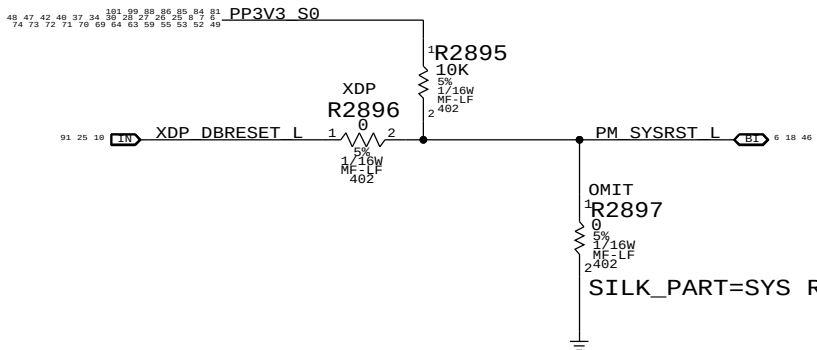
Ethernet WAKE# Isolation



PCH S0 PWRGD



PCH Reset Button



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Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

D

D

C

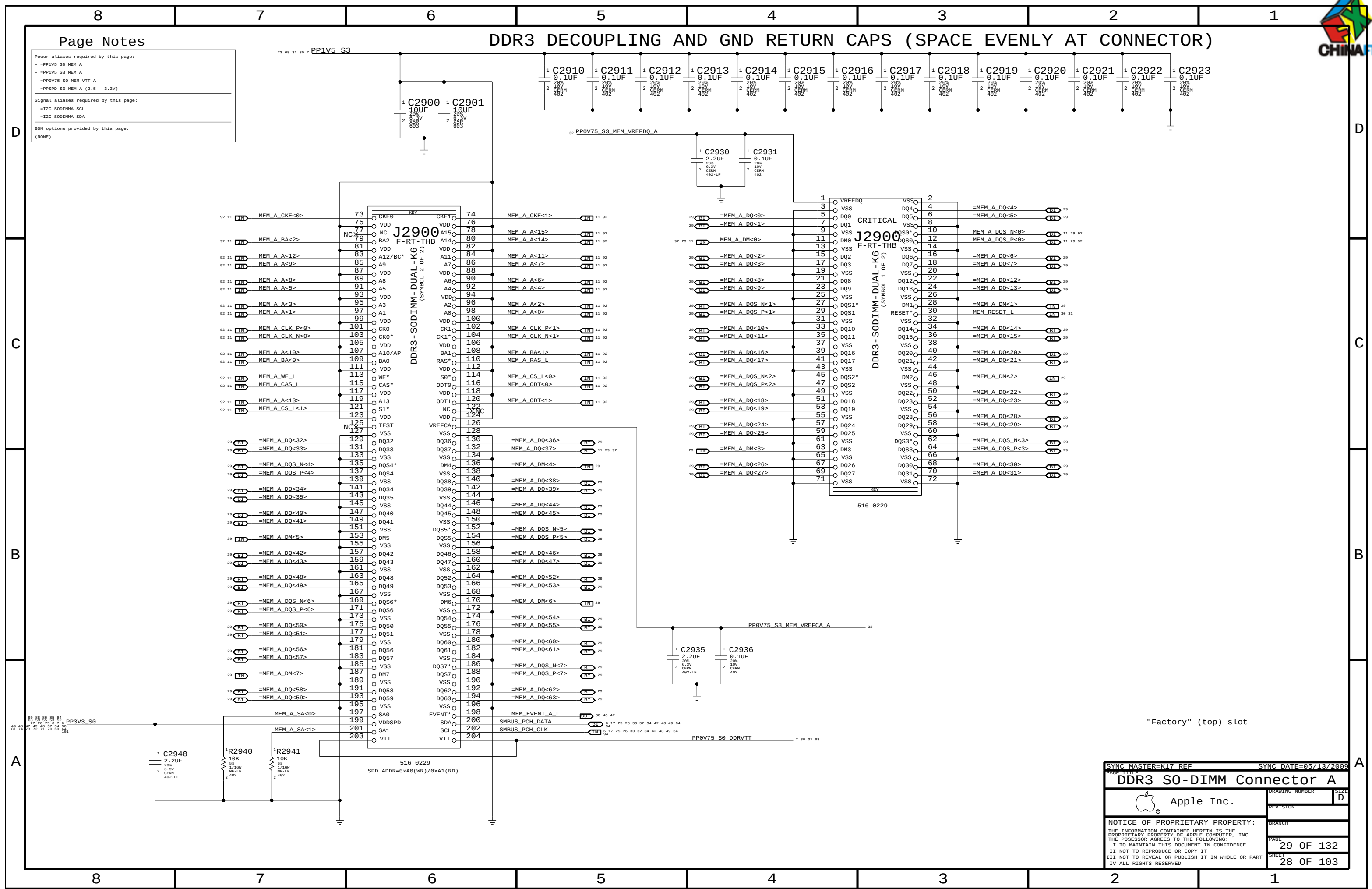
C

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"Factory" (top) slot

SYNC MASTER=K17 REF		SYNC DATE=05/13/2009	
PAGE TITLE			
DDR3 S0-DIMM Connector A			
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8		7		6		5		4		3		2		1	
D	CPU CHANNEL A DQS 0 -> DIMM A DQS 0		CPU CHANNEL B DQS 0 -> DIMM B DQS 0												
	92 29 28 11	MEM A DQS N<0>	MAKE_BASE=TRUE	MEM A DQS N<0>	11 29 29 92 92 30 29 11	MEM B DQS N<0>	MAKE_BASE=TRUE	MEM B DQS N<0>	11 29 30 92						
	92 29 28 11	MEM A DQS P<0>	MAKE_BASE=TRUE	MEM A DQS P<0>	11 29 29 92 92 30 29 11	MEM B DQS P<0>	MAKE_BASE=TRUE	MEM B DQS P<0>	11 29 30 92						
	92 29 28 11	MEM A DM<0>	MAKE_BASE=TRUE	MEM A DM<0>	11 29 29 92 92 30 29 11	MEM B DM<0>	MAKE_BASE=TRUE	MEM B DM<0>	11 29 30 92						
	92 11	MEM A DQ<7>	MAKE_BASE=TRUE	=MEM A DQ<7>	28	92 11	MEM B DQ<7>	MAKE_BASE=TRUE	=MEM B DQ<7>	30					
	92 11	MEM A DQ<6>	MAKE_BASE=TRUE	=MEM A DQ<6>	28	92 11	MEM B DQ<6>	MAKE_BASE=TRUE	=MEM B DQ<6>	30					
	92 11	MEM A DQ<5>	MAKE_BASE=TRUE	=MEM A DQ<5>	28	92 11	MEM B DQ<5>	MAKE_BASE=TRUE	=MEM B DQ<5>	30					
	92 11	MEM A DQ<4>	MAKE_BASE=TRUE	=MEM A DQ<4>	28	92 11	MEM B DQ<4>	MAKE_BASE=TRUE	=MEM B DQ<4>	30					
	92 11	MEM A DQ<3>	MAKE_BASE=TRUE	=MEM A DQ<3>	28	92 11	MEM B DQ<3>	MAKE_BASE=TRUE	=MEM B DQ<3>	30					
	92 11	MEM A DQ<2>	MAKE_BASE=TRUE	=MEM A DQ<2>	28	92 11	MEM B DQ<2>	MAKE_BASE=TRUE	=MEM B DQ<2>	30					
C	CPU CHANNEL A DQS 1 -> DIMM A DQS 1		CPU CHANNEL B DQS 1 -> DIMM B DQS 1												
	92 11	MEM A DQS N<1>	MAKE_BASE=TRUE	=MEM A DQS N<1>	28	92 11	MEM B DQS N<1>	MAKE_BASE=TRUE	=MEM B DQS N<1>	30					
	92 11	MEM A DQS P<1>	MAKE_BASE=TRUE	=MEM A DQS P<1>	28	92 11	MEM B DQS P<1>	MAKE_BASE=TRUE	=MEM B DQS P<1>	30					
	92 11	MEM A DM<1>	MAKE_BASE=TRUE	=MEM A DM<1>	28	92 11	MEM B DM<1>	MAKE_BASE=TRUE	=MEM B DM<1>	30					
	92 11	MEM A DQ<15>	MAKE_BASE=TRUE	=MEM A DQ<15>	28	92 11	MEM B DQ<15>	MAKE_BASE=TRUE	=MEM B DQ<15>	30					
	92 11	MEM A DQ<14>	MAKE_BASE=TRUE	=MEM A DQ<14>	28	92 11	MEM B DQ<14>	MAKE_BASE=TRUE	=MEM B DQ<14>	30					
	92 11	MEM A DQ<13>	MAKE_BASE=TRUE	=MEM A DQ<13>	28	92 11	MEM B DQ<13>	MAKE_BASE=TRUE	=MEM B DQ<13>	30					
	92 11	MEM A DQ<12>	MAKE_BASE=TRUE	=MEM A DQ<12>	28	92 11	MEM B DQ<12>	MAKE_BASE=TRUE	=MEM B DQ<12>	30					
	92 11	MEM A DQ<11>	MAKE_BASE=TRUE	=MEM A DQ<11>	28	92 11	MEM B DQ<11>	MAKE_BASE=TRUE	=MEM B DQ<11>	30					
	92 11	MEM A DQ<10>	MAKE_BASE=TRUE	=MEM A DQ<10>	28	92 11	MEM B DQ<10>	MAKE_BASE=TRUE	=MEM B DQ<10>	30					
B	CPU CHANNEL A DQS 2 -> DIMM A DQS 2		CPU CHANNEL B DQS 2 -> DIMM B DQS 2												
	92 11	MEM A DQS N<2>	MAKE_BASE=TRUE	=MEM A DQS N<2>	28	92 11	MEM B DQS N<2>	MAKE_BASE=TRUE	=MEM B DQS N<2>	30					
	92 11	MEM A DQS P<2>	MAKE_BASE=TRUE	=MEM A DQS P<2>	28	92 11	MEM B DQS P<2>	MAKE_BASE=TRUE	=MEM B DQS P<2>	30					
	92 11	MEM A DM<2>	MAKE_BASE=TRUE	=MEM A DM<2>	28	92 11	MEM B DM<2>	MAKE_BASE=TRUE	=MEM B DM<2>	30					
	92 11	MEM A DQ<23>	MAKE_BASE=TRUE	=MEM A DQ<23>	28	92 11	MEM B DQ<23>	MAKE_BASE=TRUE	=MEM B DQ<23>	30					
	92 11	MEM A DQ<22>	MAKE_BASE=TRUE	=MEM A DQ<22>	28	92 11	MEM B DQ<22>	MAKE_BASE=TRUE	=MEM B DQ<22>	30					
	92 11	MEM A DQ<21>	MAKE_BASE=TRUE	=MEM A DQ<21>	28	92 11	MEM B DQ<21>	MAKE_BASE=TRUE	=MEM B DQ<21>	30					
	92 11	MEM A DQ<20>	MAKE_BASE=TRUE	=MEM A DQ<20>	28	92 11	MEM B DQ<20>	MAKE_BASE=TRUE	=MEM B DQ<20>	30					
	92 11	MEM A DQ<19>	MAKE_BASE=TRUE	=MEM A DQ<19>	28	92 11	MEM B DQ<19>	MAKE_BASE=TRUE	=MEM B DQ<19>	30					
	92 11	MEM A DQ<18>	MAKE_BASE=TRUE	=MEM A DQ<18>	28	92 11	MEM B DQ<18>	MAKE_BASE=TRUE	=MEM B DQ<18>	30					
A	CPU CHANNEL A DQS 3 -> DIMM A DQS 3		CPU CHANNEL B DQS 3 -> DIMM B DQS 3												
	92 11	MEM A DQS N<3>	MAKE_BASE=TRUE	=MEM A DQS N<3>	28	92 11	MEM B DQS N<3>	MAKE_BASE=TRUE	=MEM B DQS N<3>	30					
	92 11	MEM A DQS P<3>	MAKE_BASE=TRUE	=MEM A DQS P<3>	28	92 11	MEM B DQS P<3>	MAKE_BASE=TRUE	=MEM B DQS P<3>	30					
	92 11	MEM A DM<3>	MAKE_BASE=TRUE	=MEM A DM<3>	28	92 11	MEM B DM<3>	MAKE_BASE=TRUE	=MEM B DM<3>	30					
	92 11	MEM A DQ<31>	MAKE_BASE=TRUE	=MEM A DQ<31>	28	92 11	MEM B DQ<31>	MAKE_BASE=TRUE	=MEM B DQ<31>	30					
	92 11	MEM A DQ<30>	MAKE_BASE=TRUE	=MEM A DQ<30>	28	92 11	MEM B DQ<30>	MAKE_BASE=TRUE	=MEM B DQ<30>	30					
	92 11	MEM A DQ<29>	MAKE_BASE=TRUE	=MEM A DQ<29>	28	92 11	MEM B DQ<29>	MAKE_BASE=TRUE	=MEM B DQ<29>	30					
	92 11	MEM A DQ<28>	MAKE_BASE=TRUE	=MEM A DQ<28>	28	92 11	MEM B DQ<28>	MAKE_BASE=TRUE	=MEM B DQ<28>	30					
	92 11	MEM A DQ<27>	MAKE_BASE=TRUE	=MEM A DQ<27>	28	92 11	MEM B DQ<27>	MAKE_BASE=TRUE	=MEM B DQ<27>	30					
	92 11	MEM A DQ<26>	MAKE_BASE=TRUE	=MEM A DQ<26>	28	92 11	MEM B DQ<26>	MAKE_BASE=TRUE	=MEM B DQ<26>	30					
8		7		6		5		4		3		2		1	

SYNC MASTER=K18 MLB		SYNC DATE=06/19/2009	
PAGE TITLE		SIZE	
DDR3 Byte/Bit Swaps		D	
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Page Notes

Power aliases required by this page:
- =PP1V5_S3_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_S0IMMB_SCL
- =I2C_S0IMMB_SDA

BOM options provided by this page:
(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

D

D

C

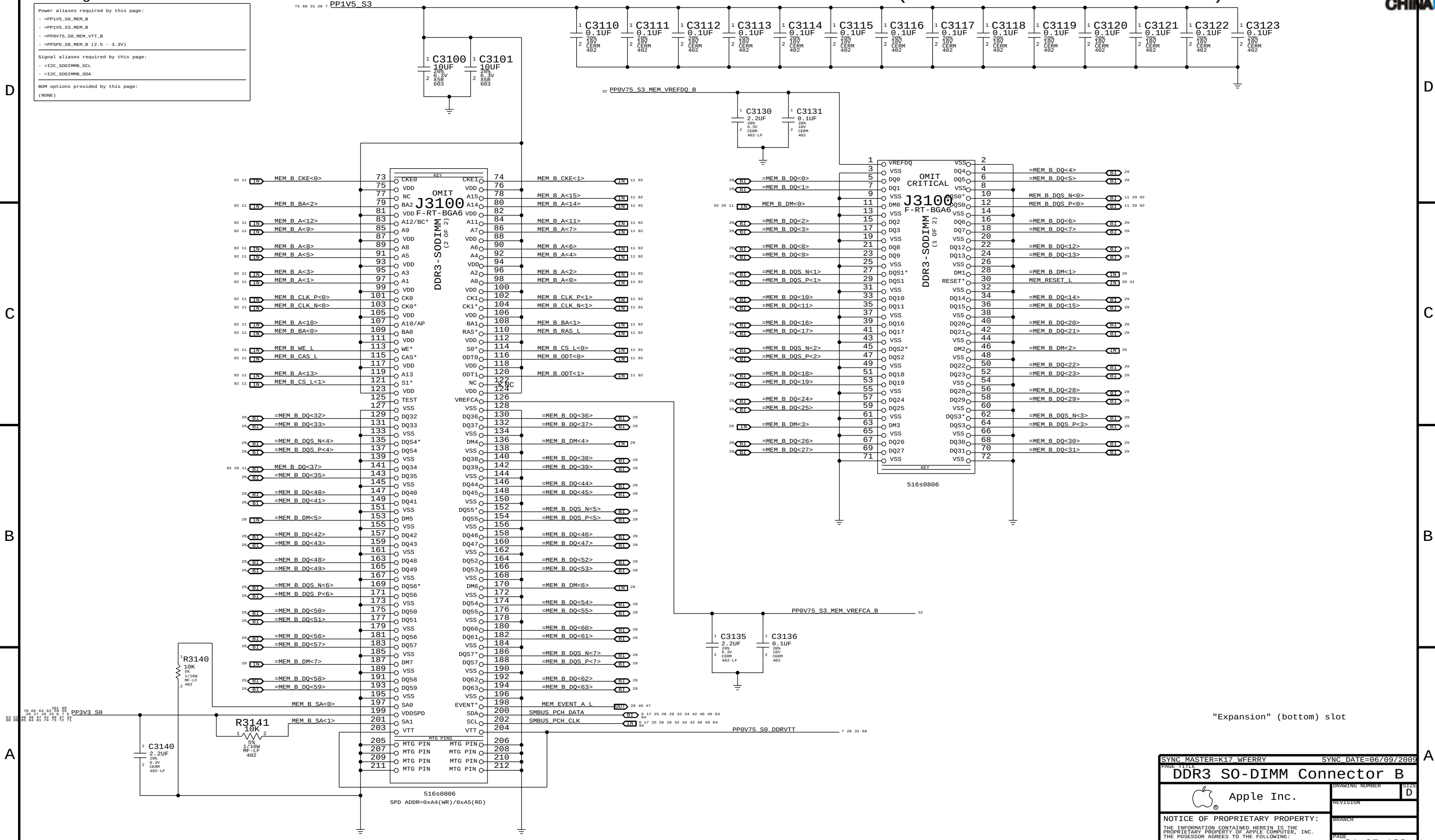
C

B

B

A

A



"Expansion" (bottom) slot

SYNC MASTER=K17 WFERRY		SYNC DATE=06/09/2009	
PAGE TITLE DDR3 S0-DIMM Connector B			
Apple Inc.		DRAWING NUMBER	SIZE D
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		PAGE 31 OF 132	SHEET 30 OF 103

The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.

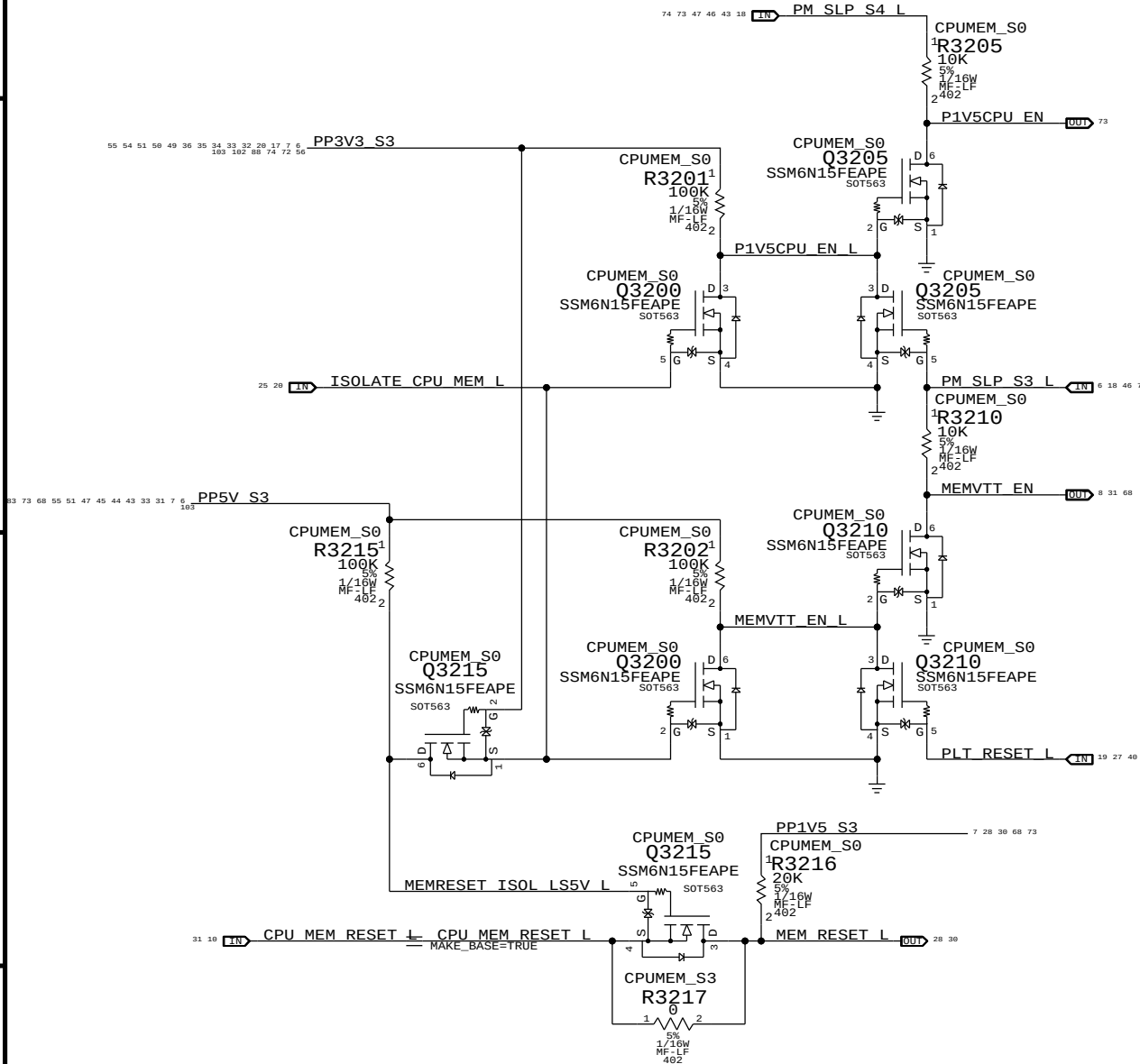
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

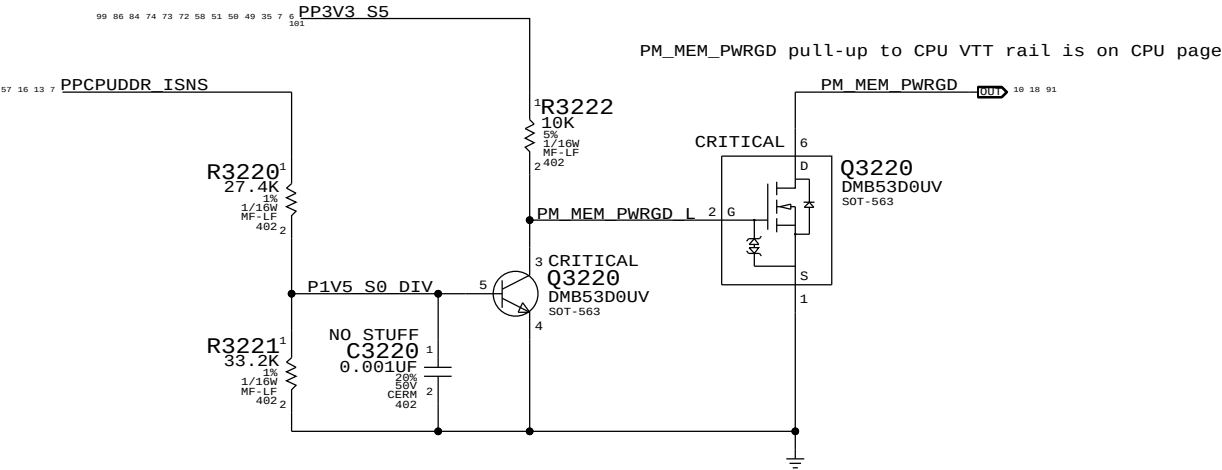
$$P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L$$

$$MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L$$

$$MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L$$

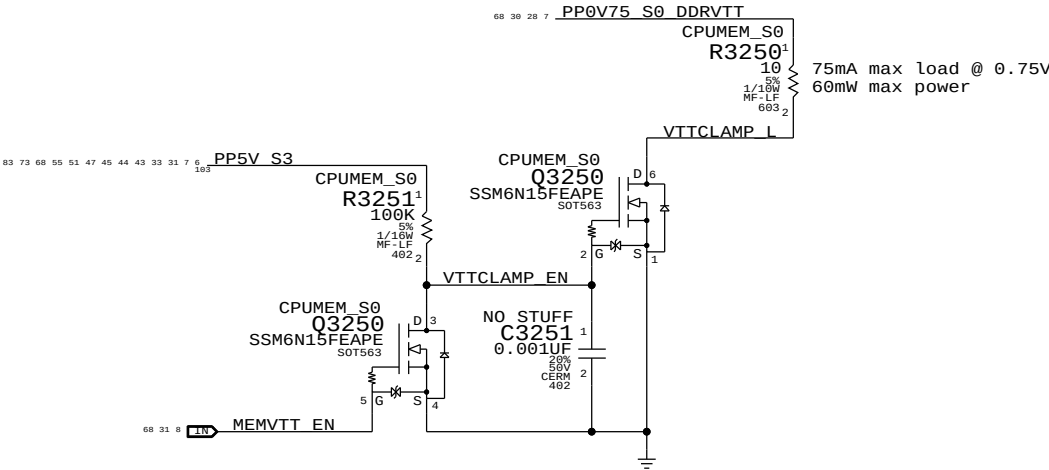


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



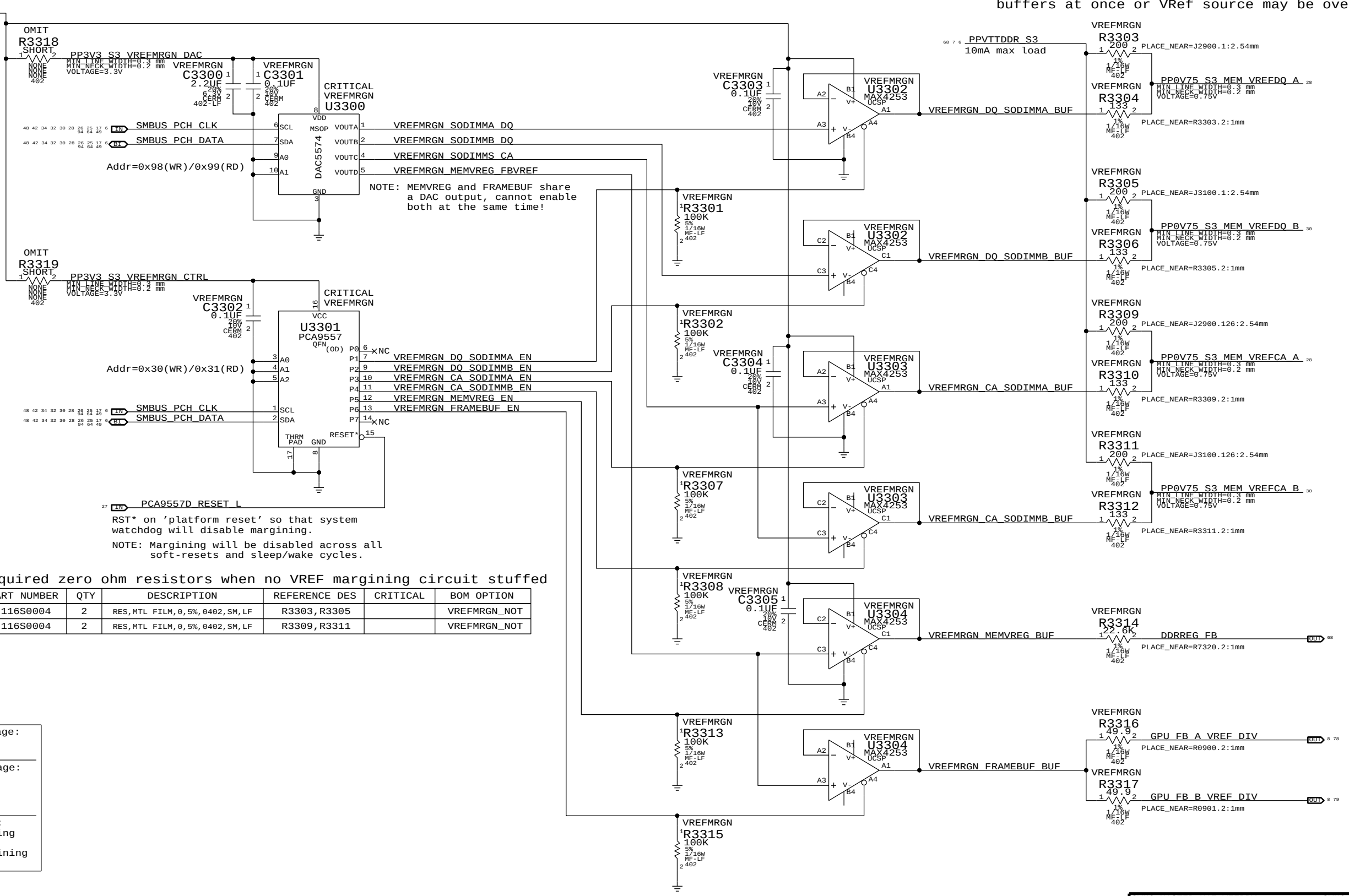
Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
to	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
to	7	1	1	1	1	CPU_MEM_RESET_L	1	1
S0								

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.



NOTE: Must not enable more than two S0-DIMM margining buffers at once or VRef source may be overloaded.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3303, R3305		VREFMRGN_NOT
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3309, R3311		VREFMRGN_NOT

Page Notes

- Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
 - =PPVTT_S3_DDR_BUF
- Signal aliases required by this page:
- =I2C_VREFDAC_SCL
 - =I2C_VREFDAC_SDA
 - =I2C_PCA9557D_SCL
 - =I2C_PCA9557D_SDA
- BOM options provided by this page:
- VREFMRGN - Stuffs VREF Margining Circuitry.
 - VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.998V - 1.002V (+/- 498mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 1.501V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
Vref current:		+3.4mA - -3.4mA (- = sourced)			+33uA - -33uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER=K17 WFERRY

SYNC DATE=06/09/2009

PAGE TITLE

FSB/DDR3/FB Vref Margining

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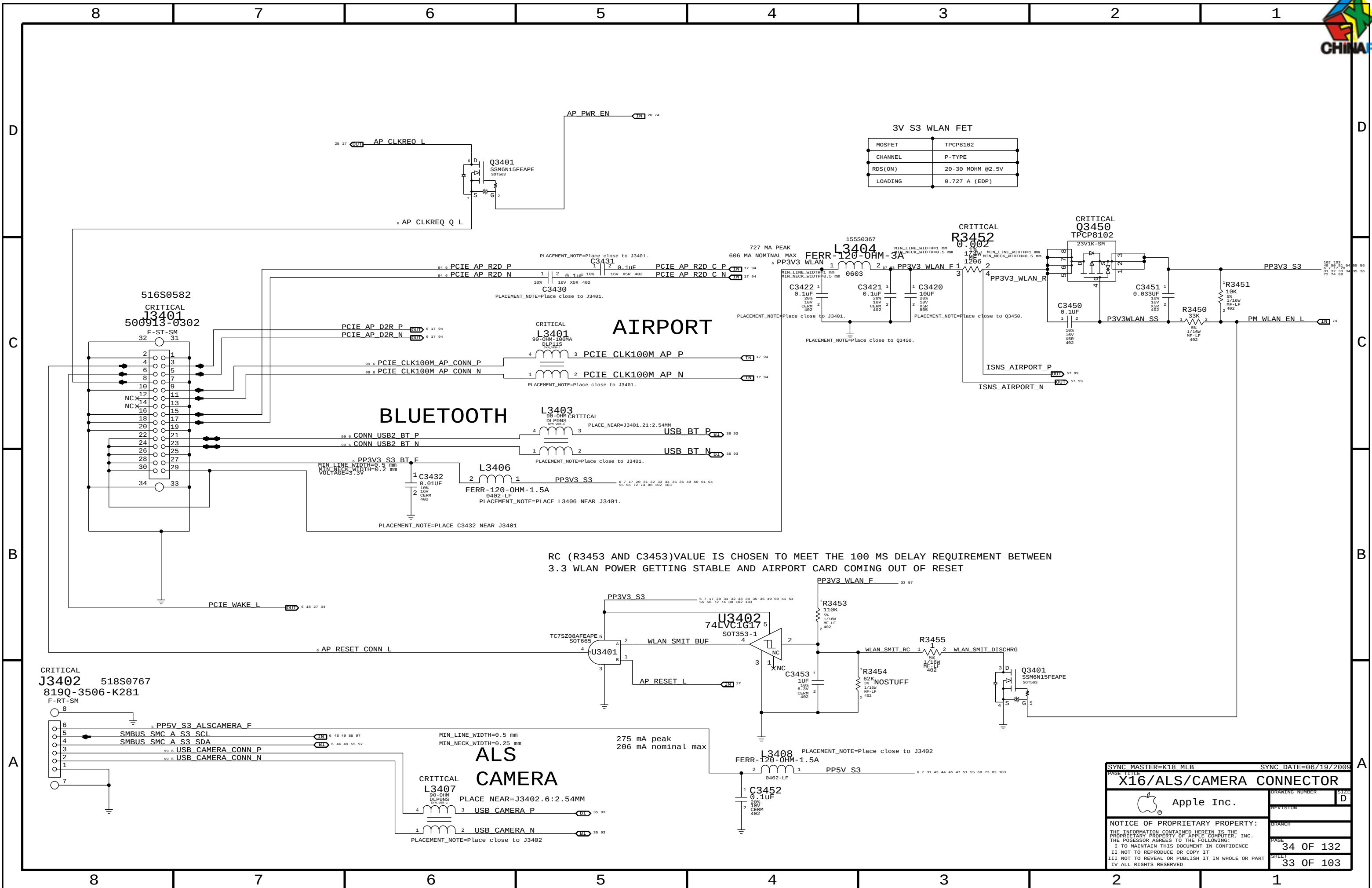
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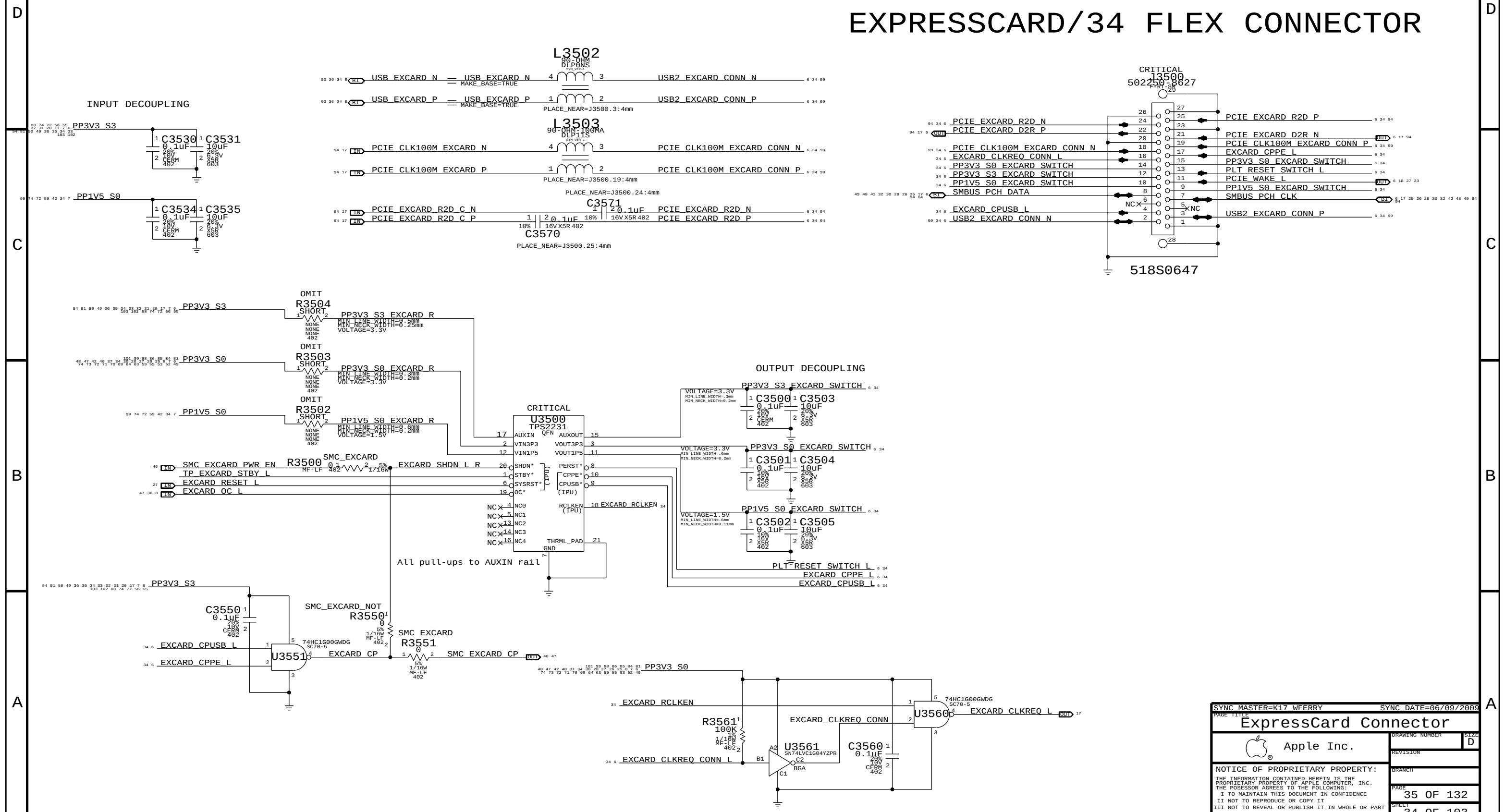
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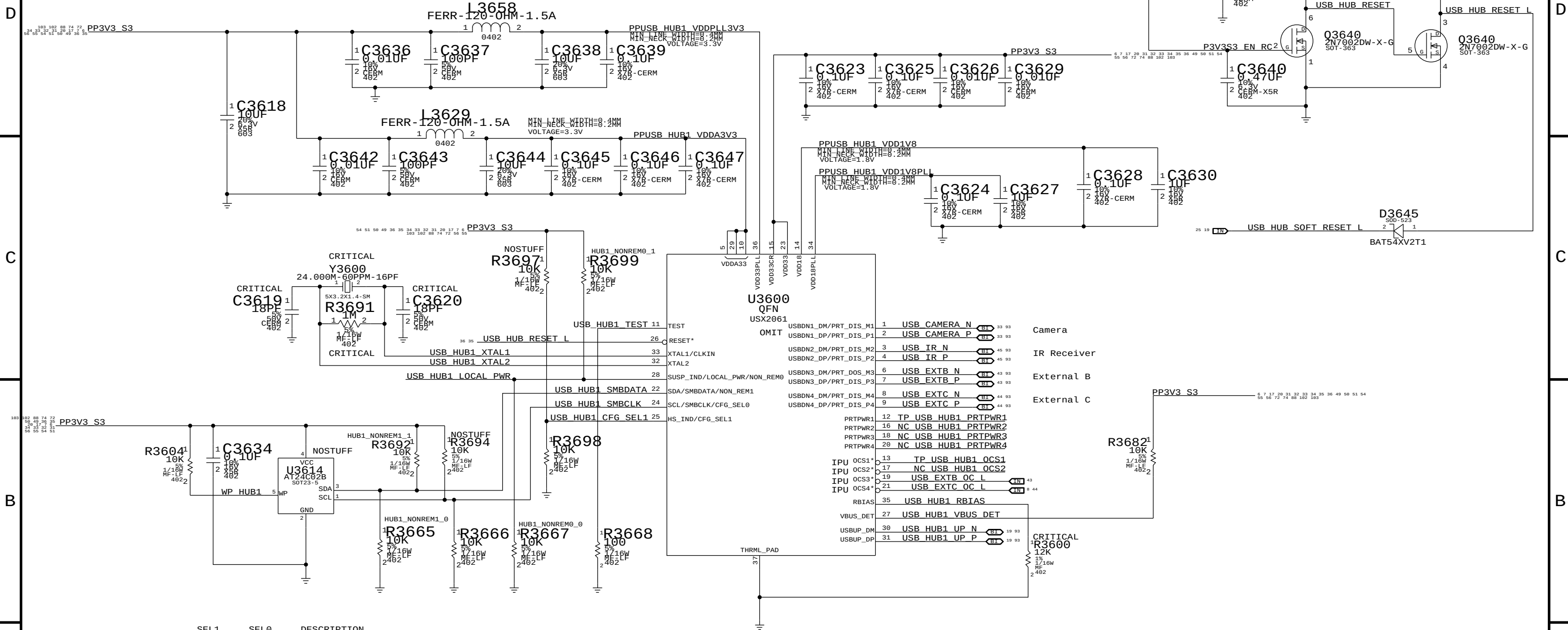
EXPRESSCARD/34 FLEX CONNECTOR



SYNC MASTER=K17_WFERRY		SYNC DATE=06/09/2009	
ExpressCard Connector		DRAWING NUMBER	SIZE D
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USB HUB - 1



SEL1	SEL0	DESCRIPTION
0	0	Internal Default with Self powered Operation
0	1	SMBUS Slave Config
1	0	Internal Default with Bus powered Operation
1	1	EEPROM Supported

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are Non removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2 and 3 are non Removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0720	2	IC,ASSP,USB2.0,HUB_CNTRL,4 PRT,360FN	U3600,U3700	CRITICAL	USBHUB_2514
338S0824	2	IC,USB2514B,USB 2.0,HUB_CNTRL,4 PRT,360FN	U3600,U3700	CRITICAL	USBHUB_2514B
338S0721	2	IC,USX2061,USB 2.0,HUB_CNTRL,4 PRT,360FN	U3600,U3700	CRITICAL	USBHUB_2061

BOM GROUP	BOM OPTIONS
HUB1_ALLREM	HUB1_NONREM1_0,HUB1_NONREM0_0
HUB1_1NONREM	HUB1_NONREM1_0,HUB1_NONREM0_1
HUB1_2NONREM	HUB1_NONREM1_1,HUB1_NONREM0_0
HUB1_3NONREM	HUB1_NONREM1_1,HUB1_NONREM0_1

SYNC MASTER=K18 MLB SYNC DATE=10/07/2009

PAGE TITLE

USB HUB 1

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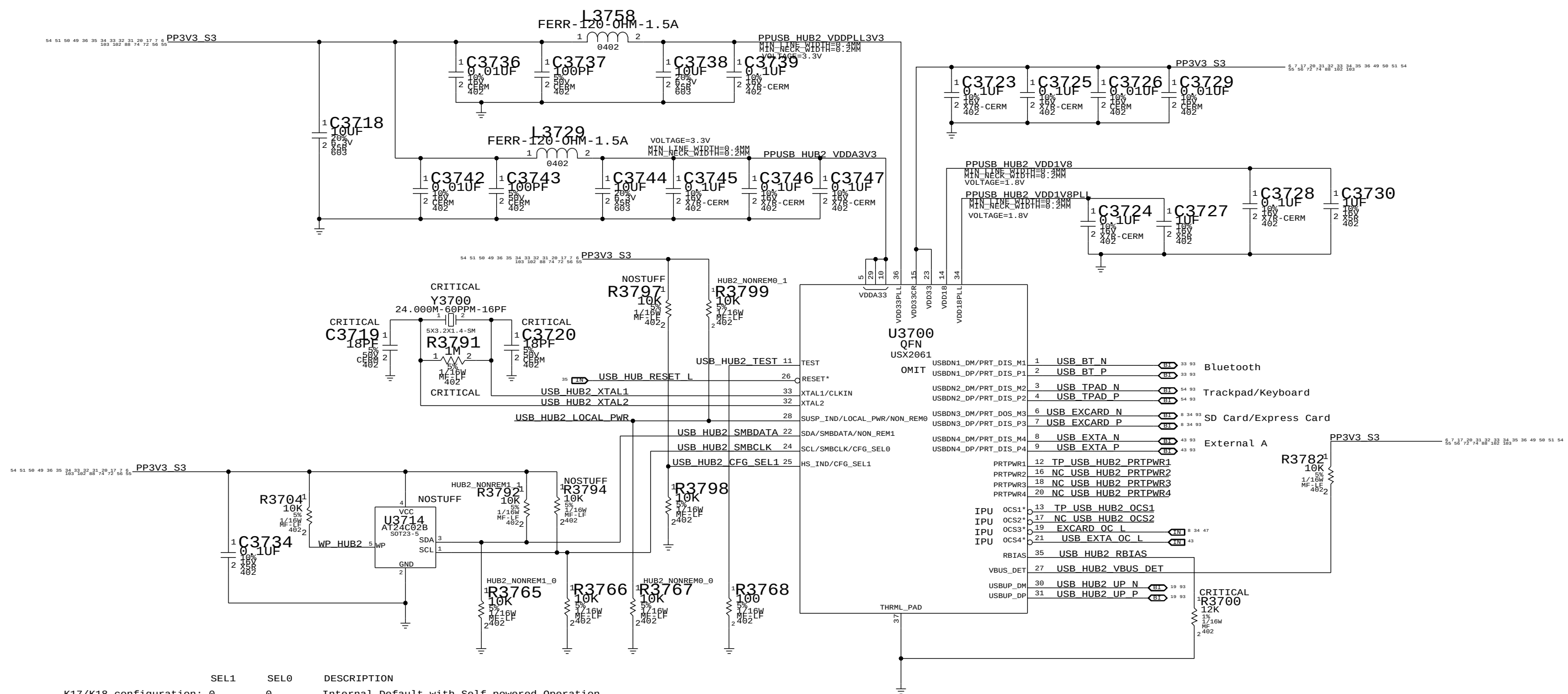
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A

A



USB HUB-2



	SEL1	SEL0	DESCRIPTION
K17/K18 configuration:	0	0	Internal Default with Self powered Operation
	0	1	SMBUS Slave Config
	1	0	Internal Default with Bus powered Operation
	1	1	EEPROM Supported

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM GROUP	BOM OPTIONS
HUB2_ALLREM	HUB2_NONREM1_0, HUB2_NONREM0_0
HUB2_1NONREM	HUB2_NONREM1_0, HUB2_NONREM0_1
HUB2_2NONREM	HUB2_NONREM1_1, HUB2_NONREM0_0
HUB2_3NONREM	HUB2_NONREM1_1, HUB2_NONREM0_1

SYNC MASTER=K18 MLB SYNC DATE=10/09/2009

PAGE TITLE

USB HUB 2

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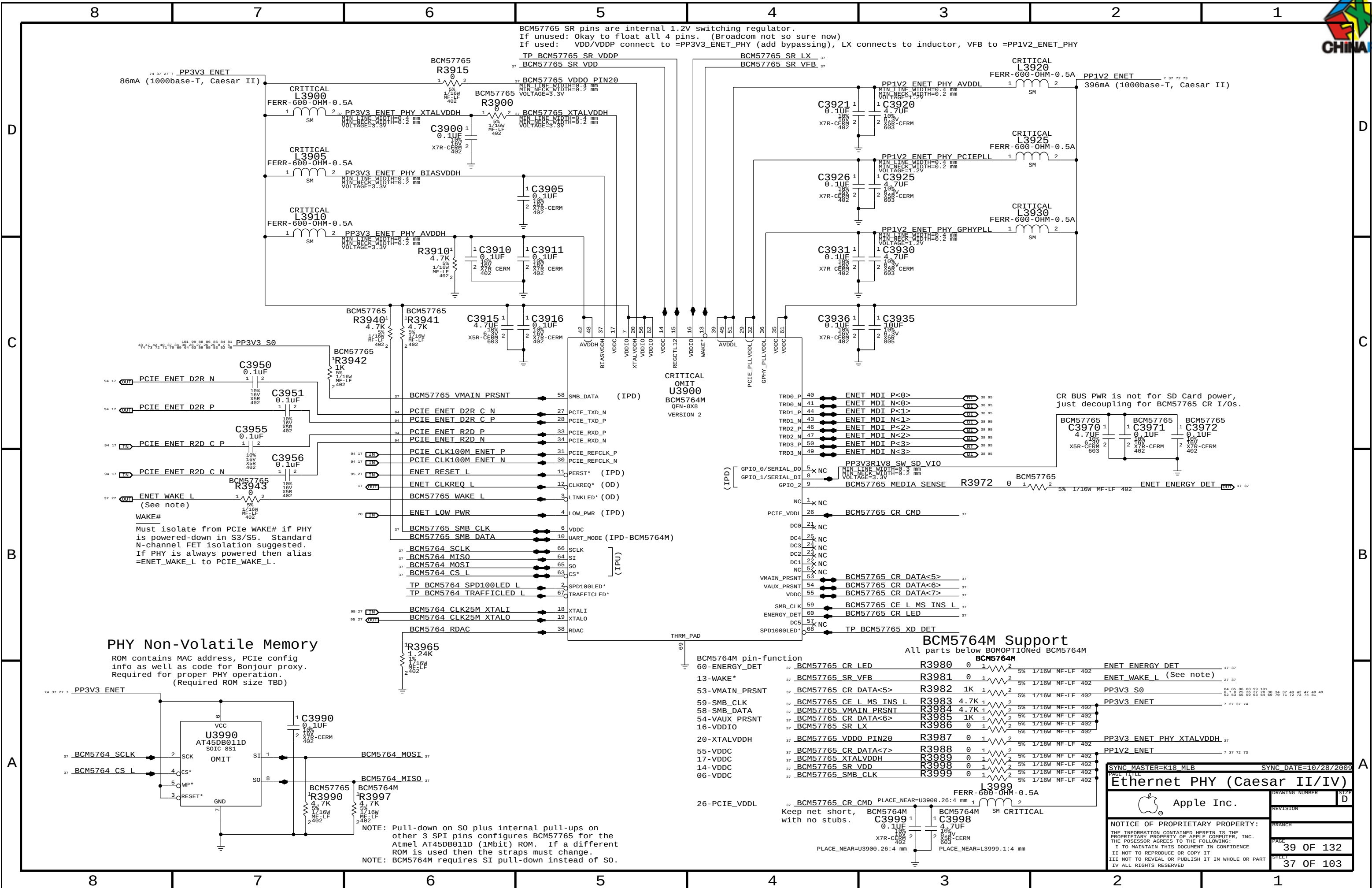
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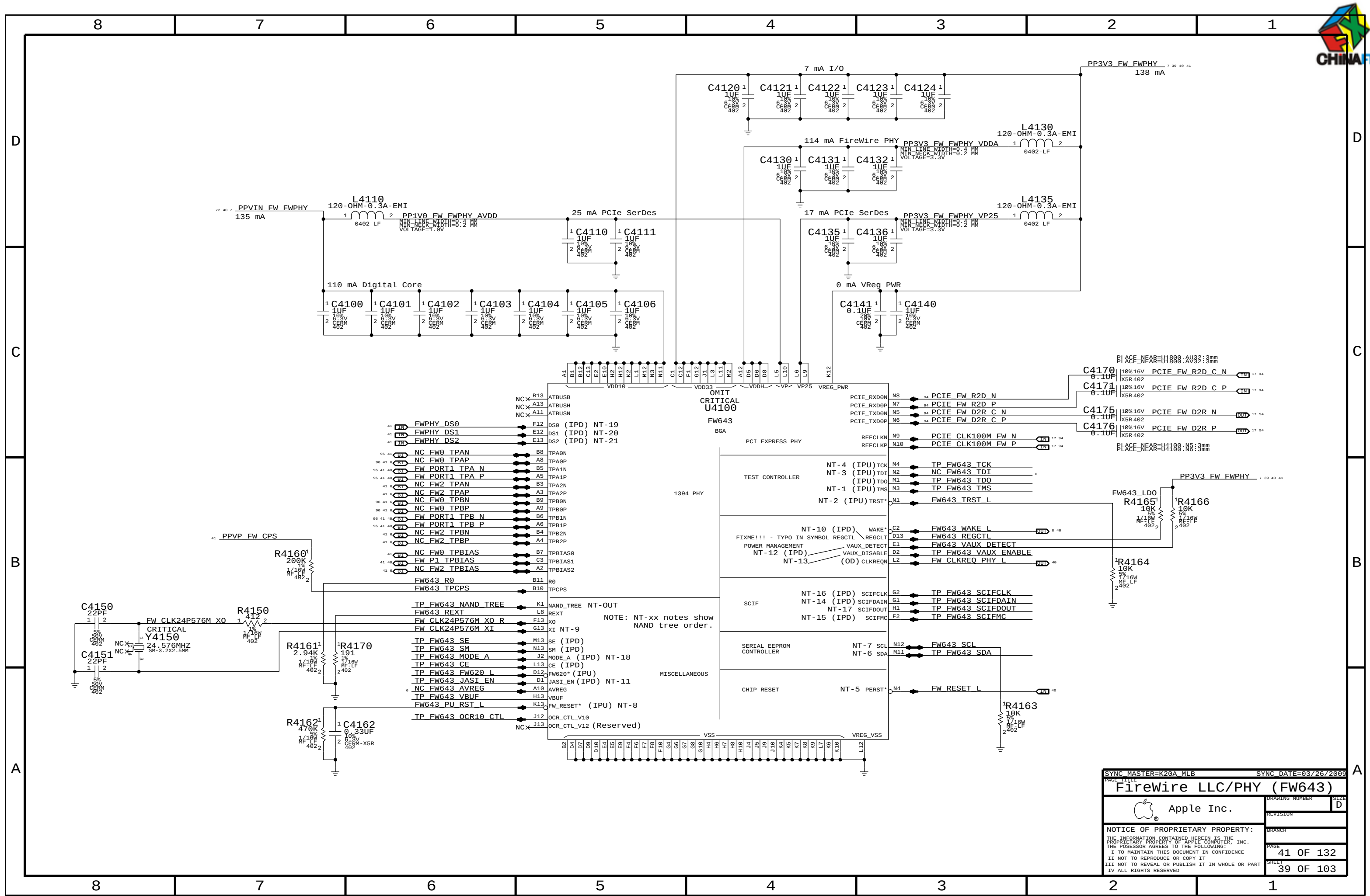
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SYNC MASTER-K17 WFERRY		SYNC DATE=06/09/2009	
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SYNC MASTER=K20A_MLB		SYNC DATE=03/26/2009	
PAGE TITLE		FireWire LLC/PHY (FW643)	
		DRAWING NUMBER	SIZE D
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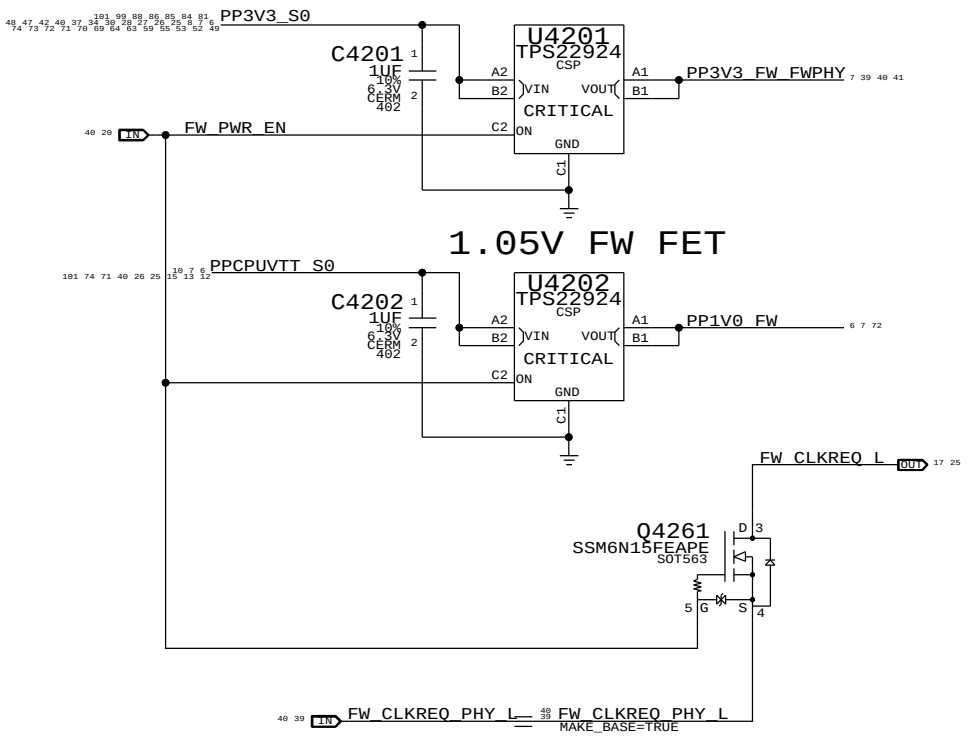


Page Notes

- Power aliases required by this page:
- =PPBUS_S5_FWPWRSW (system supply for bus power)
 - =PP3V3_FW_LATEVG_ACTIVE
 - =PPVP_FW_SUMNODE (power passthru summation node)
- Signal aliases required by this page:
- (NONE)
- BOM options provided by this page:
- FW_PORT_FAULT_PU

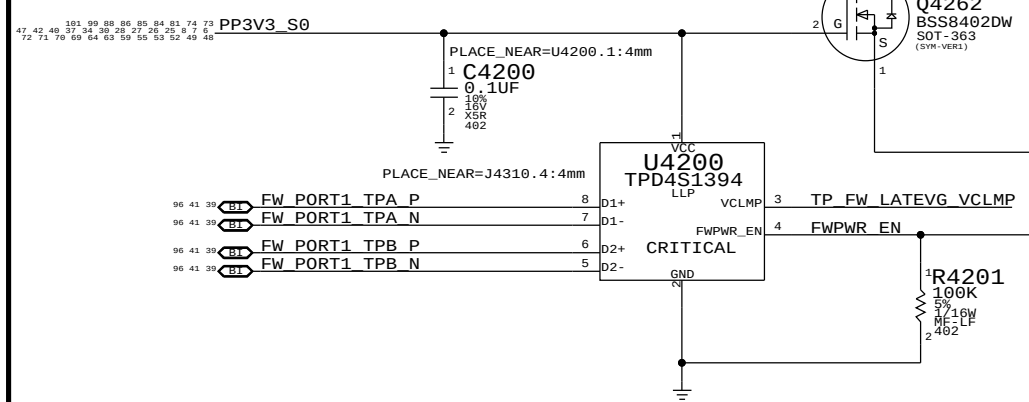
3.3V FW FET

I(max) = 1.7A (85C)



FireWire Port Power Switch

Late-VG Protection



PAGE TITLE		PAGE TITLE	
FireWire Port Power		FireWire Port Power	
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CRITICAL
J4310
1394B-M97
F-RT-TH

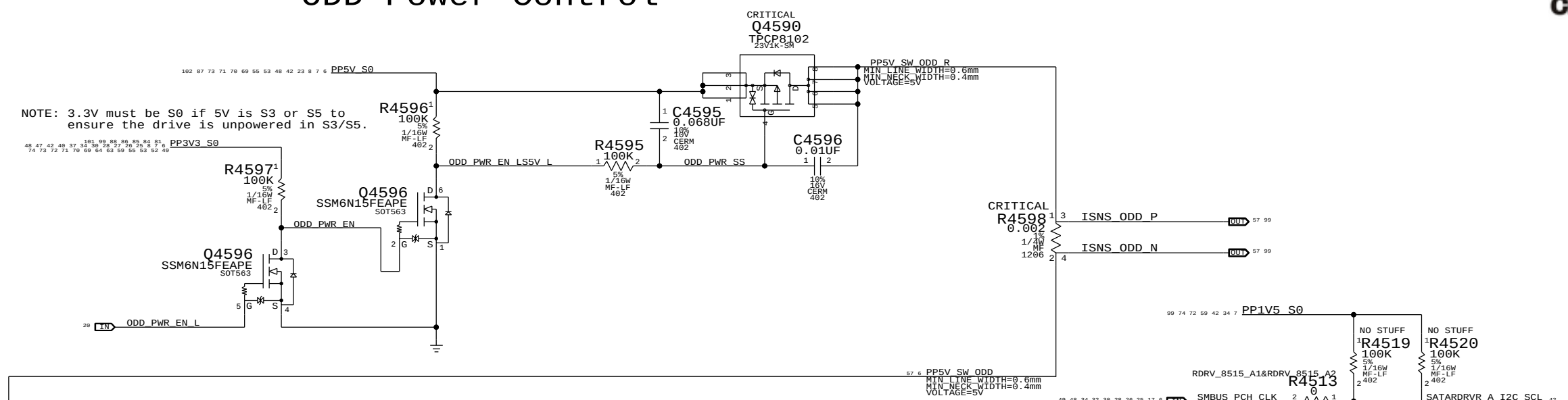
514S0605

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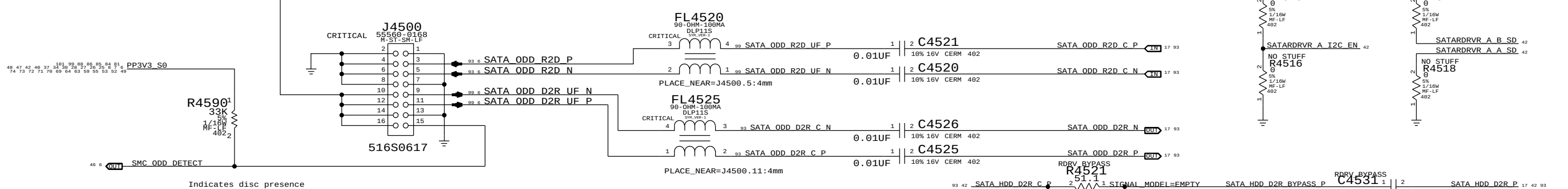


ODD Power Control

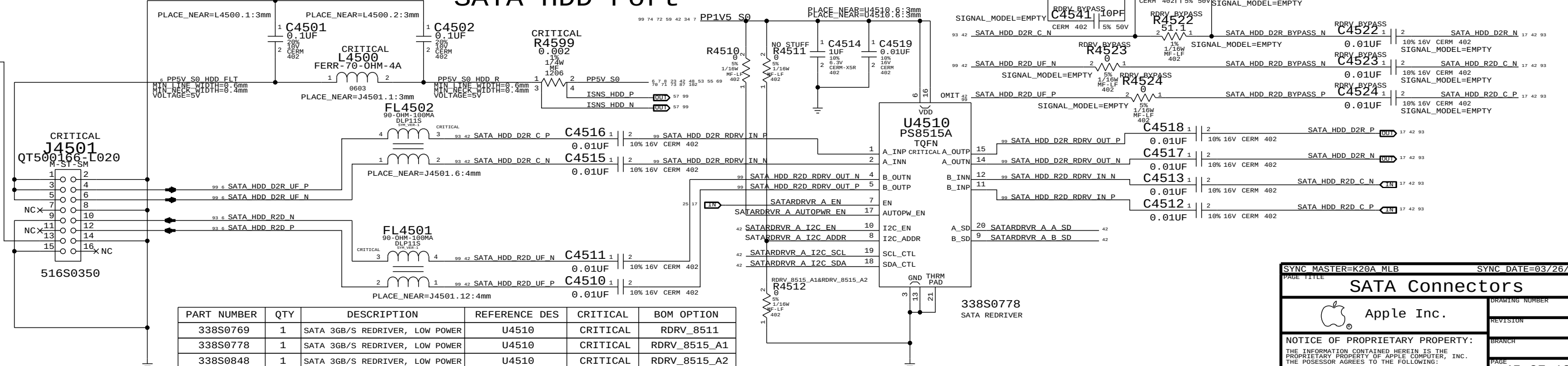
NOTE: 3.3V must be S0 if 5V is S3 or S5 to ensure the drive is unpowered in S3/S5.



SATA ODD Port



SATA HDD Port



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
338S0769	1	SATA 3GB/S REDRIVER, LOW POWER	U4510	CRITICAL	RDRV_8511
338S0778	1	SATA 3GB/S REDRIVER, LOW POWER	U4510	CRITICAL	RDRV_8515_A1
338S0848	1	SATA 3GB/S REDRIVER, LOW POWER	U4510	CRITICAL	RDRV_8515_A2

SYNC MASTER=K20A MLB

SYNC DATE=03/26/2009

SATA Connectors

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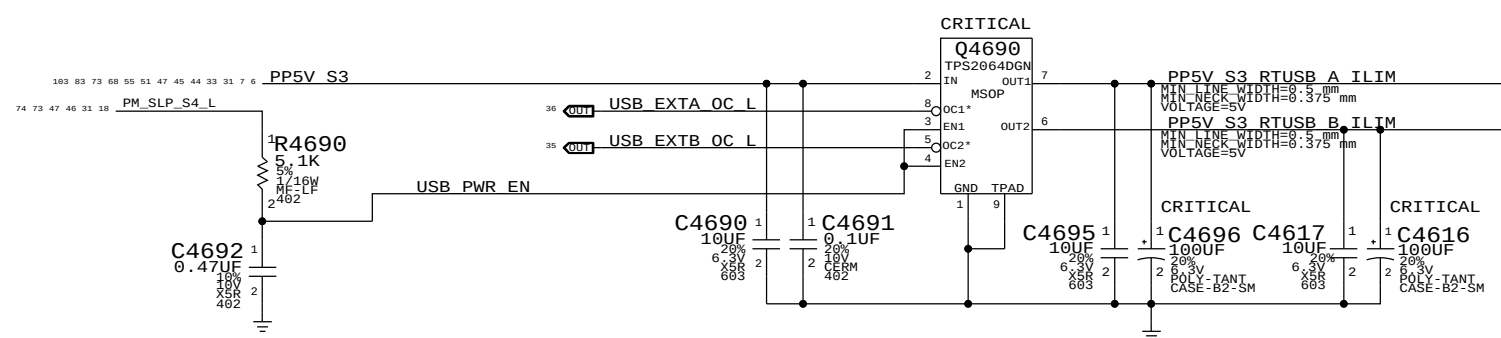
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45 OF 132

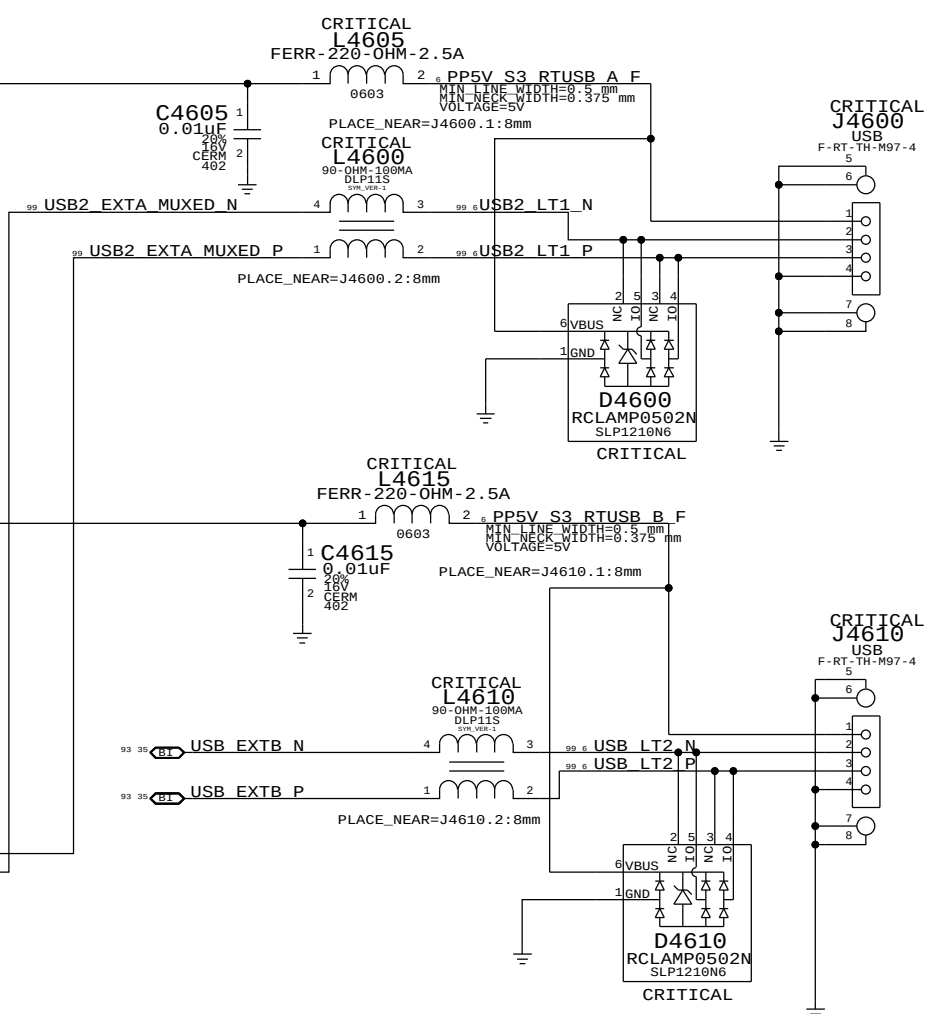
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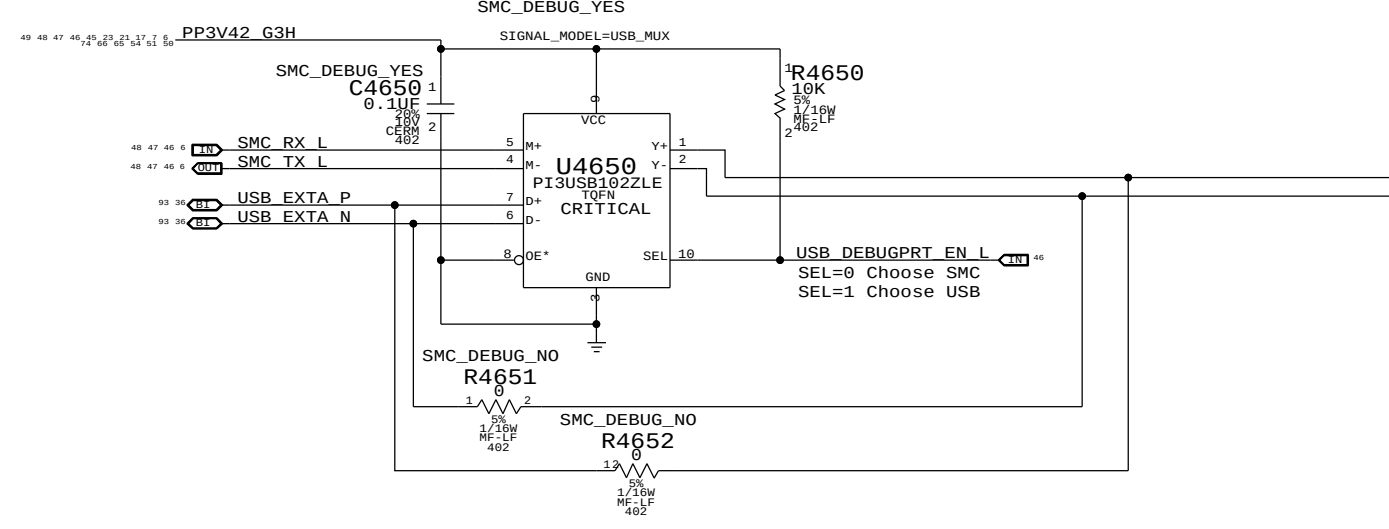
Port Power Switch



Left USB Port A



USB/SMC Debug Mux



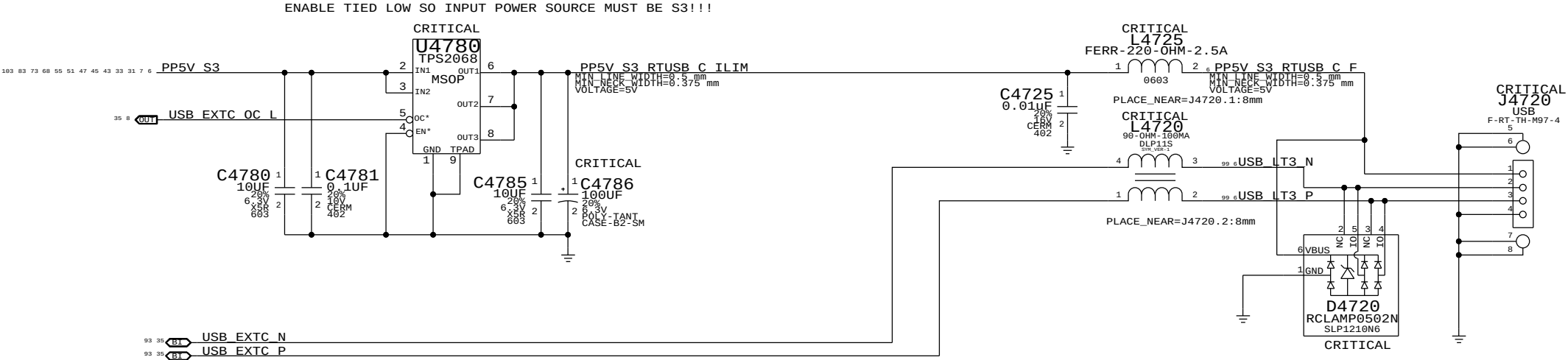
Left USB Port B

SYNC MASTER=K17_WFERRY		SYNC DATE=06/09/2009	
External USB Connectors		DRAWING NUMBER	SIZE
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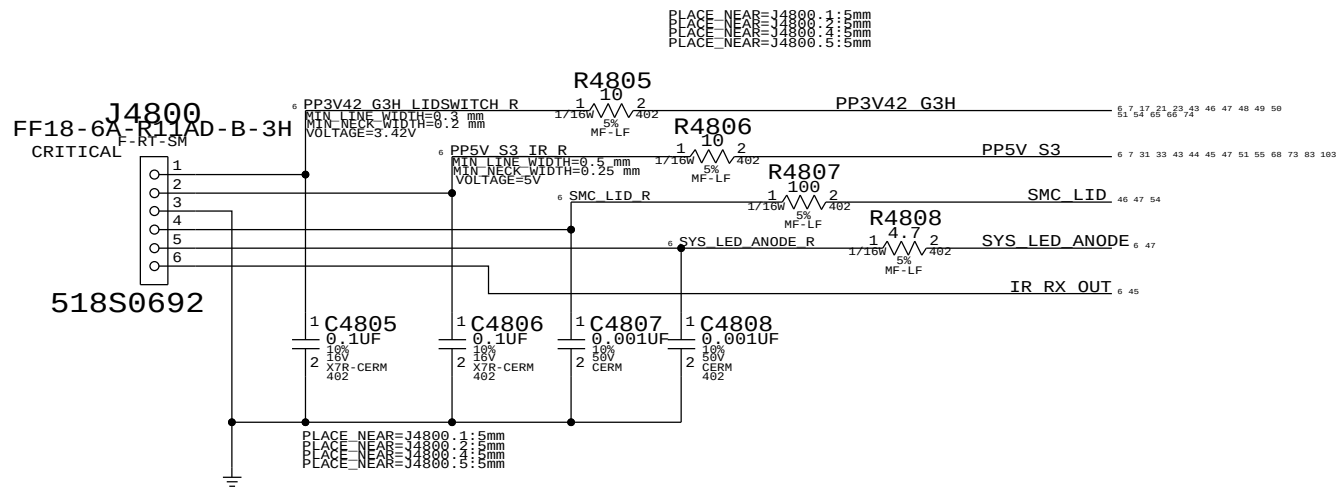
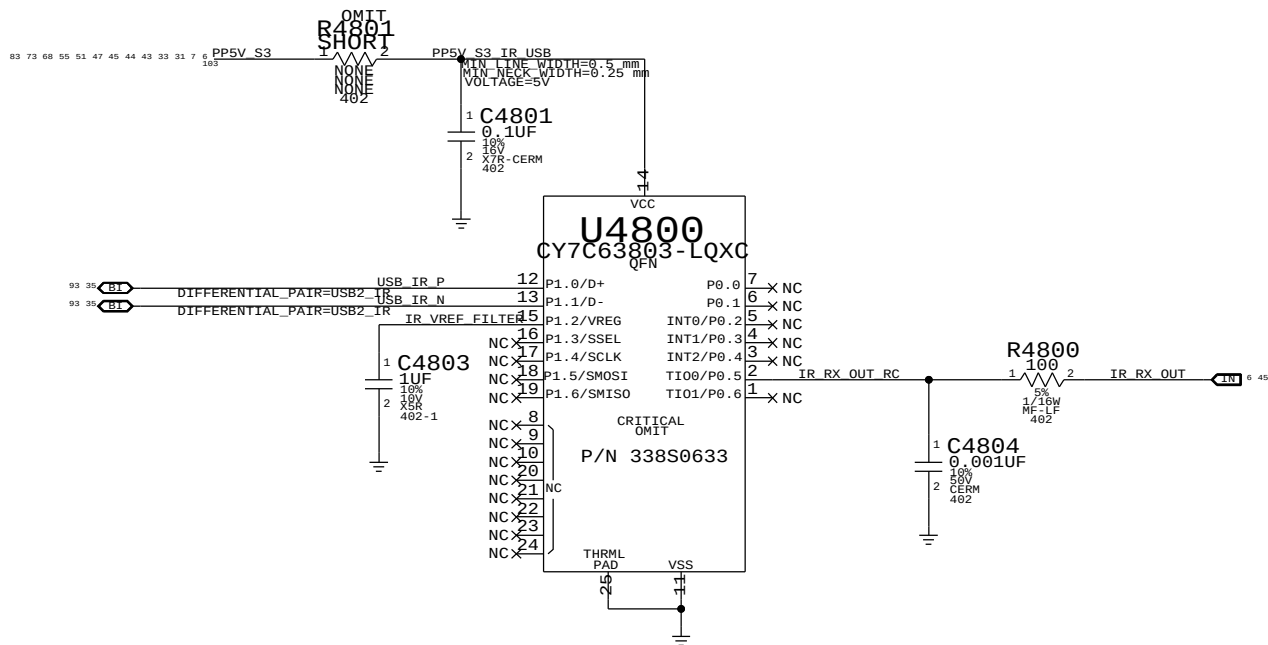


Port Power Switch

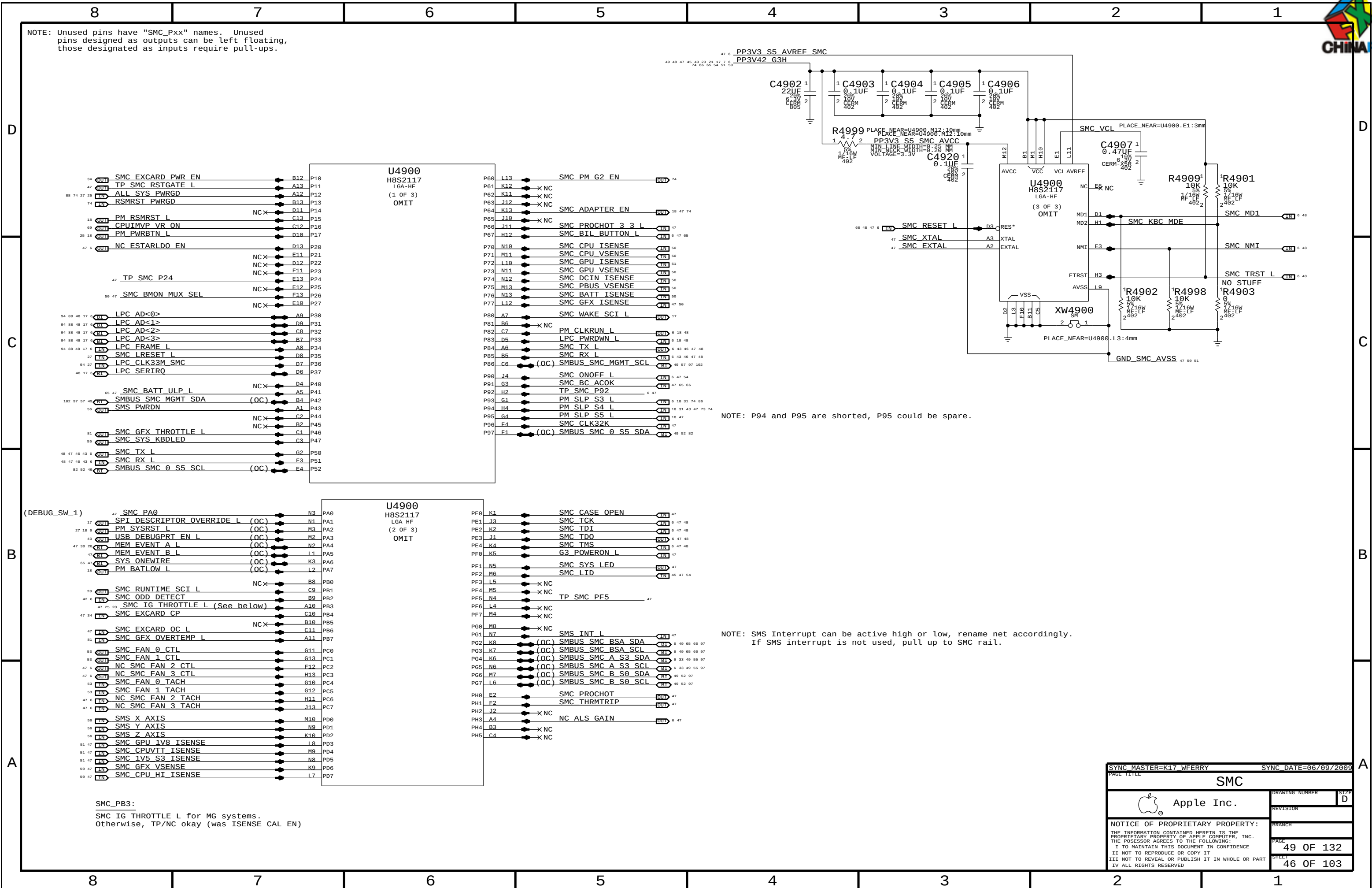
LEFT USB PORT C



SYNC MASTER=K20A_MLB		SYNC DATE=03/26/2009	
PAGE TITLE			
PROJECT SPECIFIC CONNS			
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		PAGE	47 OF 132
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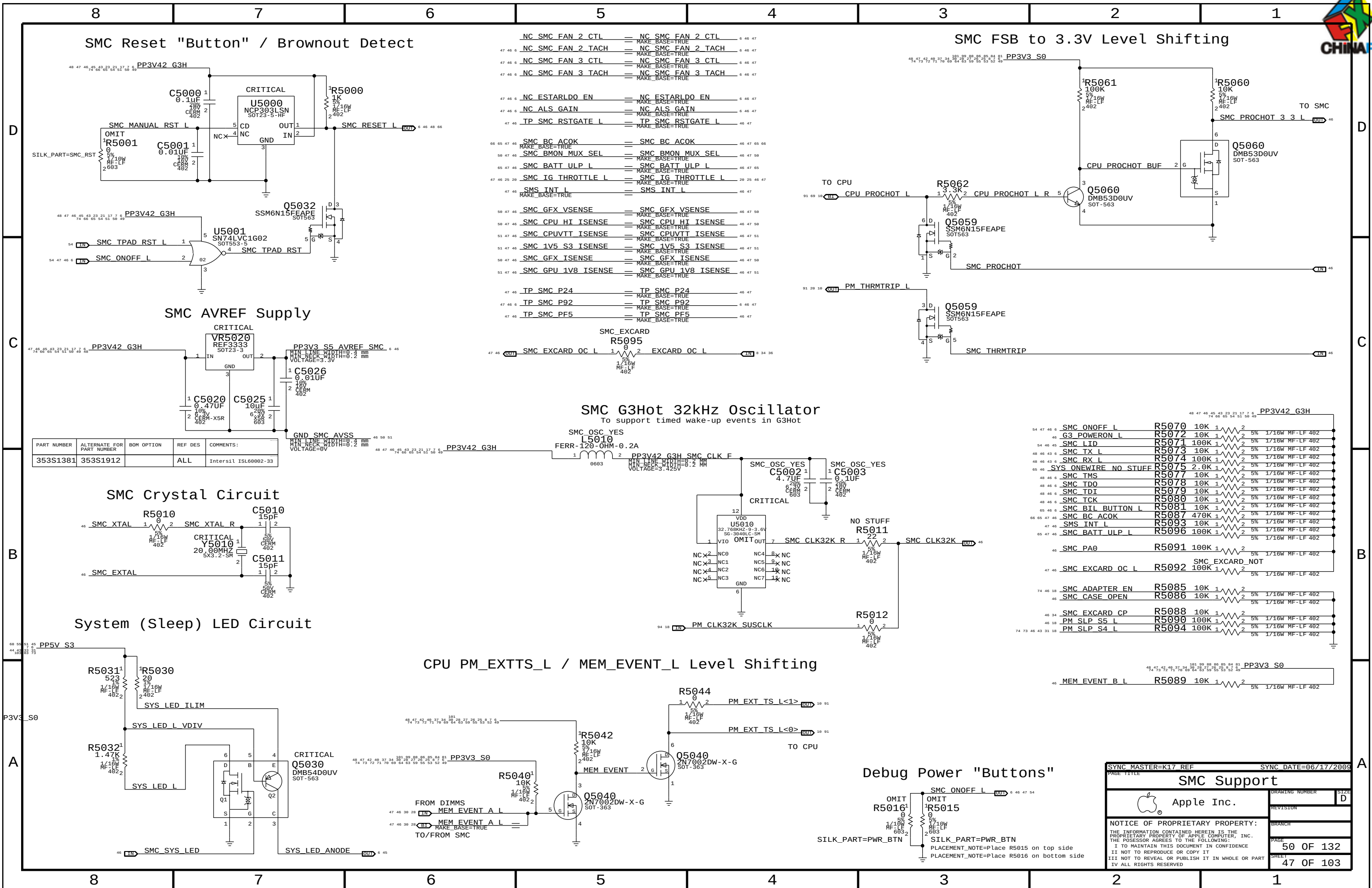


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Front Flex Support		DRAWING NUMBER	
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Debug Power "Buttons"

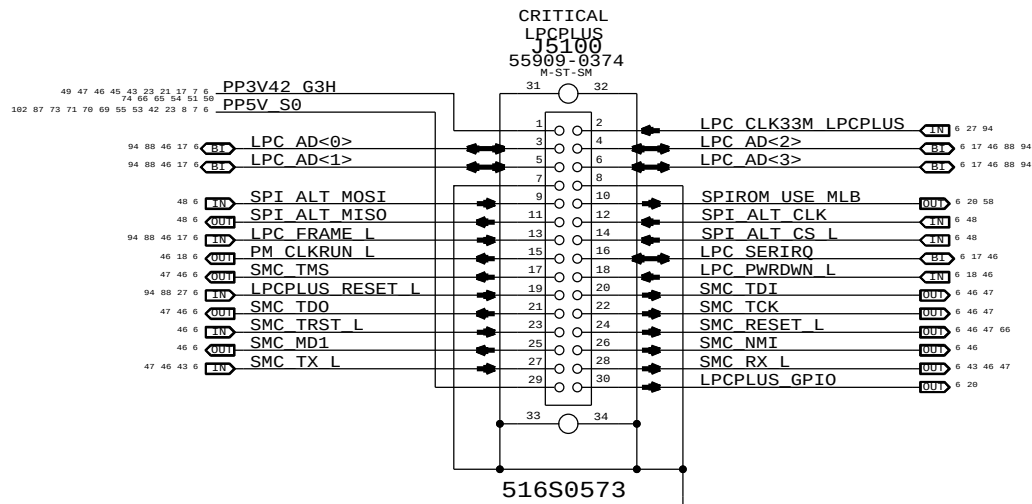


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S1381	353S1912		ALL	Intersil ISL60002-33

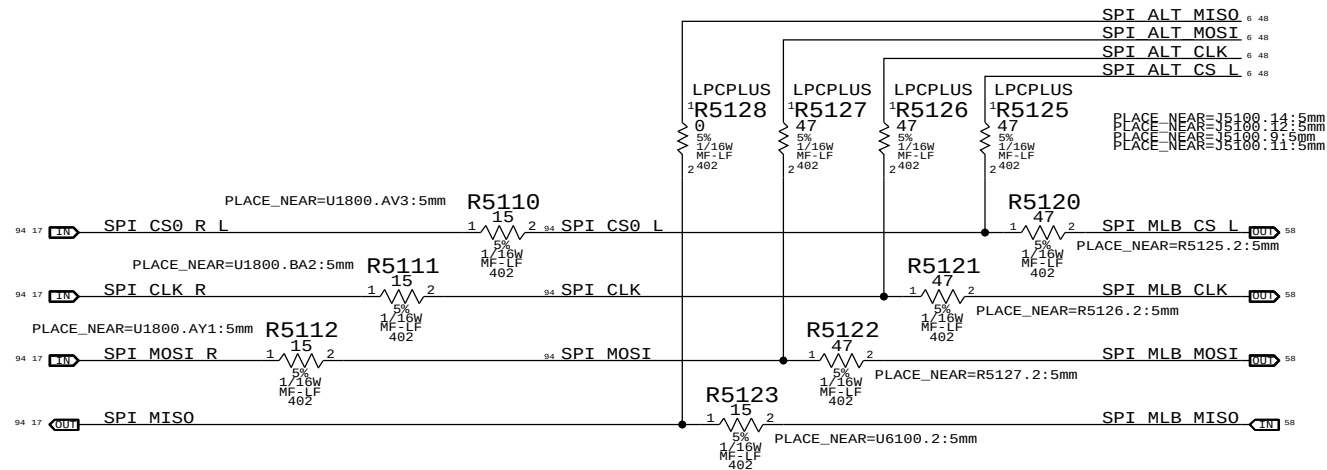
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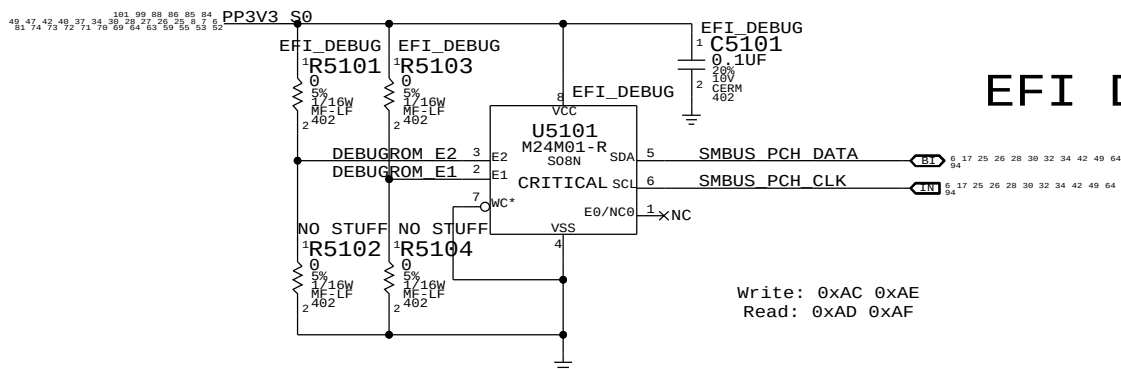
LPC+SPI Connector

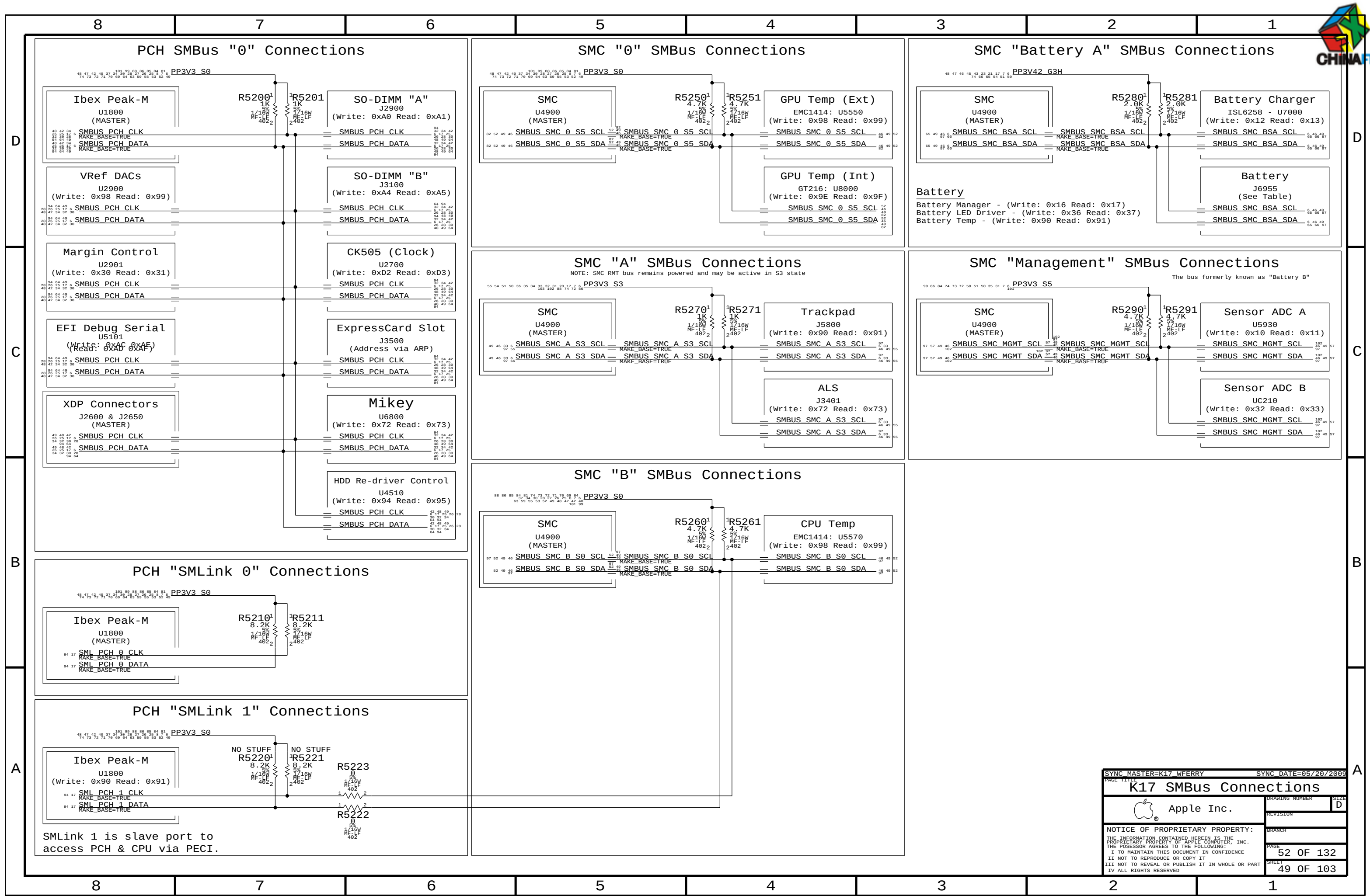


SPI Bus Series Termination

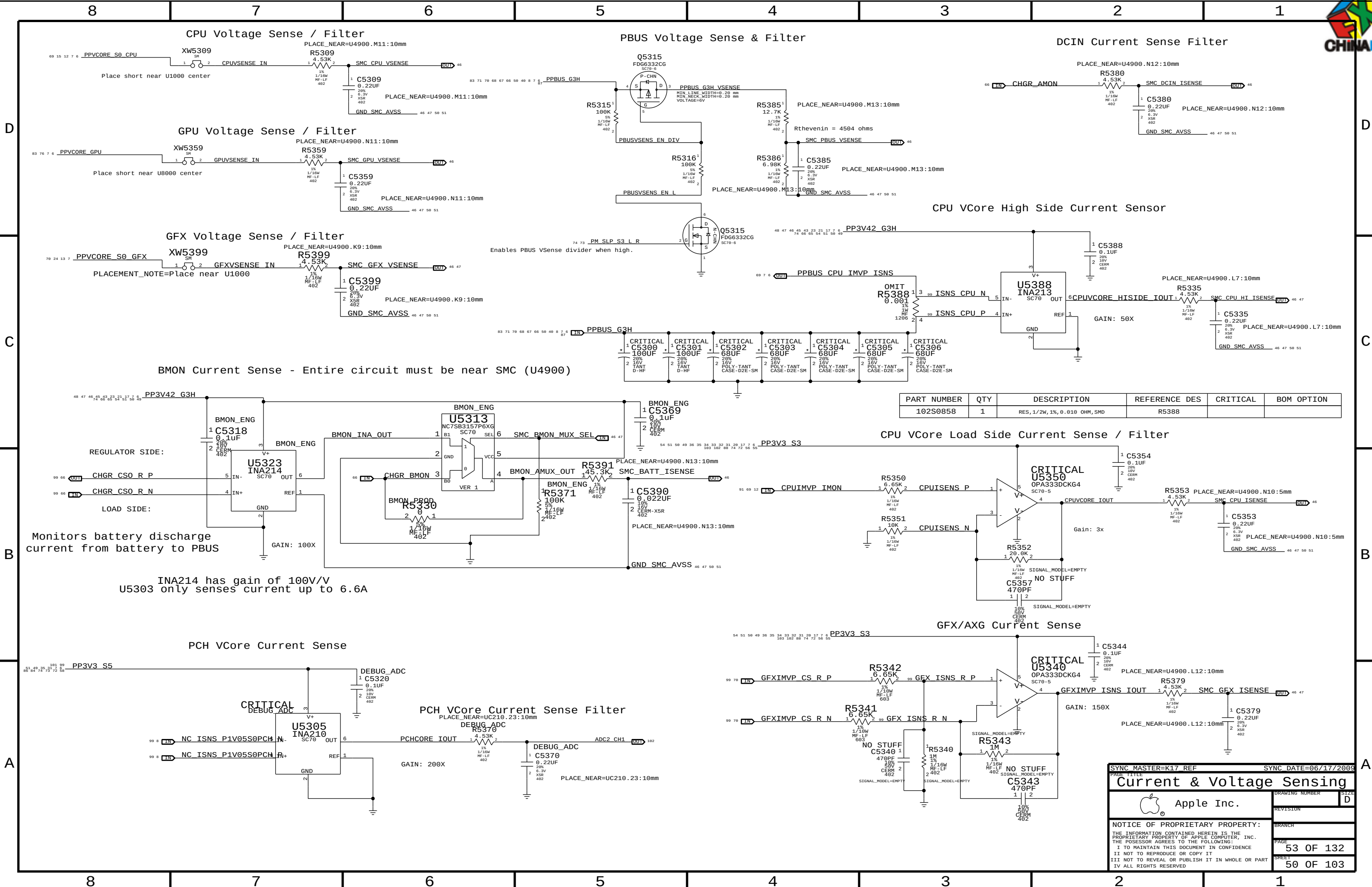


EFI Debug ROM





SYNC MASTER=K17 WFERRY		SYNC DATE=05/20/2009	
PAGE TITLE			
K17 SMBus Connections			
 Apple Inc.		DRAWING NUMBER	SIZE
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		SHEET	49 OF 103



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
102S0858	1	RES, 1/2W, 1%, 0.010 OHM, SMD	R5388		

Monitors battery discharge current from battery to PBUS

INA214 has gain of 100V/V
U5303 only senses current up to 6.6A

SYNC MASTER=K17 REF

SYNC DATE=06/17/2009

Current & Voltage Sensing

Apple Inc.

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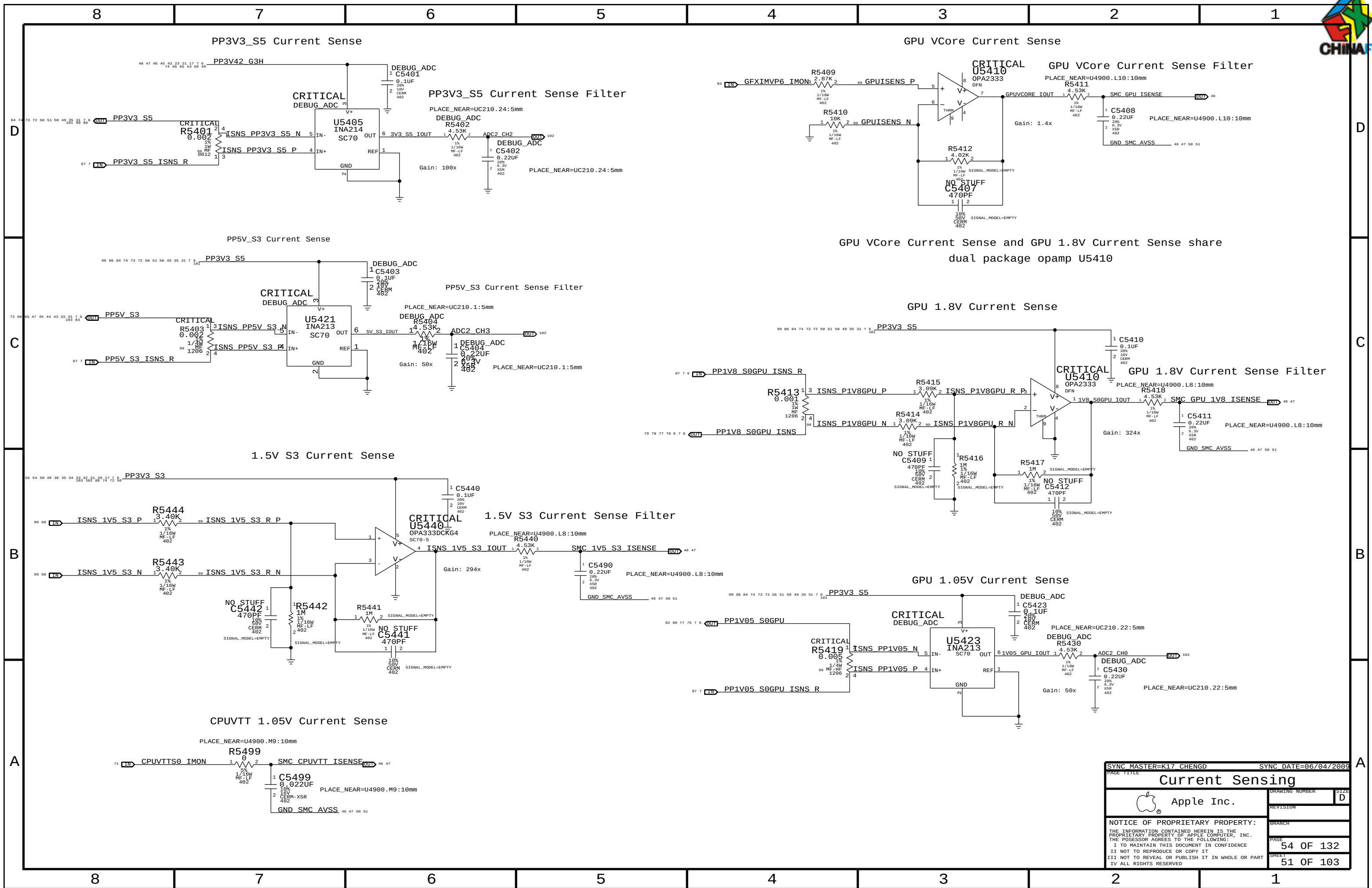
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DRAWING NUMBER

53 OF 132

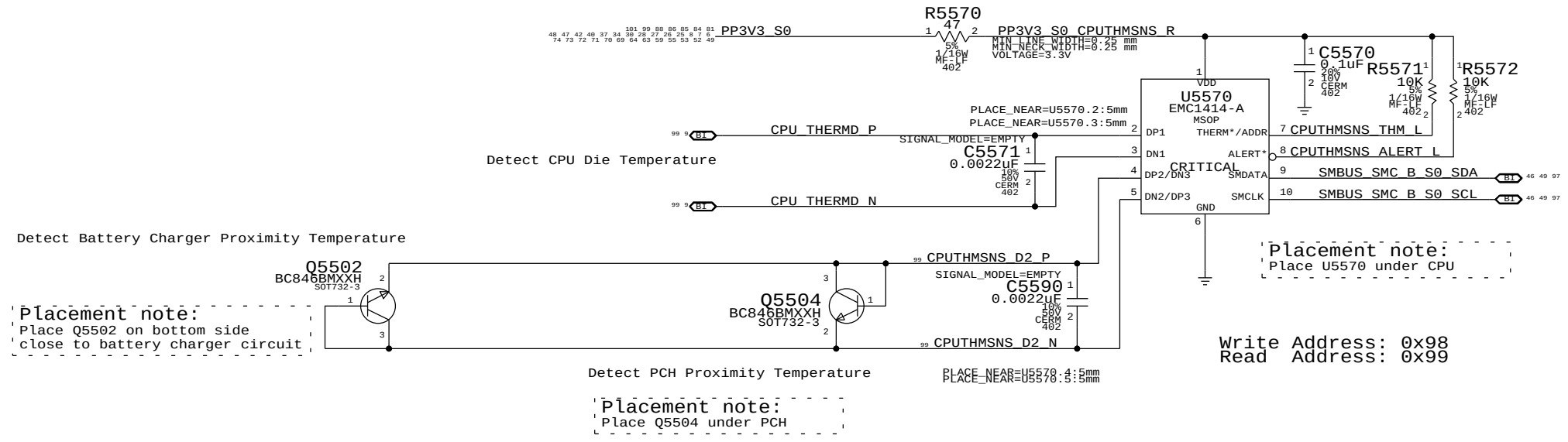
SIZE

50 OF 103



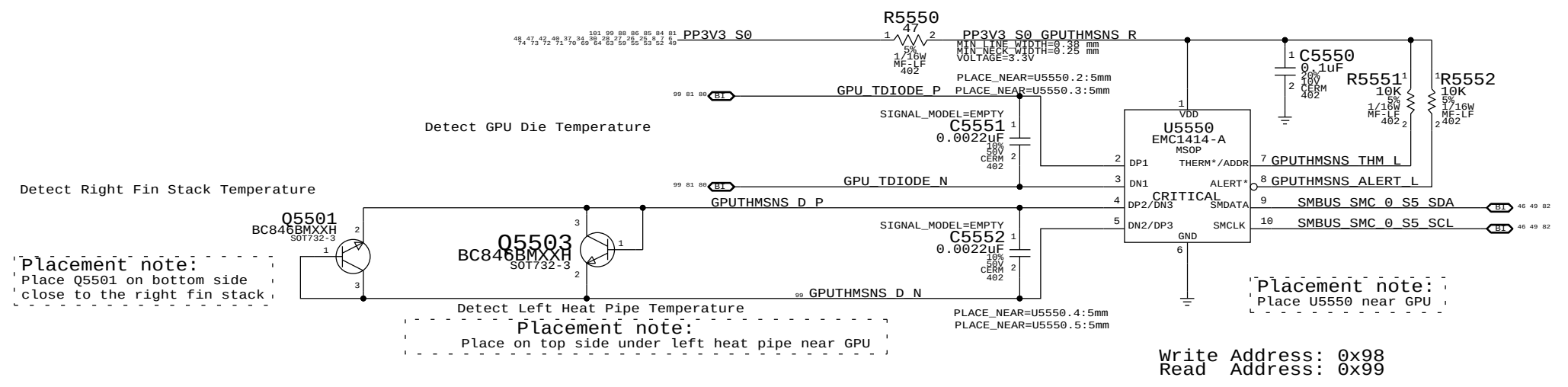


CPU Proximity/CPU Die/PCH Proximity/Battery Charger Proximity



Note: EMC1414 can perform Beta Compensation for External Diode 1 only

GPU Proximity/GPU Die/Left Heat Pipe/Right Fin Stack



SYNC MASTER=K17_CHENG D		SYNC DATE=07/08/2009	
PAGE TITLE		Thermal Sensors	
Apple Inc.		DRAWING NUMBER	SIZE D
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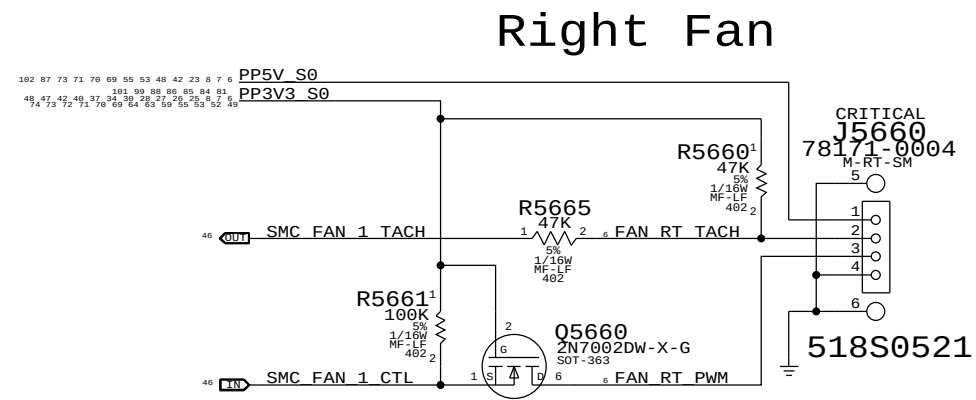
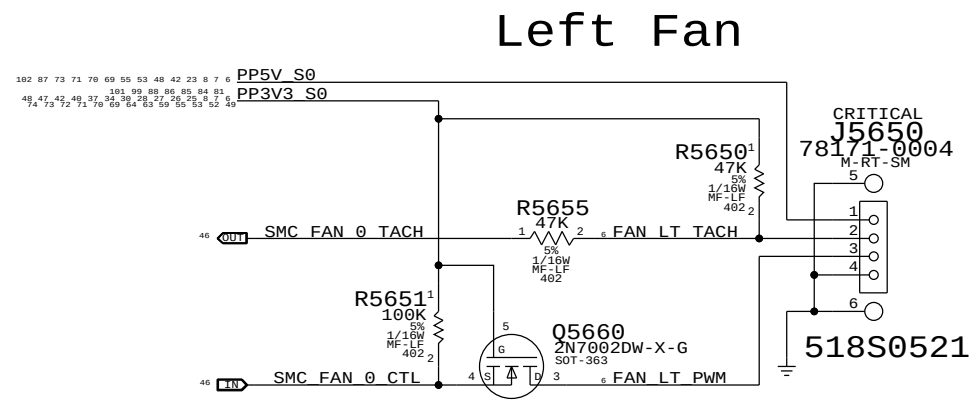
A

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SYNC MASTER=K20A MLB		SYNC DATE=03/26/2009	
PAGE TITLE			
Fan Connectors			
 Apple Inc.		DRAWING NUMBER	SIZE
			D
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		BRANCH	
		PAGE	56 OF 132
		SHEET	53 OF 103

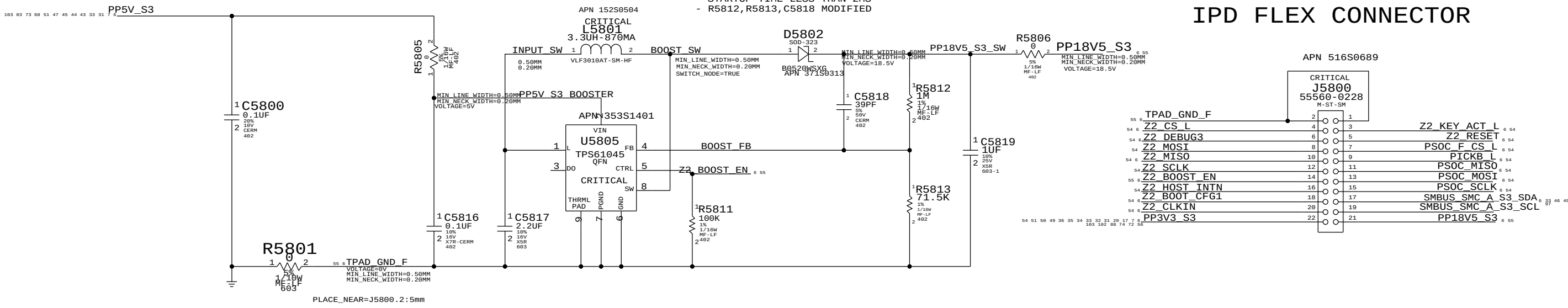




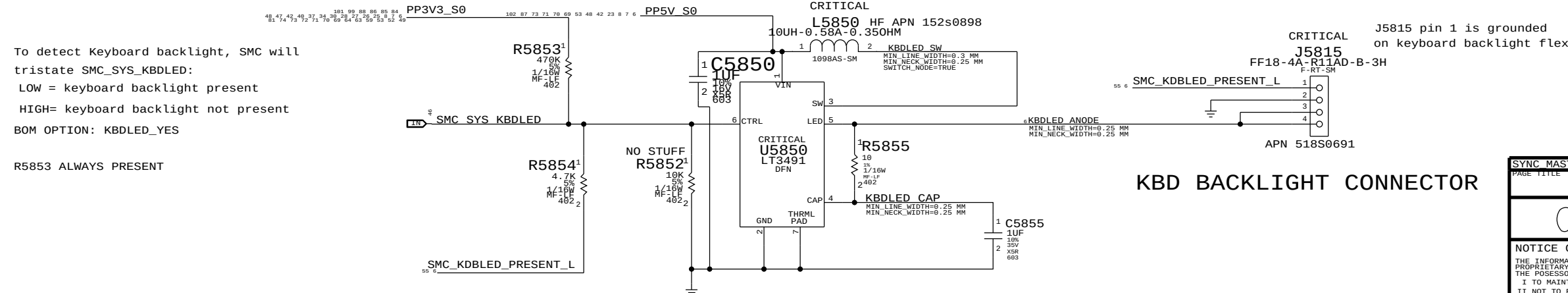
BOOSTER +18.5VDC FOR SENSORS

- BOOSTER DESIGN CONSIDERATION:
- POWER CONSUMPTION
 - DROOP LINE REGULATION
 - RIPPLE TO MEET ERS
 - 100-300 KHZ CLEAN SPECTRUM
 - STARTUP TIME LESS THAN 2MS
 - R5812,R5813,C5818 MODIFIED

IPD FLEX CONNECTOR



Keyboard LED Driver



KBD BACKLIGHT CONNECTOR

SYNC MASTER=K17 WFERRY		SYNC DATE=06/09/2009	
PAGE TITLE		WELLSPRING 2	
Apple Inc.		DRAWING NUMBER	SIZE
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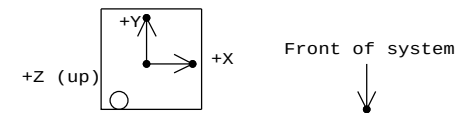
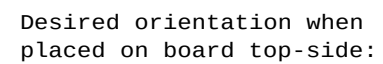
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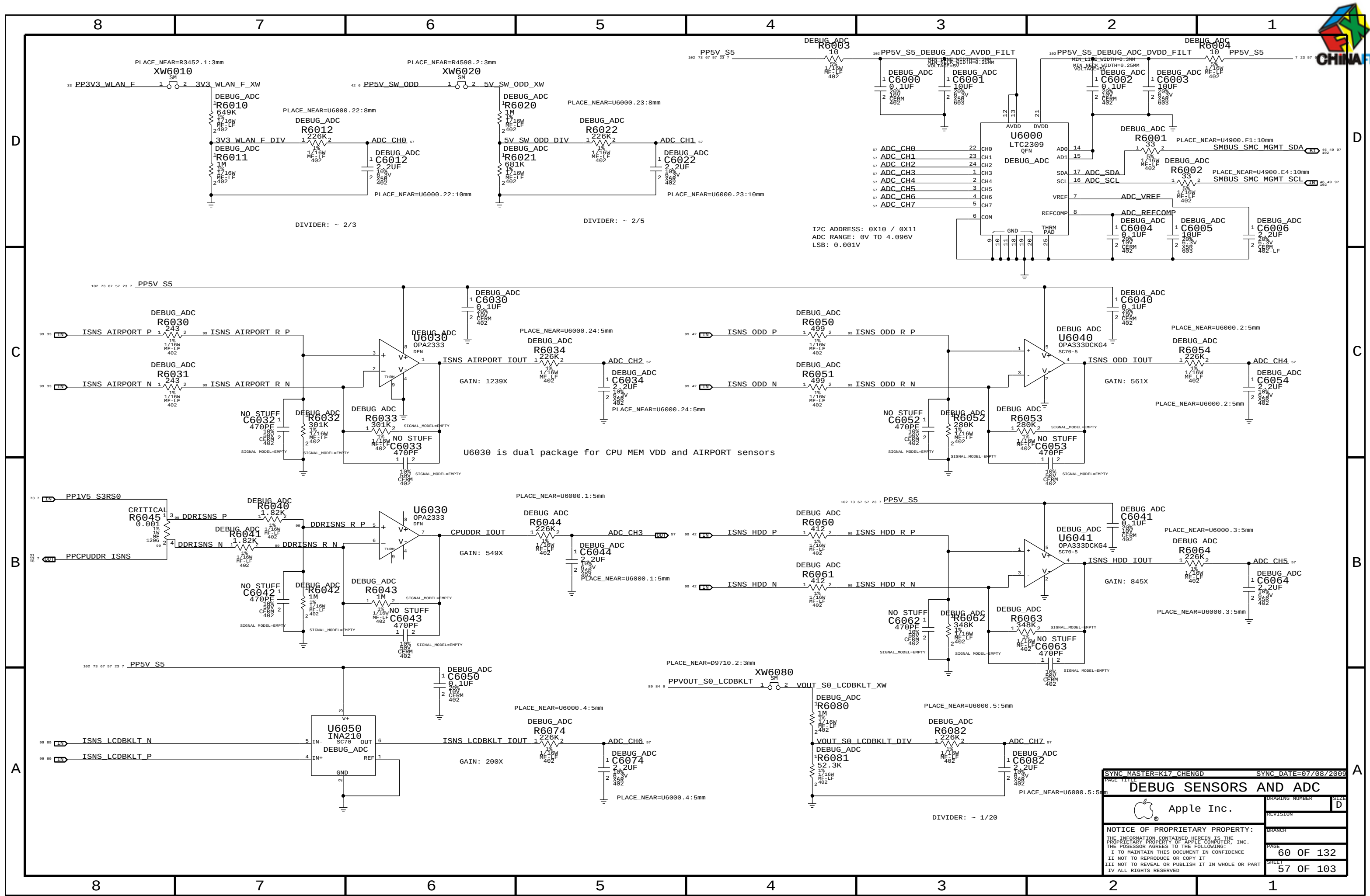
A |

R5921 PULLS UP SMS_PWRDN TO TURN OFF SMS WHEN PIN IS NOT BEING DRIVEN BY SMC

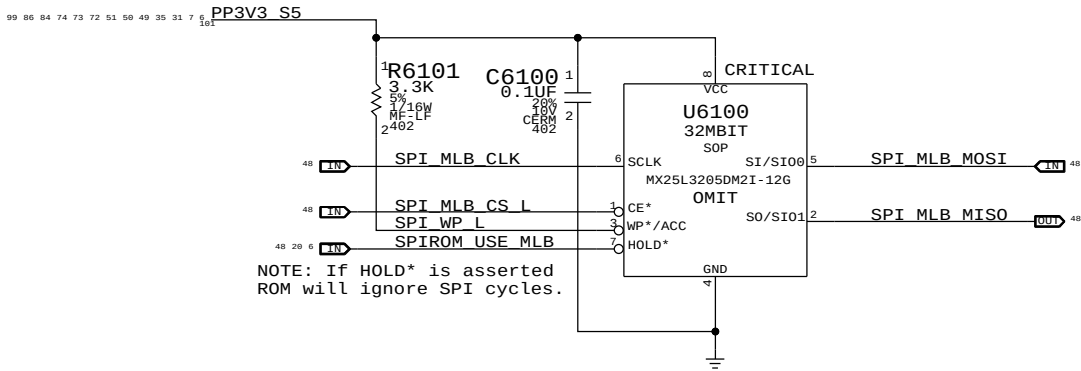


Circle indicates pin 1 location when placed
in correct orientation

SYNC MASTER=K20A MLB		SYNC DATE=03/26/2009	
PAGE TITLE		SIZE	
Sudden Motion Sensor (SMS)		D	
 Apple Inc.		DRAWING NUMBER	
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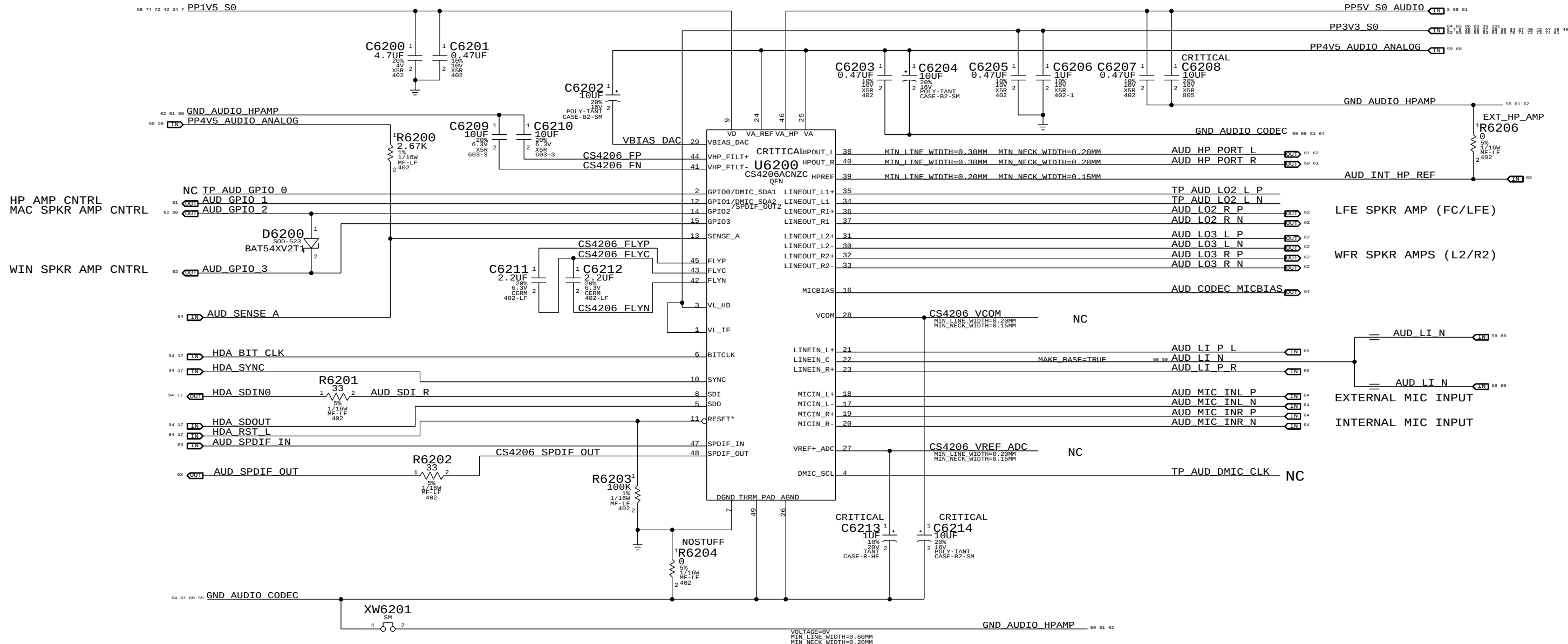
SYNC MASTER=K17 CHENG D		SYNC DATE=07/08/2009	
PAGE TITLE		SIZE	
DEBUG SENSORS AND ADC		D	
Apple Inc.		REVISION	
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SYNC MASTER=K17 WFERRY		SYNC DATE=06/09/2009	
PAGE TITLE		SPI ROM	
		DRAWING NUMBER	SIZE
		REVISION	D
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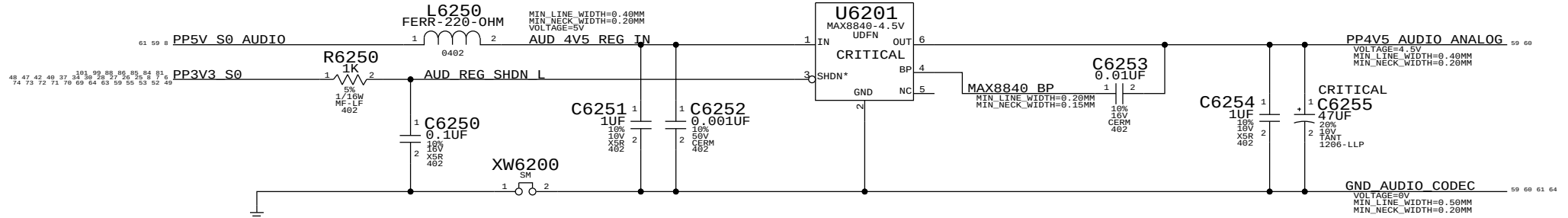


AUDIO CODEC
APPLE P/N 353S2592



DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

AUDIO 4.5V REGULATOR
APPLE P/N 353S2234



SYNC MASTER=K17 REF		SYNC DATE=05/30/2009	
PAGE TITLE		AUDIO: CODEC	
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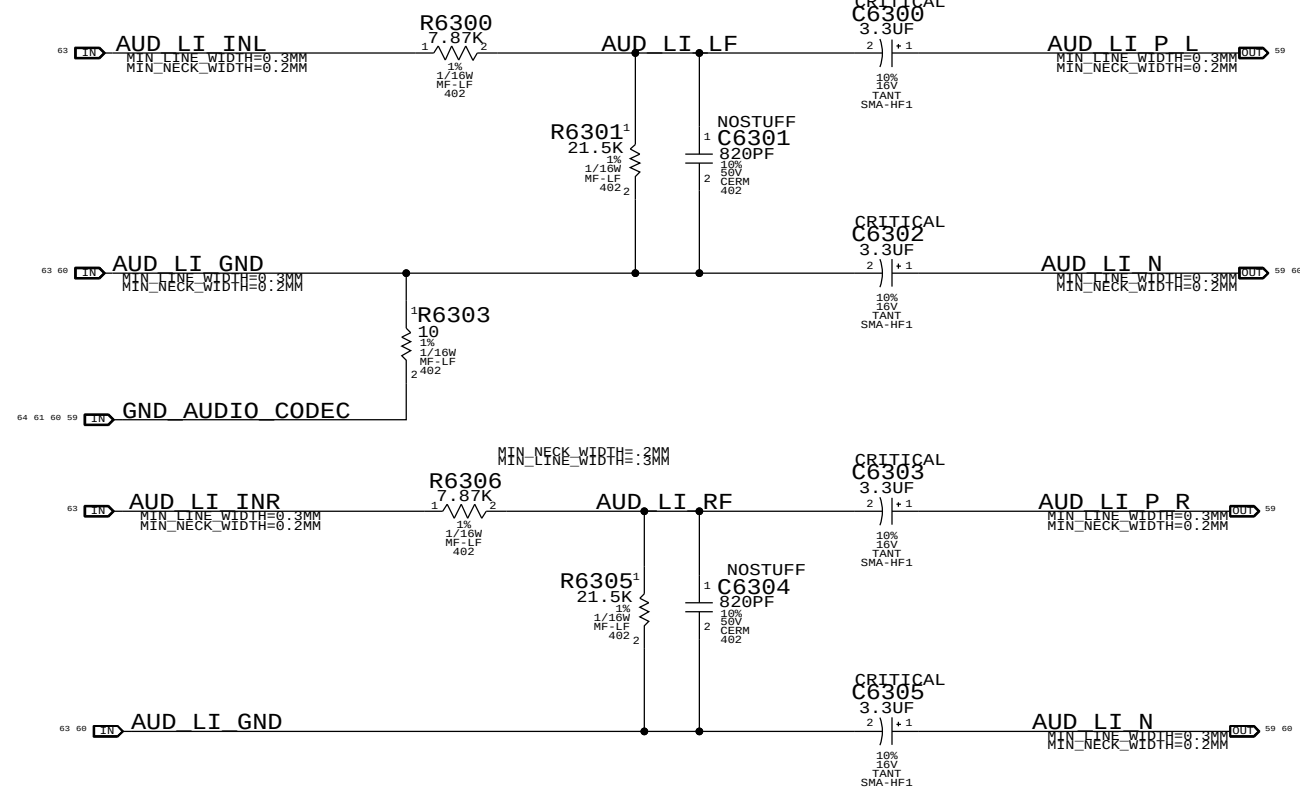
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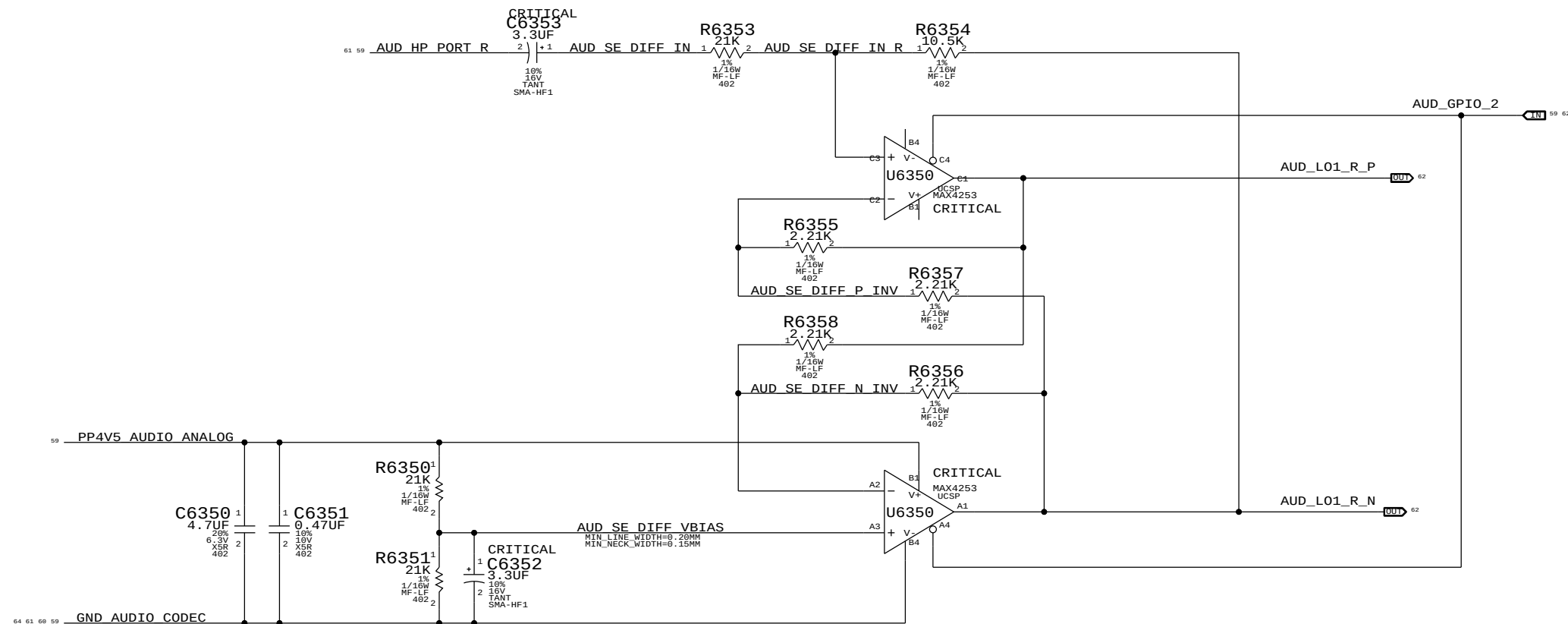
A

CODEC_Nom SE_RIN = 20K OHMS
FC = 5 Hz Max
VIN = 2VRMS CODEC VIN = 1.14 VRMS
NET_RIN = 18K OHMS

MIN_NECK_WIDTH=0.3MM



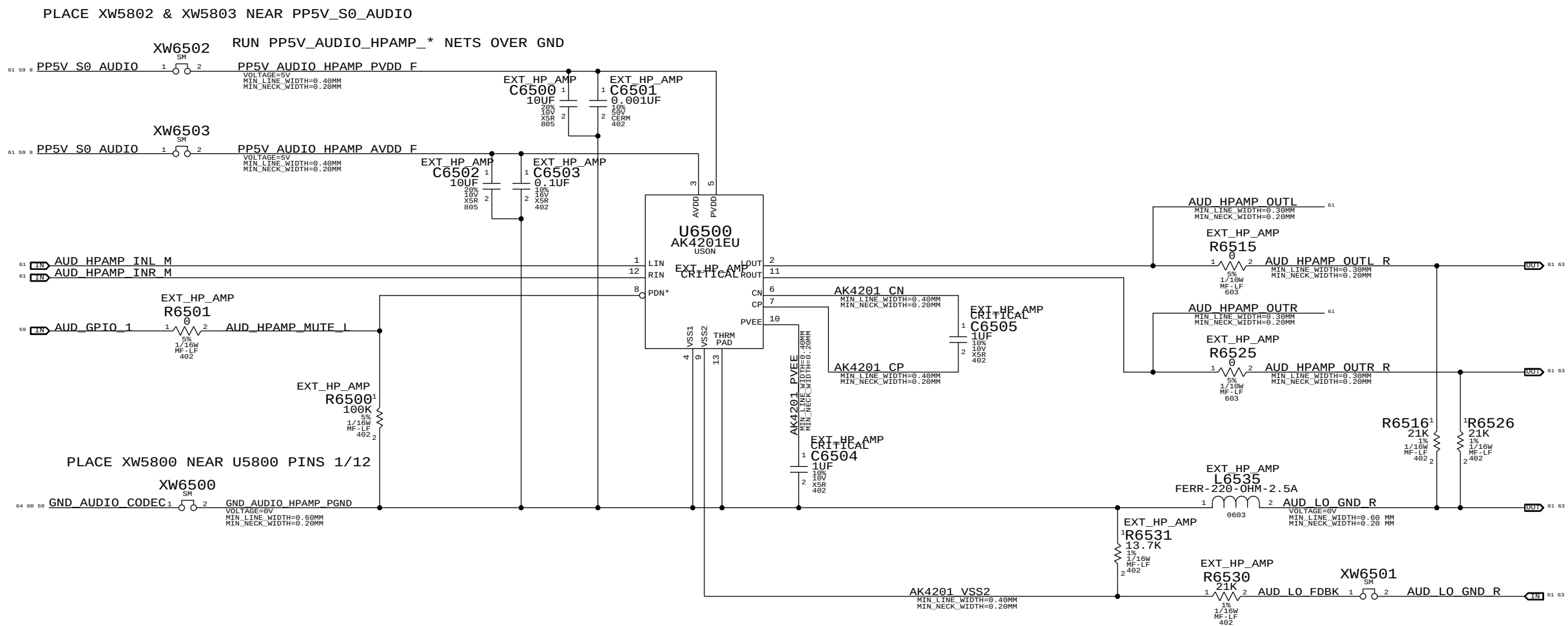
SE-TO-DIFF CONVERTER



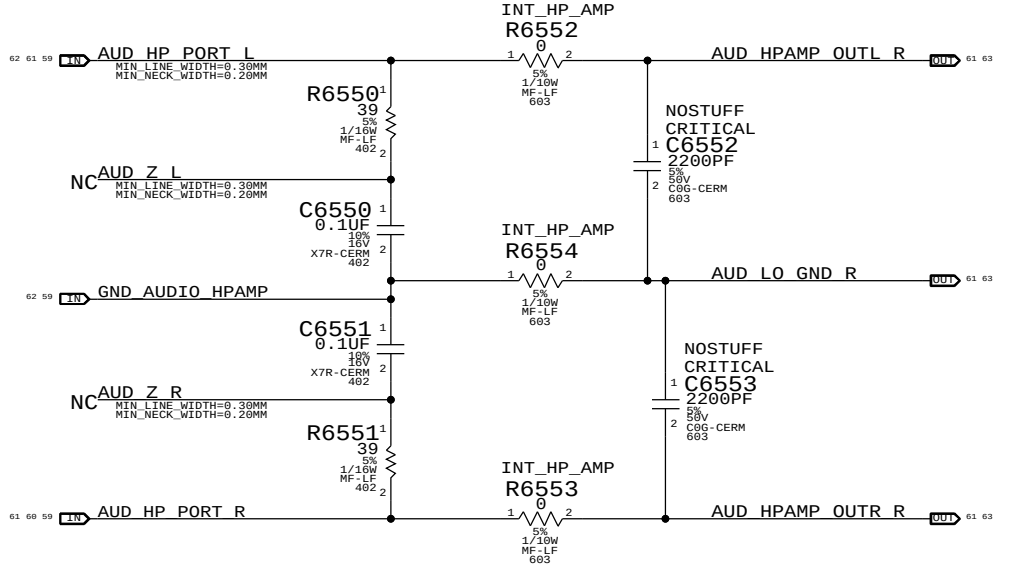
PAGE TITLE		SYNC DATE=05/30/2009	
AUDIO: LINE IN		DRAWING NUMBER	SIZE
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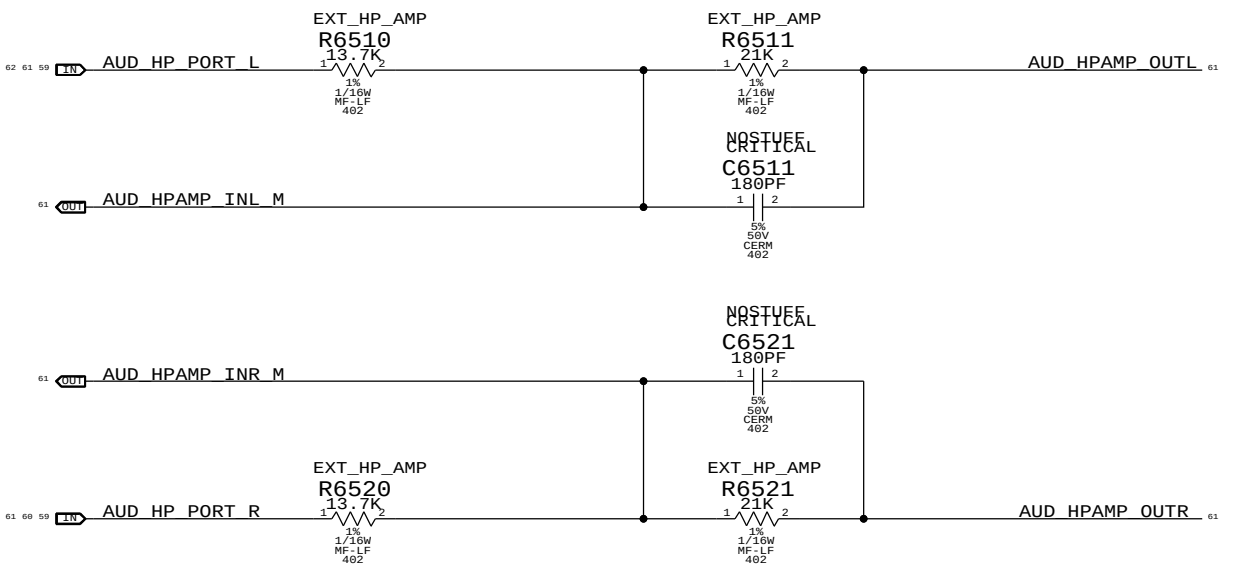
HEADPHONE AMPLIFIER (AK4201)
APN:353S2347
VOLTAGE GAIN:1.53



CS4206A HP OUT ZOBEL NETWORK &
1ST ORDER DAC FILTER PLACEHOLDER



1ST ORDER DAC FILTER
LP:42.10 KHZ



SYNC MASTER=K17 REF		SYNC DATE=05/30/2009	
PAGE TITLE		AUDIO: HEADPHONE OUT	
Apple Inc.		DRAWING NUMBER	SIZE D
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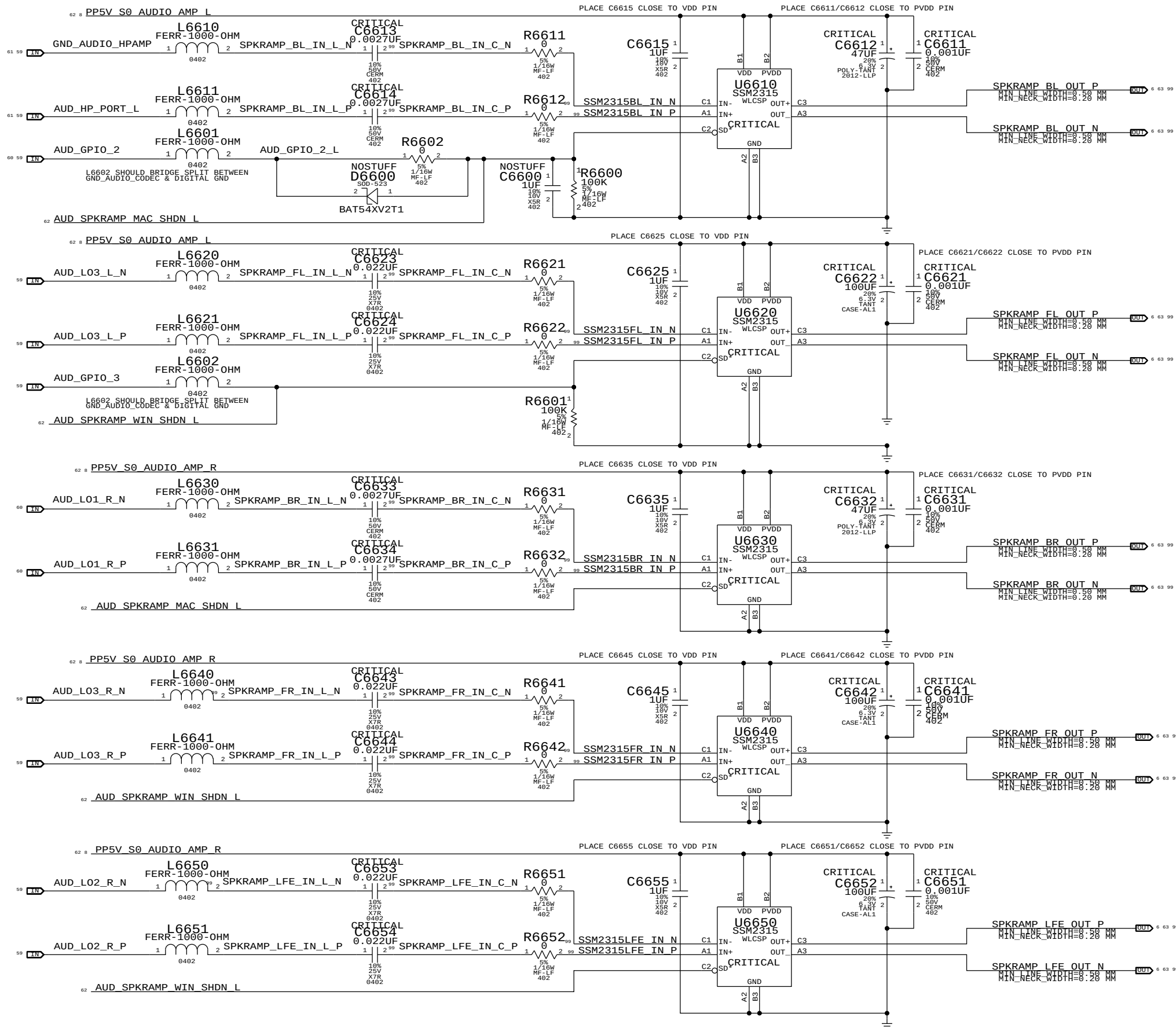
5X MONO SPEAKER AMPLIFIERS (SSM2315)

APN: 353S2500

GAIN = +6 DB

FC (SPEAKERS BL/BR) = ~737 HZ

FC (SPEAKERS FL/FR/LFE) = ~90 HZ

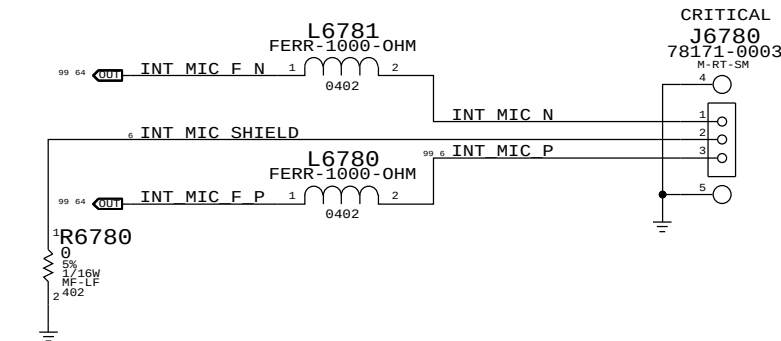


SYNC MASTER=K17 REF		SYNC DATE=05/30/2009	
PAGE TITLE		DRAWING NUMBER	
AUDIO:SPEAKER AMP		D	
Apple Inc.		REVISION	
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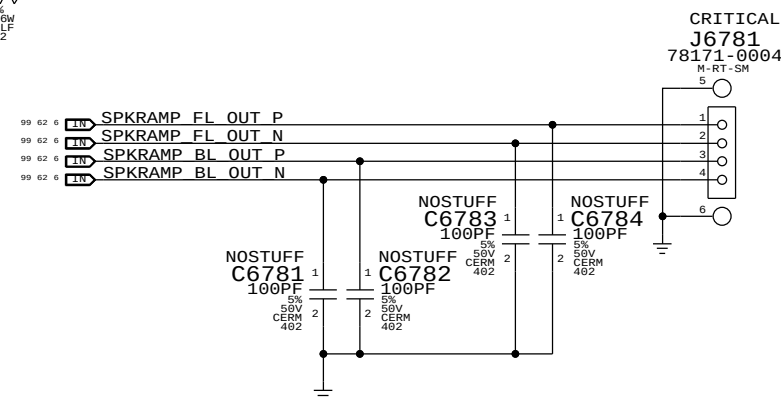


AUDIO JACK 1 LO/HP JACK, SPDIF TX

MIC CONNECTOR
APN: 518S0520

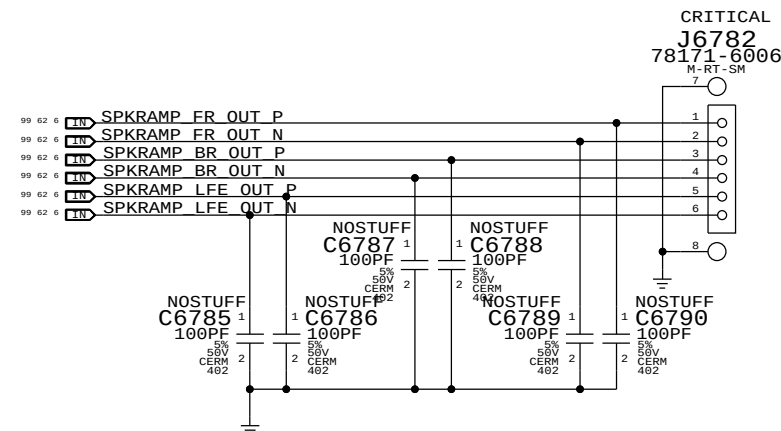


SPEAKER CONNECTORS
APN: 518S0521



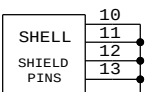
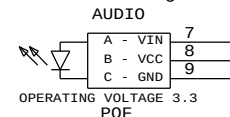
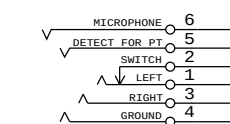
AUDIO JACK 2 LINE IN JACK, SPDIF RX

APN: 518S0672



APN: 514-0632

CRITICAL
J6700
SPDIF-TX-K20
F-RT-TH



C6700
10UF
18V
X5R
402

CRITICAL
DZ6700
6.8V-100PF
402

CRITICAL
DZ6702
6.8V-100PF
402

CRITICAL
DZ6704
6.8V-100PF
402

CRITICAL
DZ6701
6.8V-100PF
402

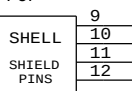
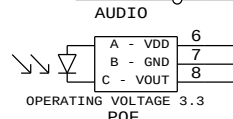
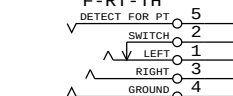
CRITICAL
DZ6703
6.8V-100PF
402

CRITICAL
DZ6705
6.8V-100PF
402

GND CHASSIS AUDIO JACK
MIN LINE WIDTH=0.30MM
MIN NECK WIDTH=0.10MM
VOLTAGE=8V

APN: 514-0633

CRITICAL
J6750
SPDIF-RX-K20
F-RT-TH



C6750
10UF
18V
X5R
402

CRITICAL
DZ6753
ESDLC5-1BM2
S0D882

CRITICAL
DZ6750
ESDLC5-1BM2
S0D882

CRITICAL
DZ6751
6.8V-100PF
402

CRITICAL
DZ6752
6.8V-100PF
402

RETURN FOR HF NOISE

R6750
10K
5%
1/10W
MF-LF
402

SYNC MASTER=K17 LENGO		SYNC DATE=11/24/2009	
PAGE TITLE		AUDIO: JACKS	
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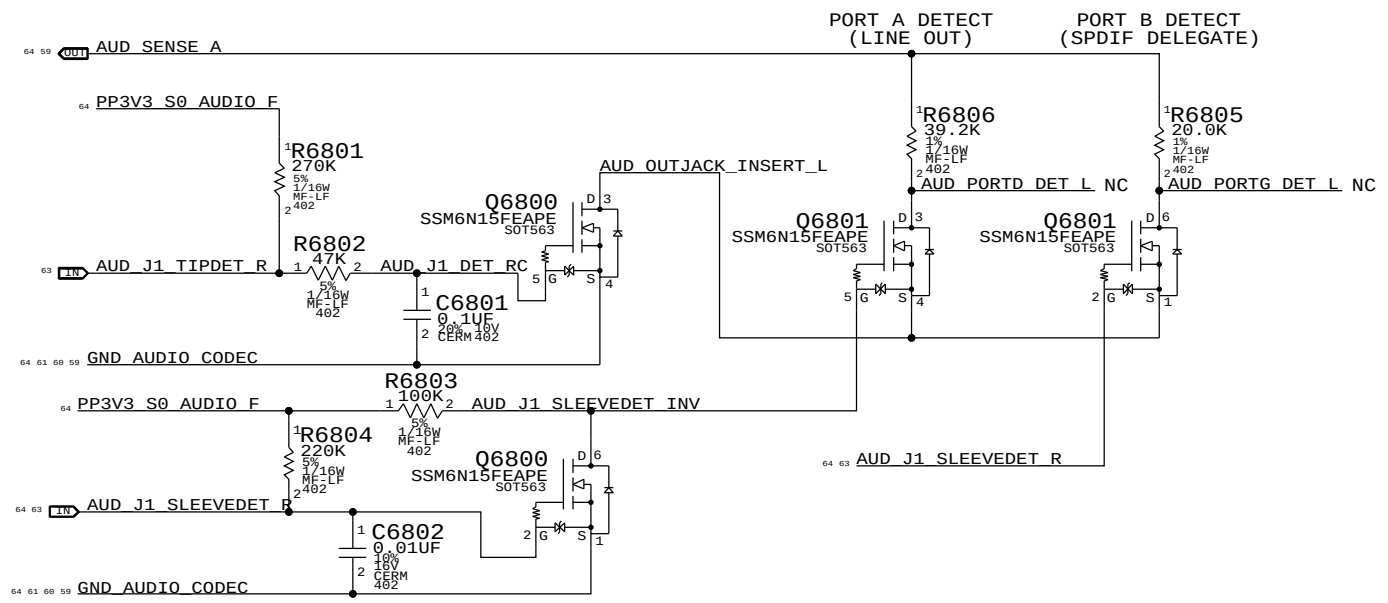
D
C
B
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CODEC OUTPUT SIGNAL PATHS

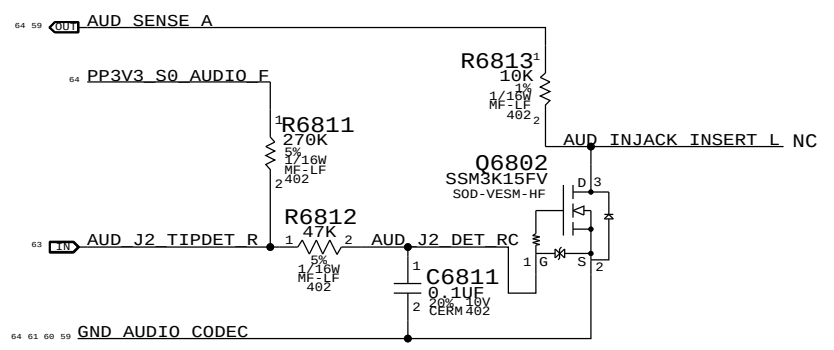
FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC OS SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)	N/A	N/A	0X09 (A)
SPEAKERS BL/BR	0X02 (2)	0X02 (2)	0X09 (9,V23)	GPIO_2	N/A	N/A
SPEAKERS FL/FR	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_2	GPIO_3	N/A
SPEAKER LFE	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0X10 (16)	N/A	N/A	0X0D (B)

CODEC INPUT SIGNAL PATHS

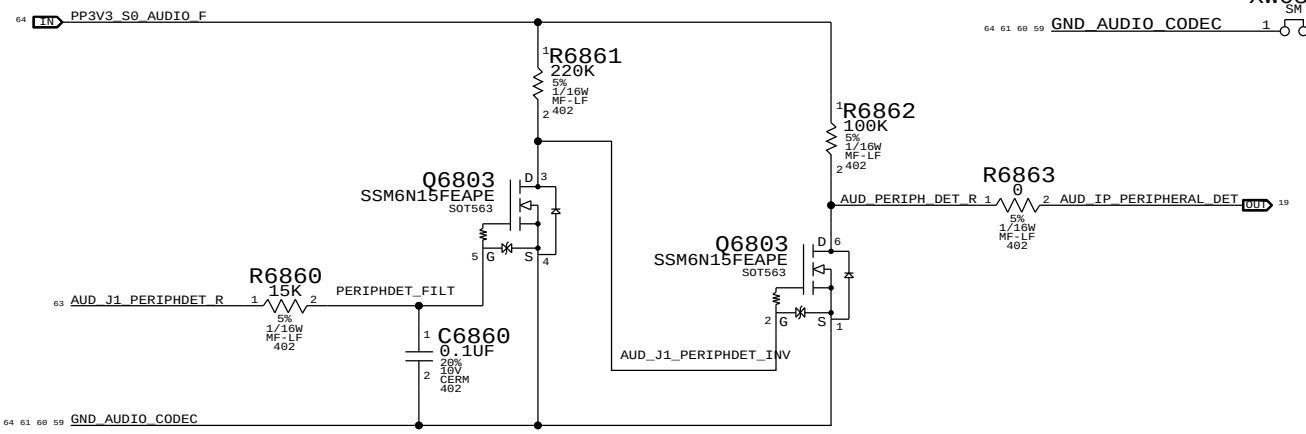
FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X12 (12,C)	N/A	0X12 (C)
SPDIF IN	0X07 (7)	0X0F (15)	N/A	N/A
INTERNAL MIC	0X06 (6)	0X0D (13,B,RIGHT)	MICBIAS (80%)	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY



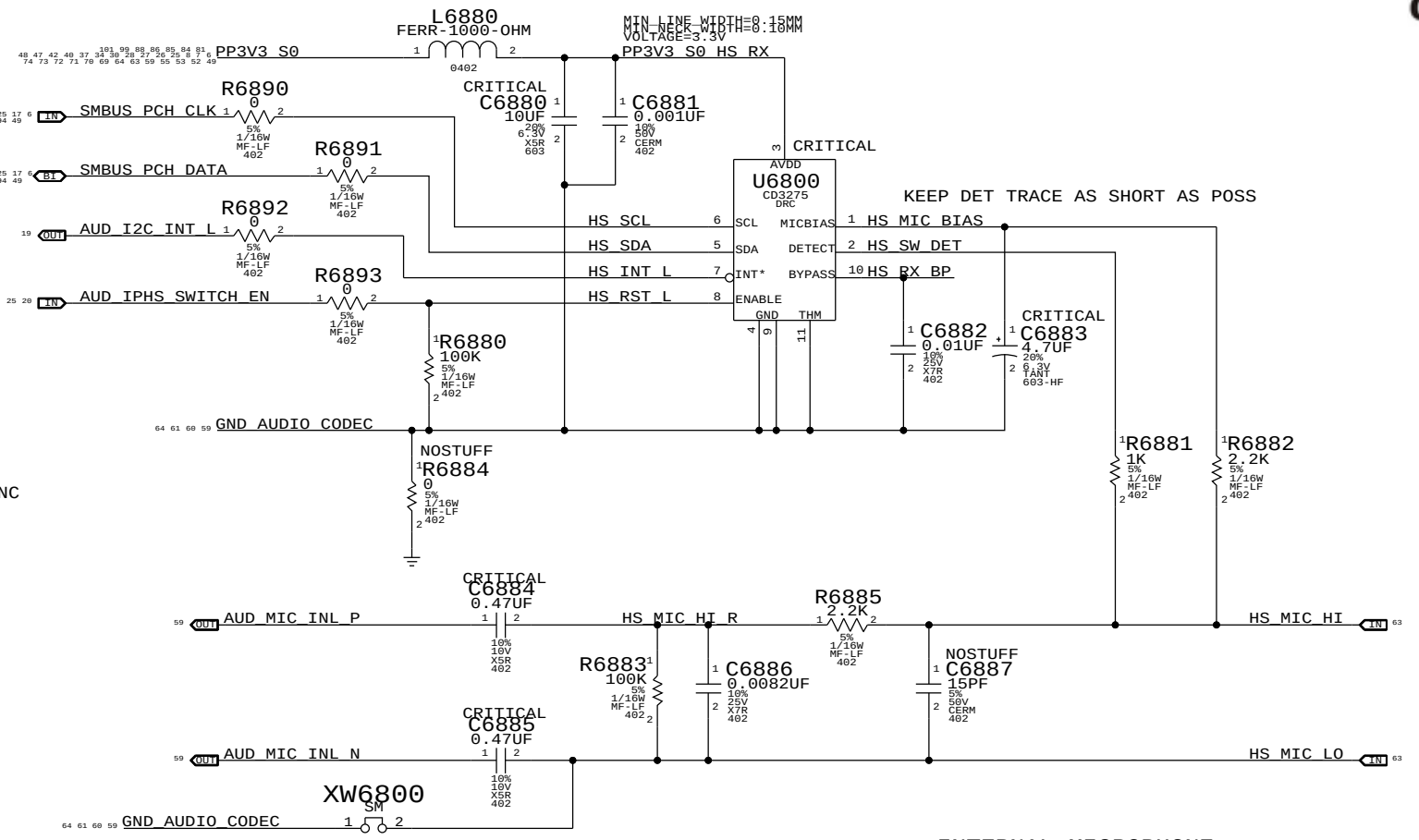
PORT C DETECT (LINE IN)



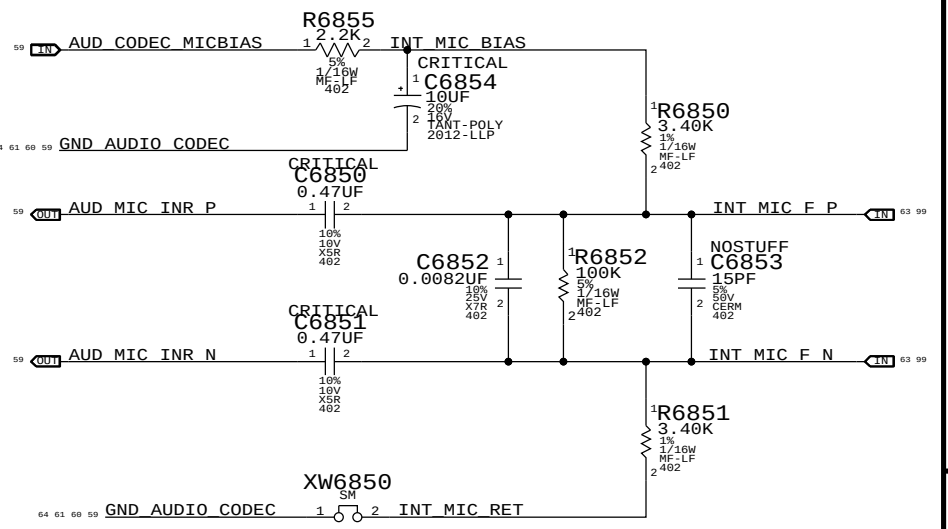
EXTRACTION NOTIFICATION CKT



"MIKEY"/EXTERNAL MICROPHONE



INTERNAL MICROPHONE

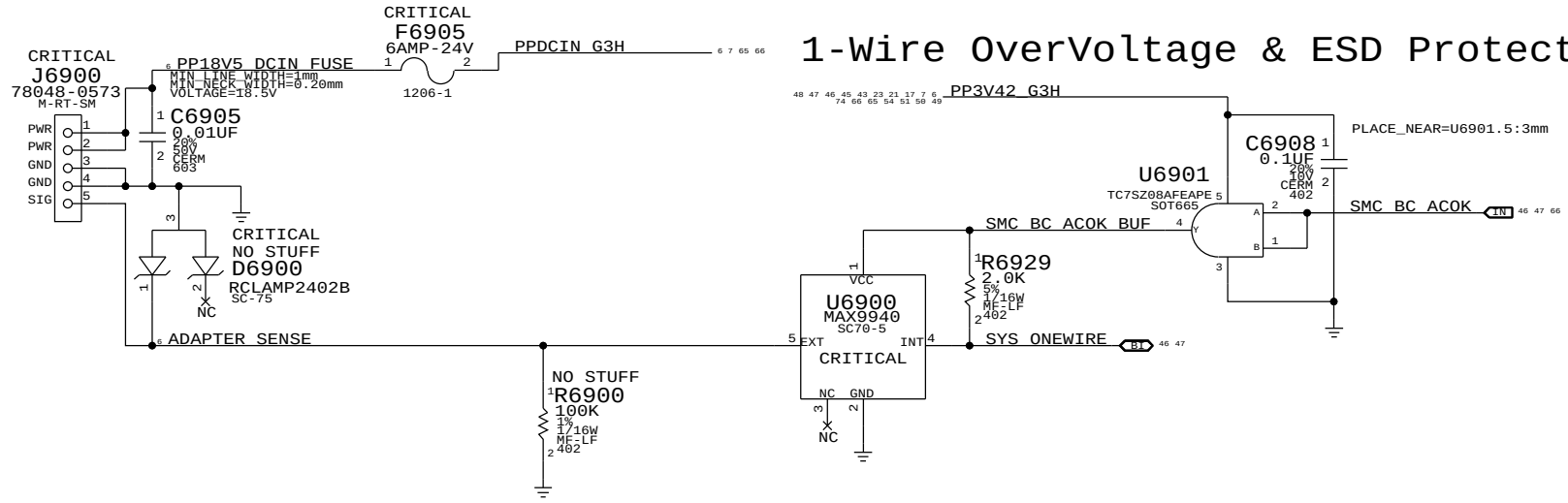


PAGE TITLE		SYNC DATE=05/30/2009	
AUDIO: JACK TRANSLATORS		DRAWING NUMBER	SIZE
Apple Inc.		REVISION	D
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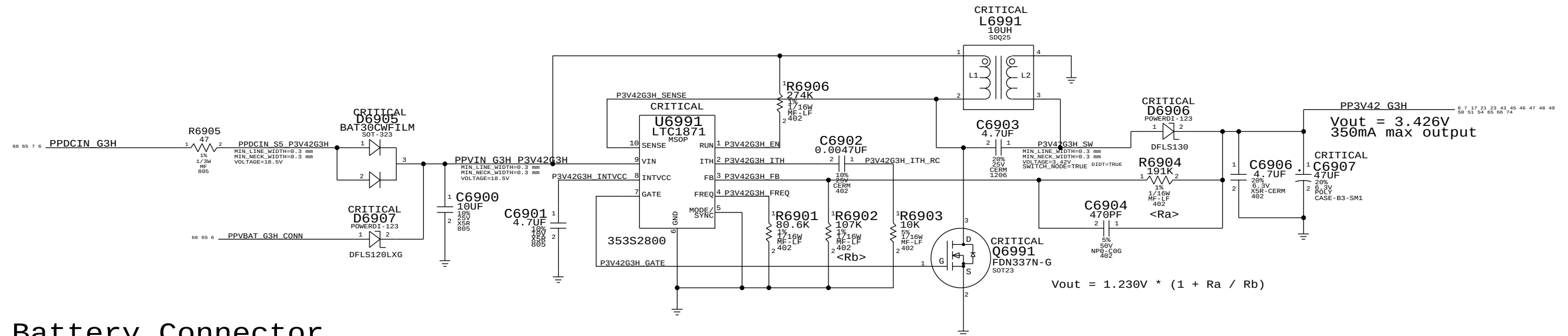
MagSafe DC Power Jack

1-Wire OverVoltage & ESD Protection

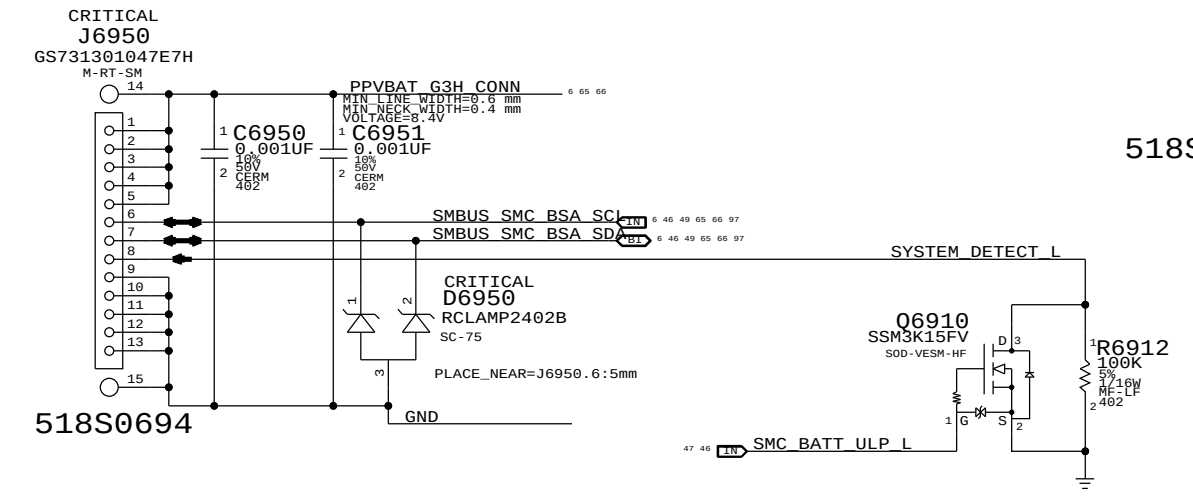


3.425V "G3Hot" Supply

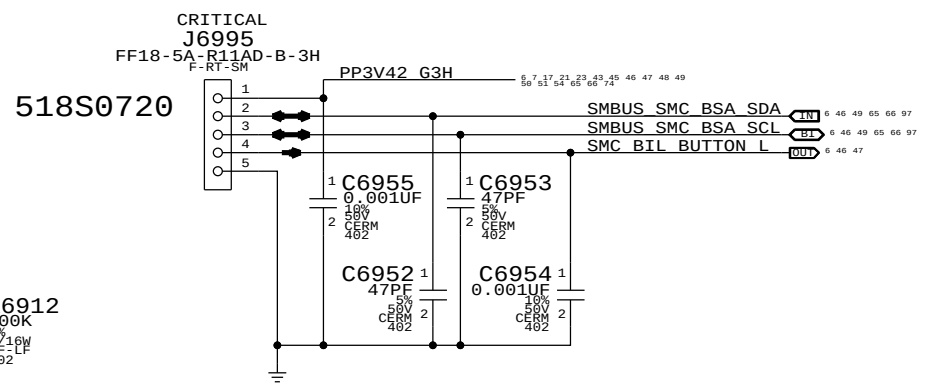
Supply needs to guarantee 3.31V delivered to SMC VRef generator



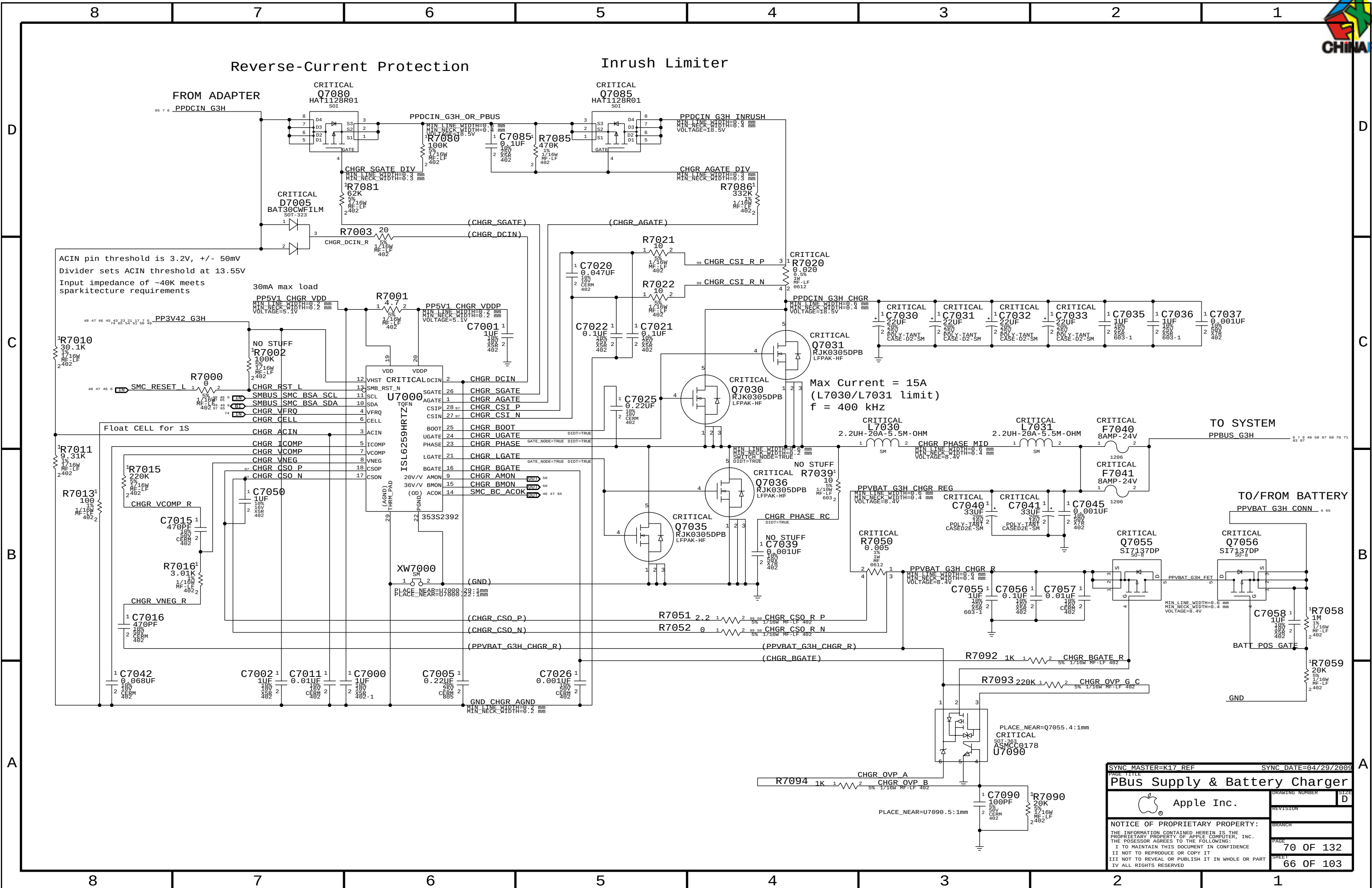
Battery Connector

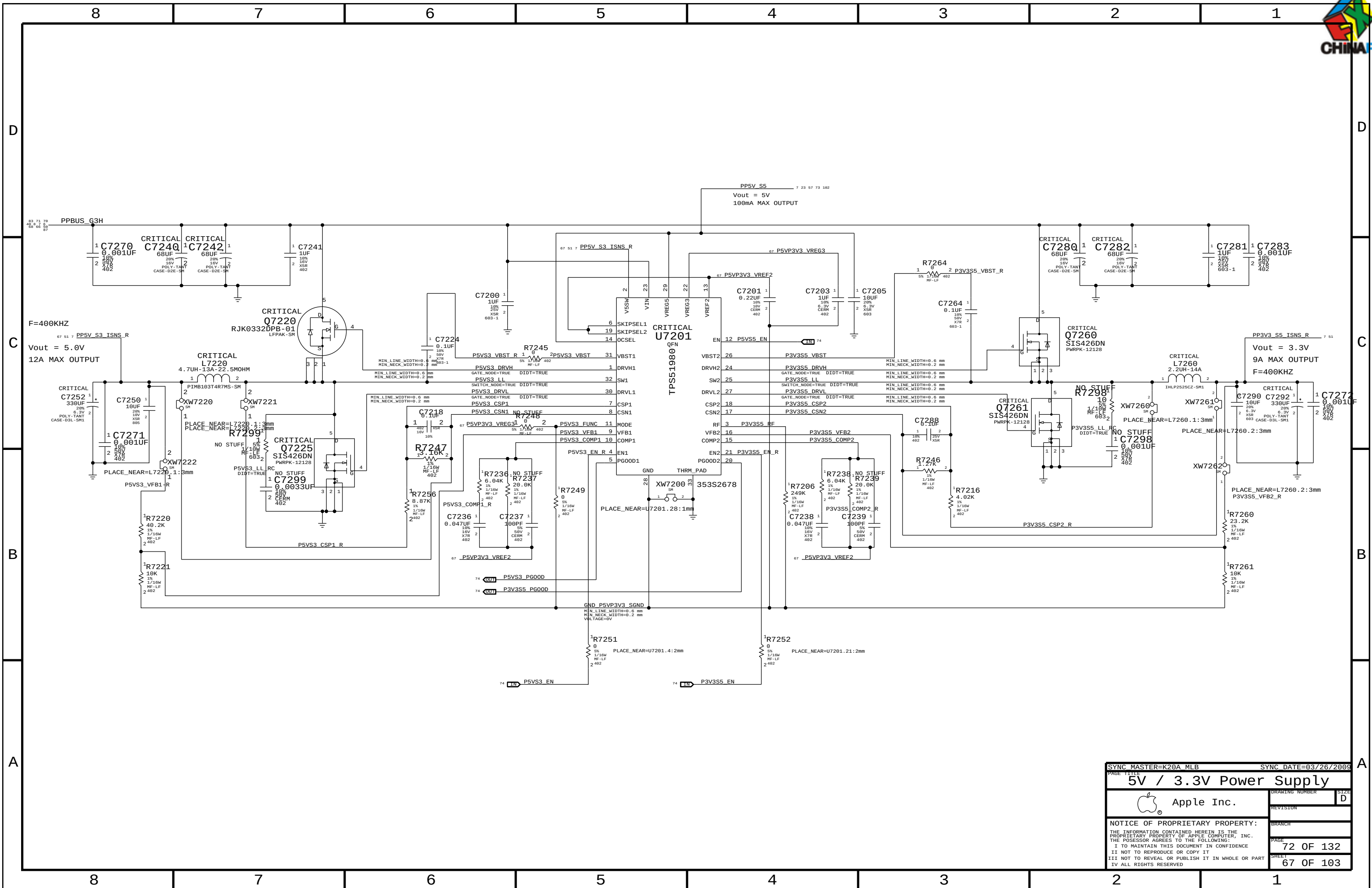


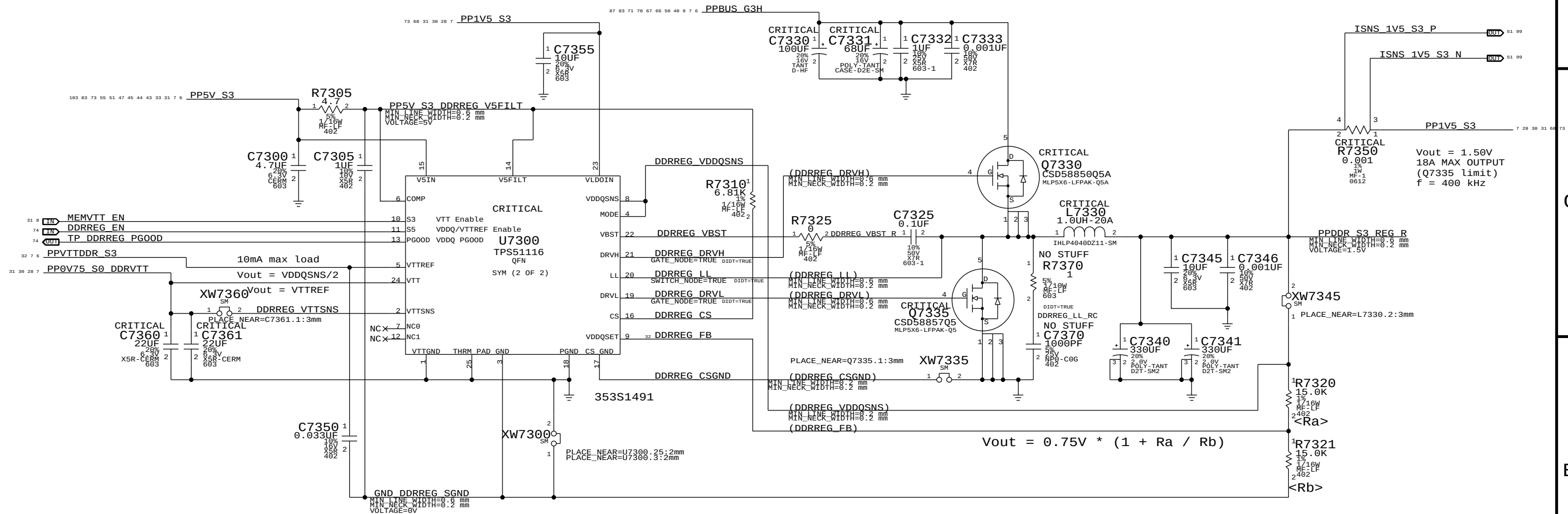
BIL Connector



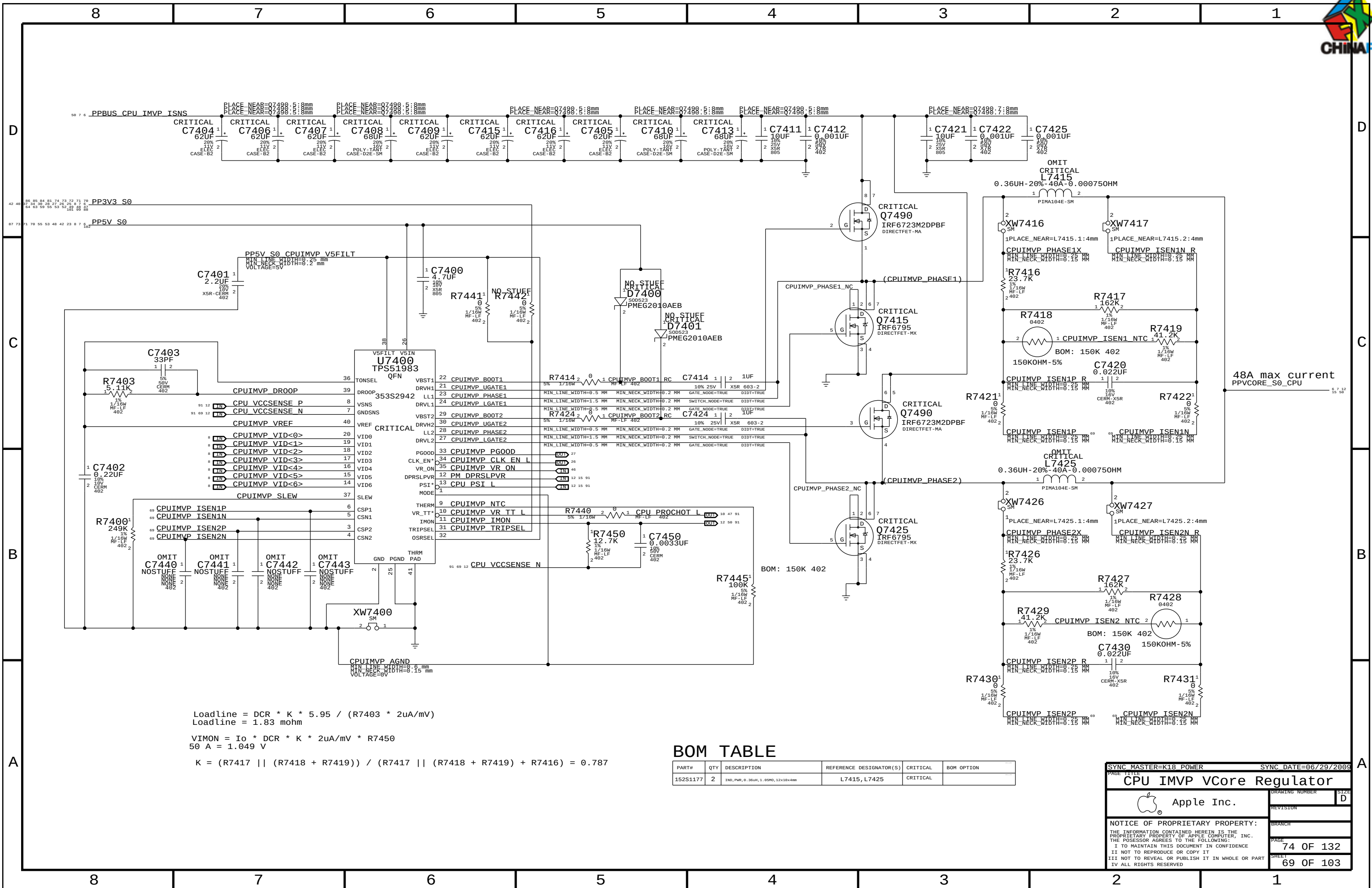
PAGE TITLE		SYNC DATE=04/29/2009	
DC-In & Battery Connectors		DRAWING NUMBER	
Apple Inc.		SIZE	
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SYNC MASTER=K17 REF		SYNC DATE=06/24/2009	
PAGE TITLE		1.5V DDR3 Supply	
Apple Inc.		DRAWING NUMBER	SIZE
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Loadline = $DCR * K * 5.95 / (R7403 * 2\mu A/mV)$
Loadline = 1.83 mohm

VIMON = $I_o * DCR * K * 2\mu A/mV * R7450$
50 A = 1.049 V

$K = (R7417 || (R7418 + R7419)) / (R7417 || (R7418 + R7419) + R7416) = 0.787$

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1177	2	IND, PWR, 0.36uH, 1.05NO, 12x18x4mm	L7415, L7425	CRITICAL	

SYNC MASTER=K18 POWER

SYNC DATE=06/29/2009

CPU IMVP VCore Regulator

Apple Inc.

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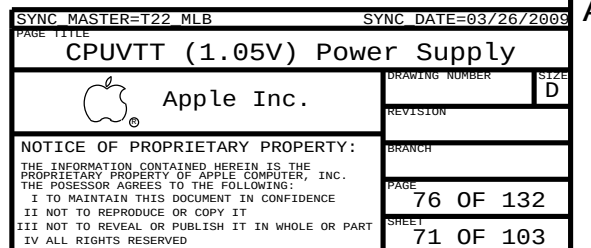
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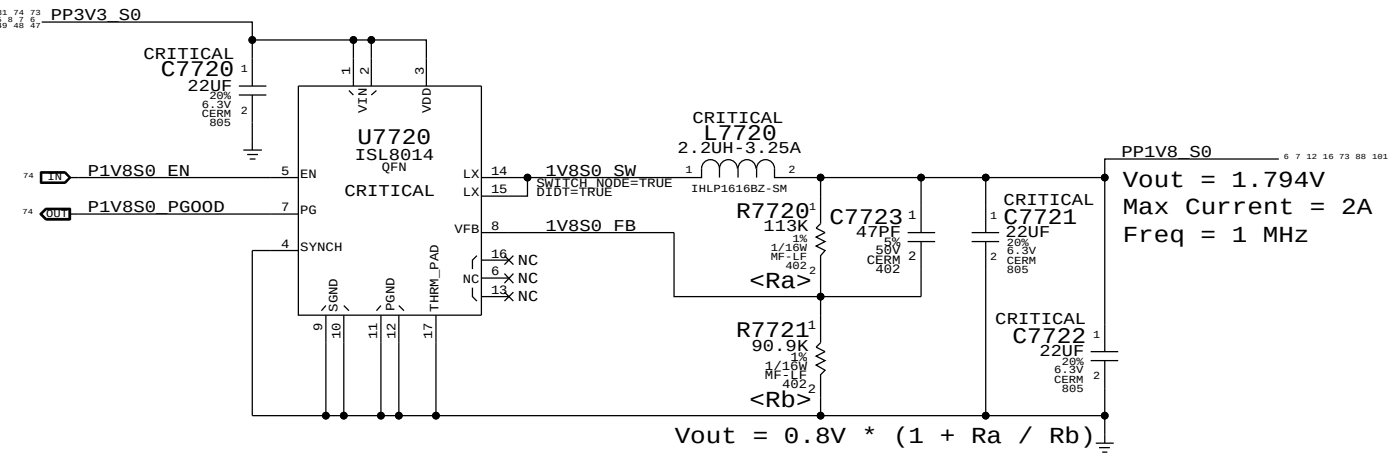


SYNCH MASTER=T22 MLB		SYNCH DATE=03/26/2009	
PAGE TITLE		PAGE NO.	
GFX IMVP VCore Regulator			
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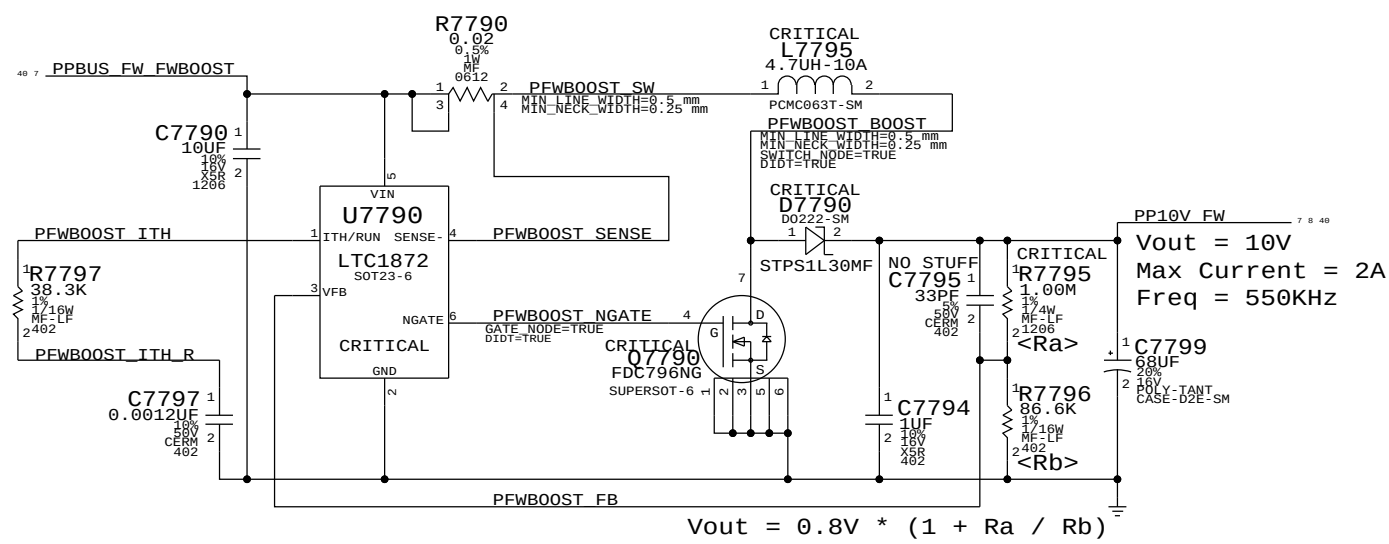




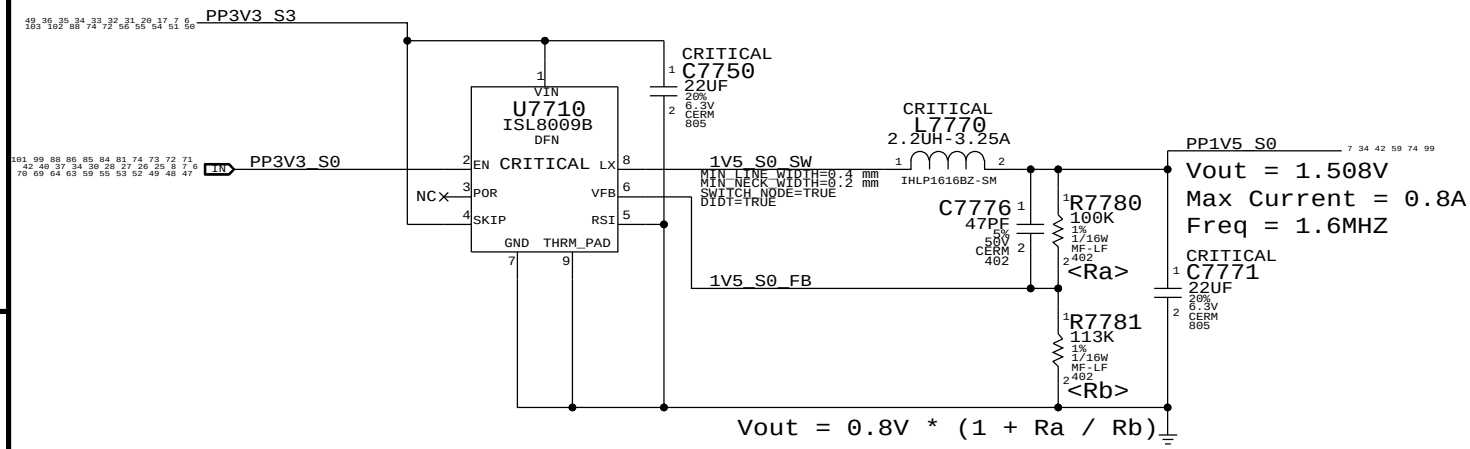
1.8V S0 Regulator



FW 10V Boost Regulator

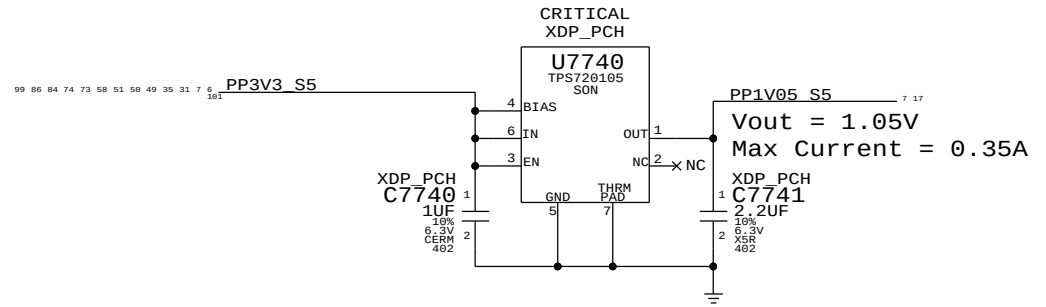


1.5V S0 Regulator

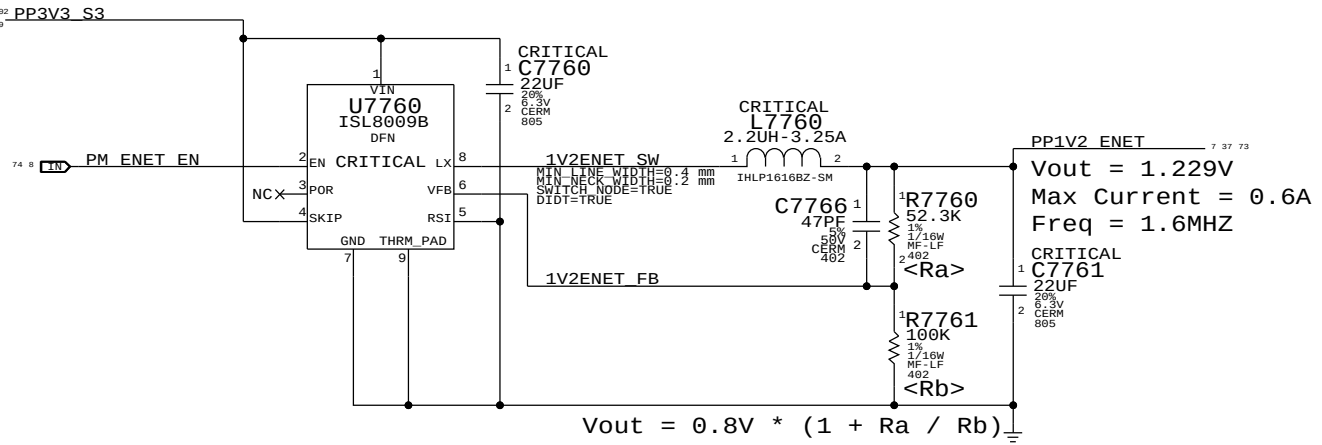


1.05V S5 LDO

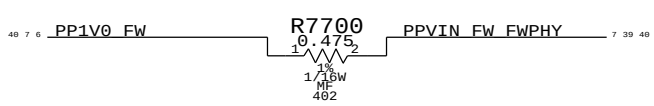
Ibex Peak-M requires JTAG pull-ups to be powered at 1.05V in S5. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



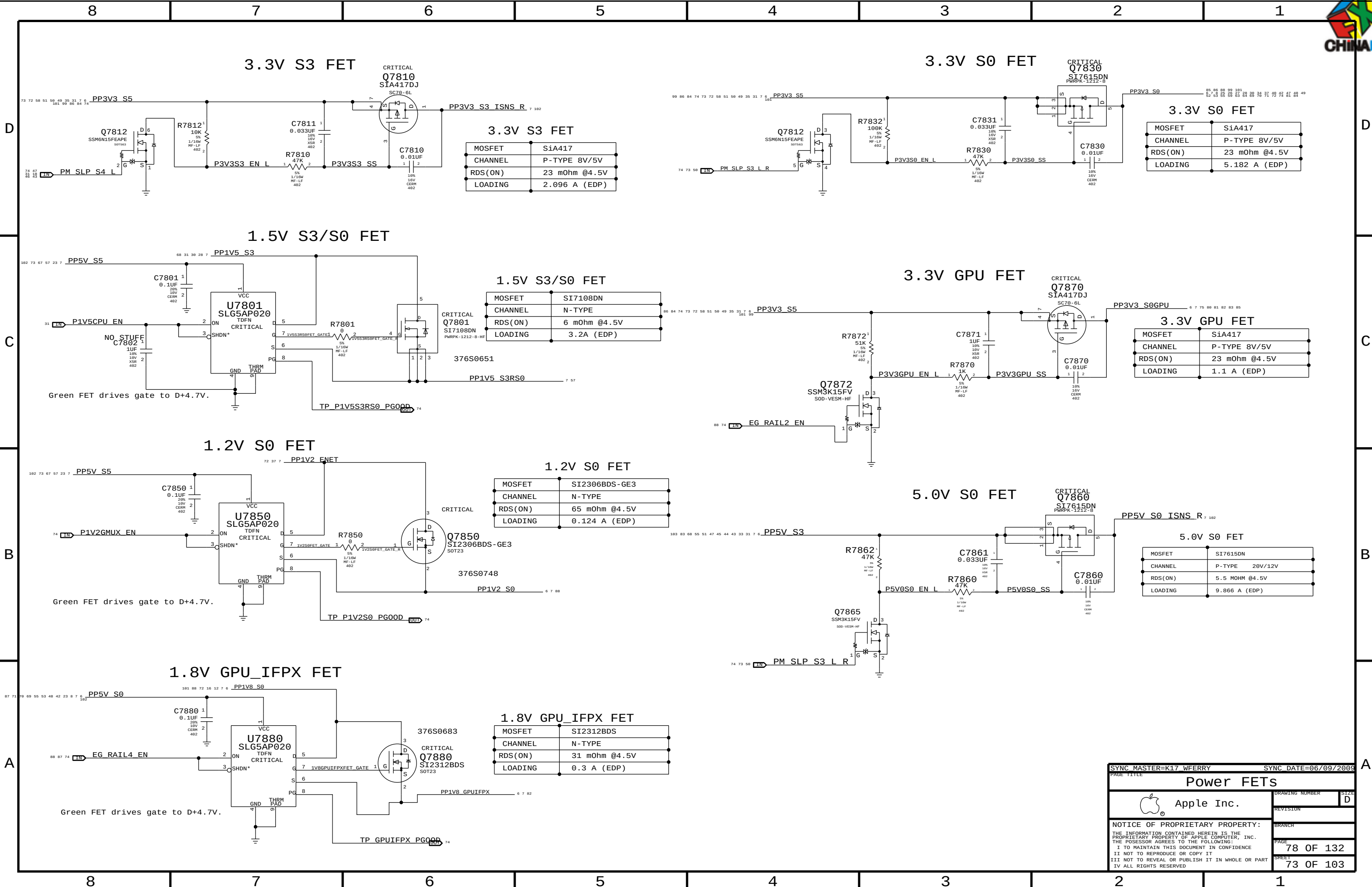
1.2V ENET Regulator



1.05V to 1.0V FW Drop

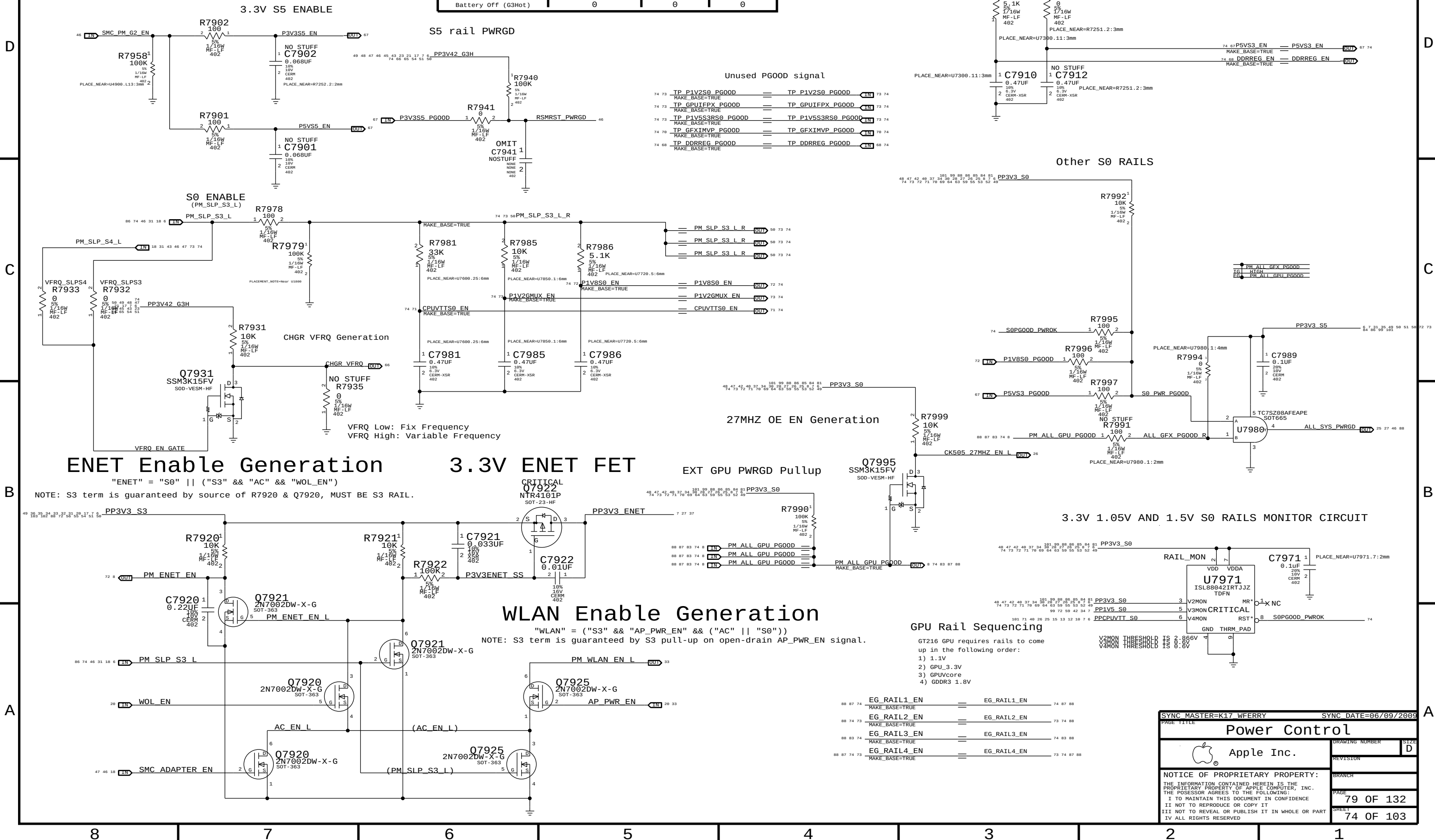


SYNC MASTER=K17 WFERRY		SYNC DATE=06/09/2009	
PAGE TITLE		Misc Power Supplies	
Apple Inc.		DRAWING NUMBER	SIZE D
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State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3Hot)	0	0	0



SYNC MASTER=K17_WFERRY

SYNC DATE=06/09/2009

Power Control

Apple Inc.

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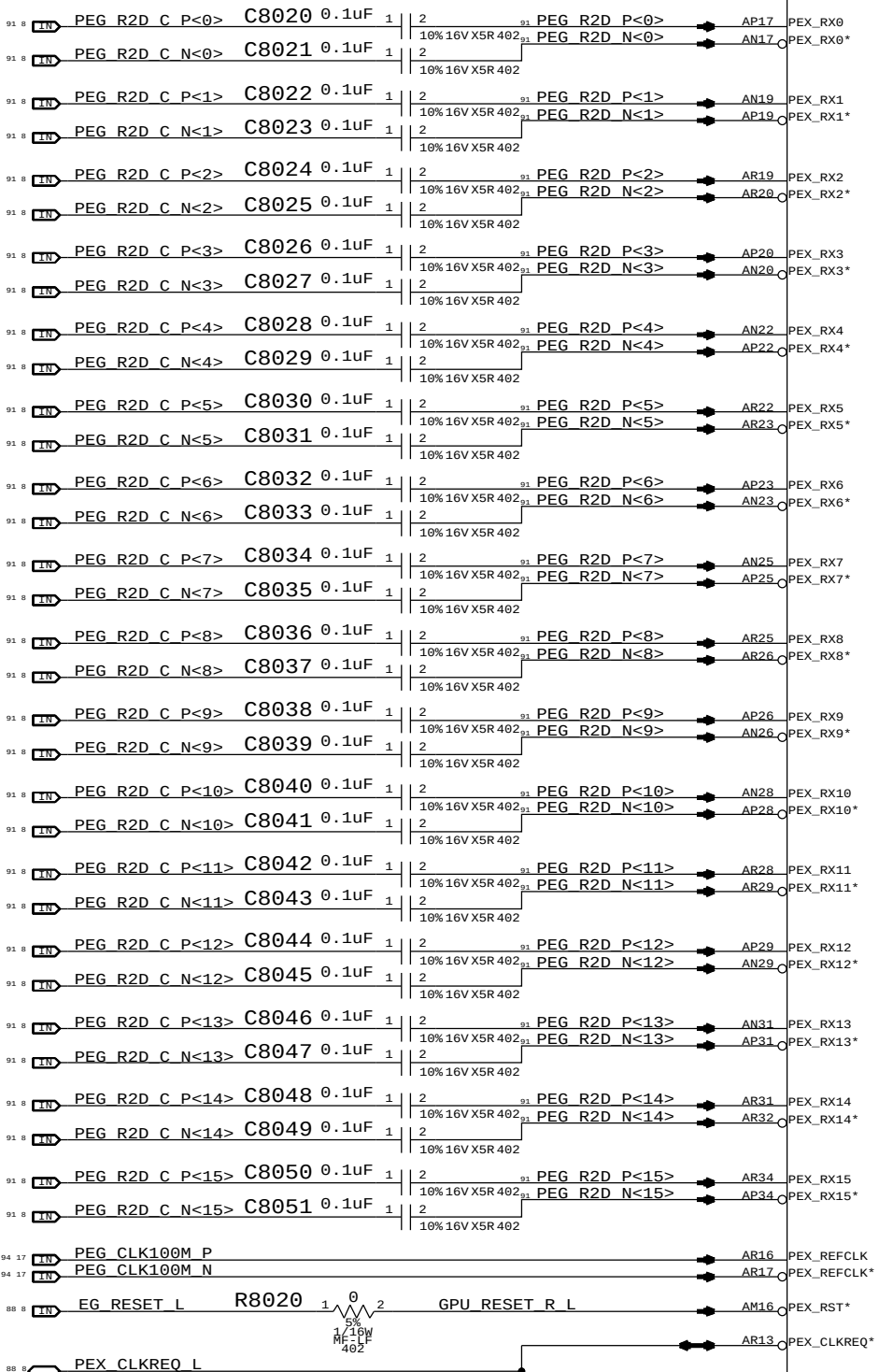
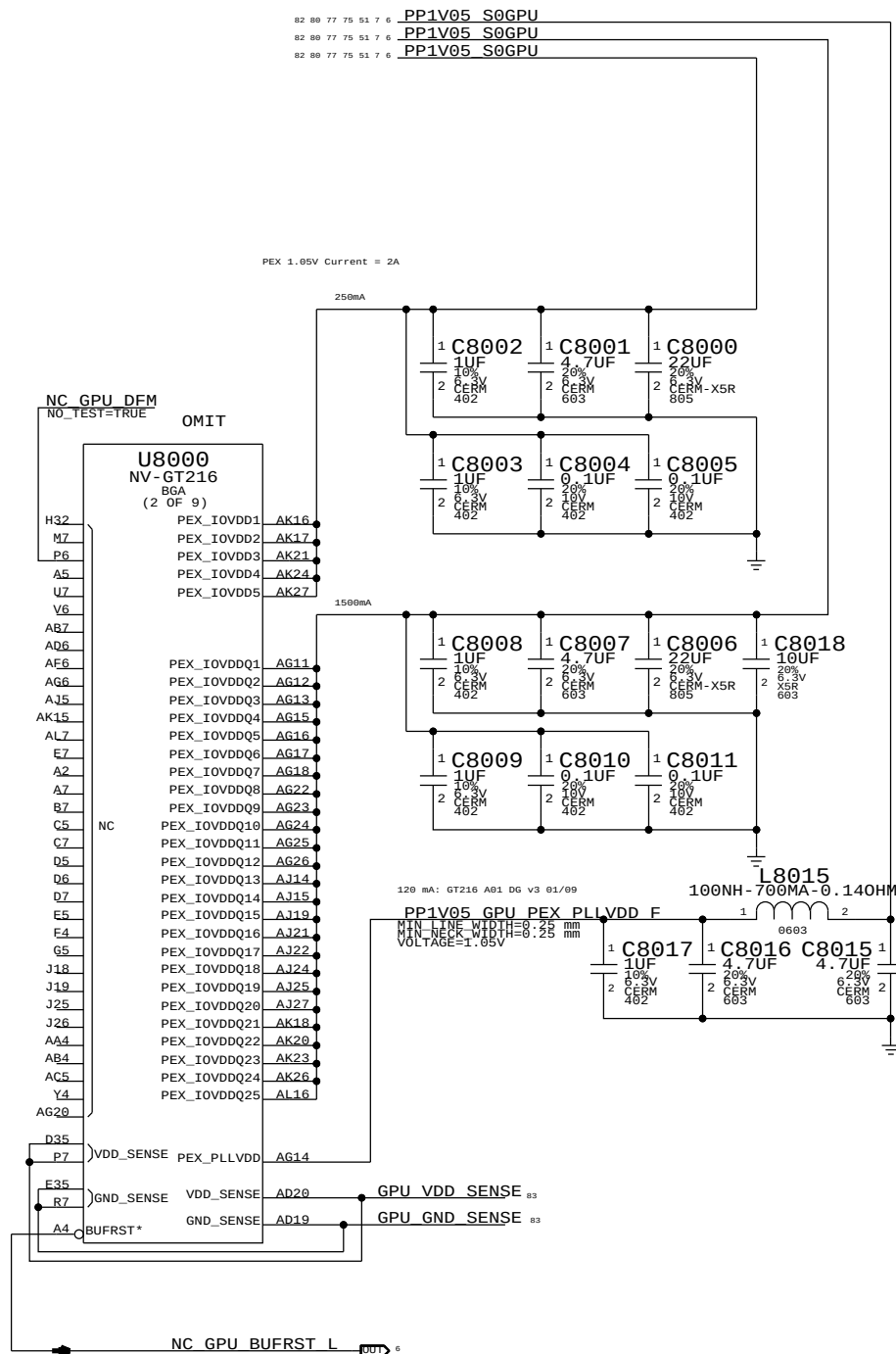


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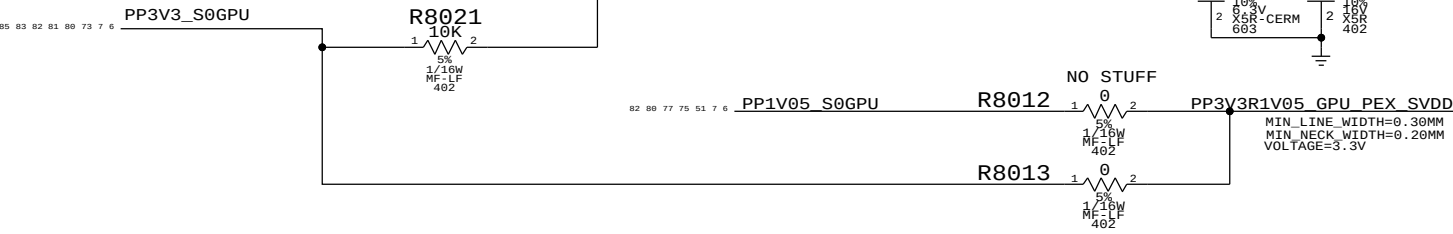
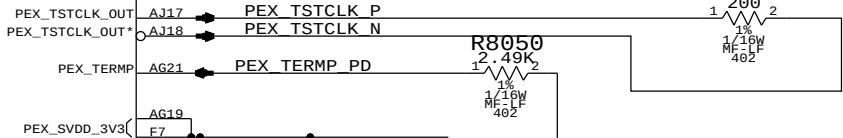
Power aliases required by this page:
- =PP1V2_GPU_PEX_PLLXVDD
- =PP1V2_GPU_PEX_IOVDDQ
- =PP1V2_GPU_PEX_IOVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



OMIT
U8000
NV-GT216
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SYNC MASTER=K18 MLB		SYNC DATE=06/29/2009	
PAGE TITLE			
NV GT216 PCI-E			
Apple Inc.		DRAWING NUMBER	SIZE D
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1

Page Notes

Power aliases required by this page:

- =PPVCORE_GPU
- =PP1V8_GPU_FBVDDQ

Signal aliases required by this page:

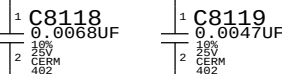
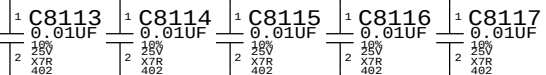
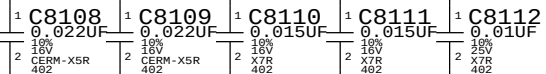
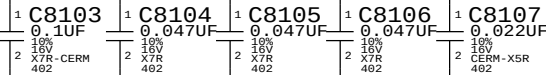
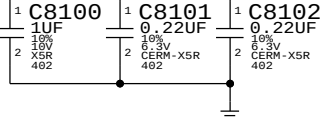
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BOM options provided by this page:

(NONE)

PPVCORE_GPU

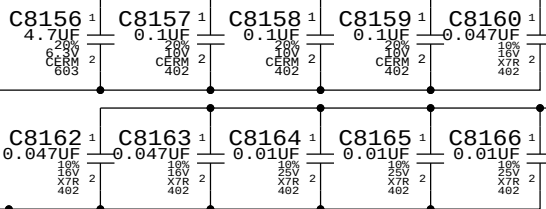
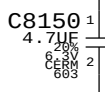
???A @ ???/???MHz Core/Mem Clk for VDD



PP1V8 S0GPU ISNS

Nvidia PRD for GB-128 uses 4x4.7uF, 8x0.47uF, 16x0.1uF

???A @ ???MHz 1.8V GDDR3

U8000
NV-GT216
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VDD

VDD

OMIT

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FBVDDQ

FBVDDQ

B18
J17
J27
AB27
AB29
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AJ28
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J15
J16

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AA29
AA31

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OMIT

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GND

GND

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AL27
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AN2
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AP3
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AP27
AP30
K9
AK14

SYNC_MASTER=GT216 SYNC_DATE=03/26/2009

PAGE TITLE

NV GT216 CORE/FB POWER



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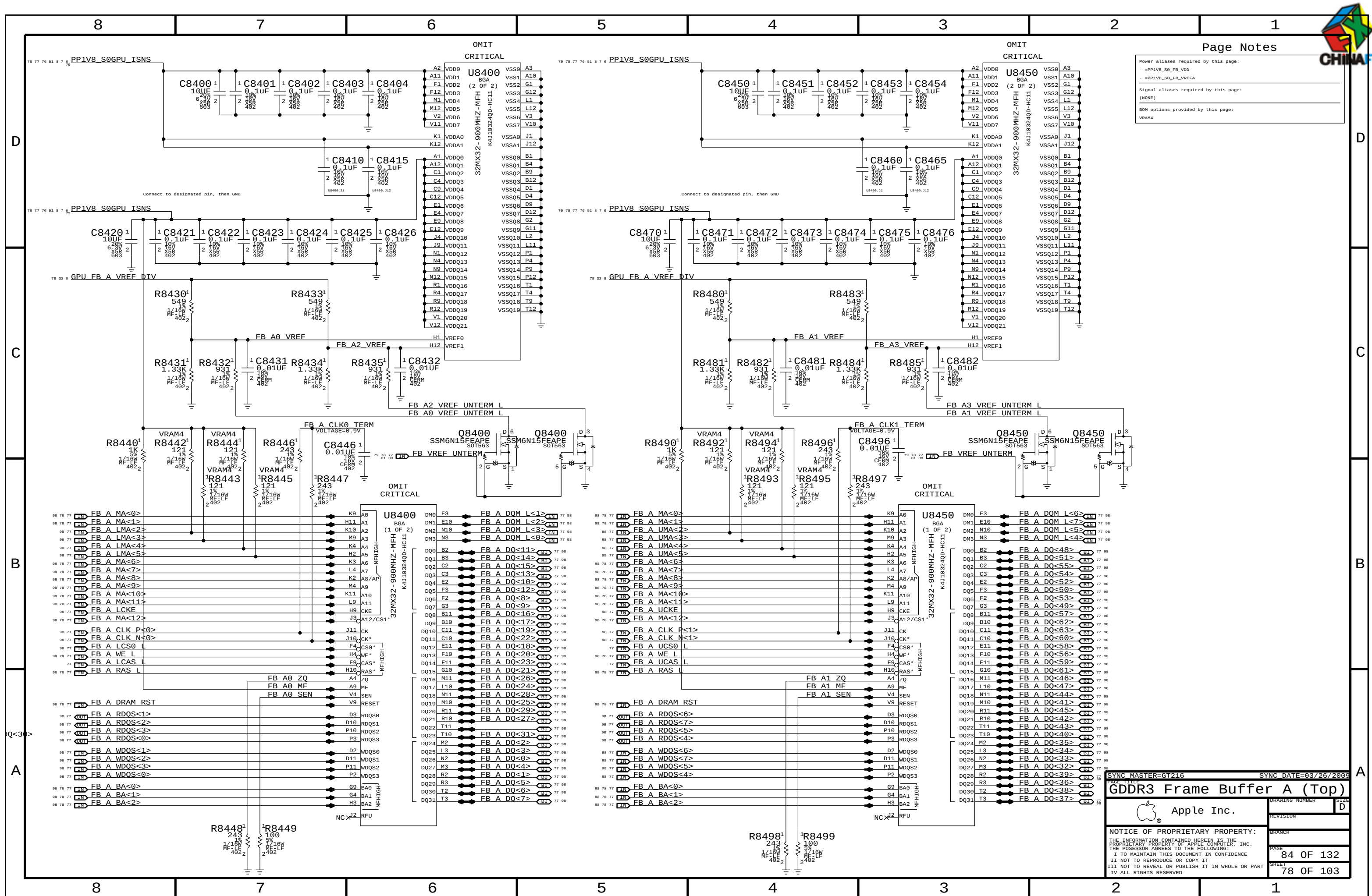
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NV GT216 FRAME BUFFER I/F			
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Power aliases required by this page:
- PP1V8_S0GPU_ISNS
- PP1V8_S0_FB_VDD
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
VRAM4







Page Notes

Power aliases required by this page:
- =PP3V3_GPU_VDDQ3
- =PP3V3_GPU_MIO
- =PP1V2_GPU_PLLVDD
- =PP1V2_GPU_H_PLLVDD
- =PP1V2_GPU_VID_PLLVDD

Signal aliases required by this page:
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BOM options provided by this page:
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D

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C

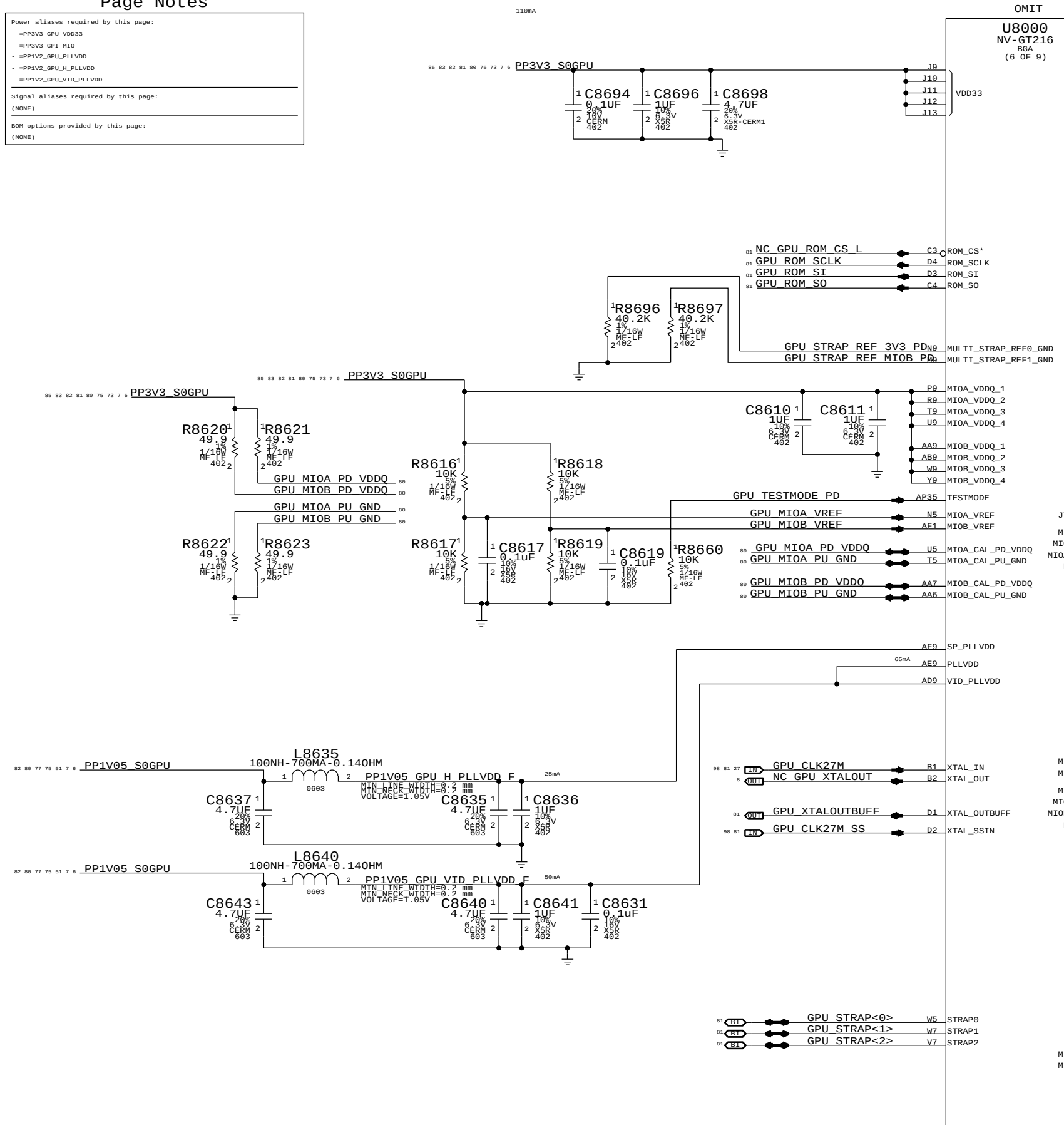
C

B

B

A

A



GPIO0	K1	GPU_VCORE_VID3	B1	B1
GPIO1	K2	DP_EG_HPD	B1	B1
GPIO2	K3	GPU_VCORE_VID4	B1	B1
GPIO3	H3	EG_LCD_PWR_EN	B1	B1
GPIO4	H2	EG_BKLT_EN	B1	B1
GPIO5	H1	GPU_VCORE_VID0	B1	B1
GPIO6	H4	GPU_VCORE_VID1	B1	B1
GPIO7	H5	GPU_VCORE_VID2	B1	B1
GPIO8	H6	SMC_GFX_OVERTEMP_R	B1	B1
GPIO9	J7	TP_GPU_GSTATE<0>	B1	B1
GPIO10	K4	FB_VREF_UNTERM	B1	B1
GPIO11	K5	GPU_GPIO_11	B1	B1
GPIO12	H7	SMC_GFX_THROTTLE_R	B1	B1
GPIO13	J4	FBVDD_ALTVO	B1	B1
GPIO14	J6	NC_GPU_GPIO_14	B1	B1
GPIO15	L1	NC_GPU_GPIO_15	B1	B1
GPIO16	L2	GPU_GPIO_16	B1	B1
GPIO17	L4	NC_GPU_GPIO_17	B1	B1
GPIO18	M4	NC_GPU_GPIO_18	B1	B1
GPIO19	L7	NC_GPU_GPIO_19	B1	B1
GPIO20	L5	NC_GPU_GPIO_20	B1	B1
GPIO21	K6	NC_GPU_GPIO_21	B1	B1
GPIO22	L6	NC_GPU_GPIO_22	B1	B1
GPIO23	M6	NC_GPU_GPIO_23	B1	B1

JTAG_TCK	AP14	TP_GPU_JTAG_TCK	B1	B1
JTAG_TDI	AN14	TP_GPU_JTAG_TDI	B1	B1
JTAG_TDO	AN16	TP_GPU_JTAG_TDO	B1	B1
JTAG_TMS	AR14	TP_GPU_JTAG_TMS	B1	B1
JTAG_TRST*	AP16	TP_GPU_JTAG_TRST_L	B1	B1
MIOA_CLKIN	N4	NC_GPU_MIOA_CLKIN	B1	B1
MIOA_CLKOUT	R4	NC_GPU_MIOA_CLKOUT	B1	B1
MIOA_CLKOUT*	T4	NC_GPU_MIOA_CLKOUT	B1	B1
MIOA_CTL3	P5	NC_GPU_MIOA_CTL3	B1	B1
MIOA_DE	N2	TP_GPU_MIOA_DE	B1	B1
MIOA_D0	N1	TP_GPU_MIOA_D<0>	B1	B1
MIOA_D1	P4	TP_GPU_MIOA_D<1>	B1	B1
MIOA_D2	P1	TP_GPU_MIOA_D<2>	B1	B1
MIOA_D3	P2	TP_GPU_MIOA_D<3>	B1	B1
MIOA_D4	P3	TP_GPU_MIOA_D<4>	B1	B1
MIOA_D5	T3	TP_GPU_MIOA_D<5>	B1	B1
MIOA_D6	T2	TP_GPU_MIOA_D<6>	B1	B1
MIOA_D7	T1	TP_GPU_MIOA_D<7>	B1	B1
MIOA_D8	U4	TP_GPU_MIOA_D<8>	B1	B1
MIOA_D9	U1	TP_GPU_MIOA_D<9>	B1	B1
MIOA_D10	U2	GPU_MIOA_D<10>	B1	B1
MIOA_D11	U3	GPU_MIOA_D<11>	B1	B1
MIOA_D12	R6	GPU_MIOA_D<12>	B1	B1
MIOA_D13	T6	GPU_MIOA_D<13>	B1	B1
MIOA_D14	N6	GPU_MIOA_D<14>	B1	B1
MIOA_HSYNC	N3	NC_GPU_MIOA_HSYNC	B1	B1
MIOA_VSYNC	L3	NC_GPU_MIOA_VSYNC	B1	B1
MIOB_CLKIN	AE1	NC_GPU_MIOB_CLKIN	B1	B1
MIOB_CLKOUT	V4	NC_GPU_MIOB_CLKOUT	B1	B1
MIOB_CLKOUT*	W4	NC_GPU_MIOB_CLKOUT	B1	B1
MIOB_CTL3	W3	NC_GPU_MIOB_CTL3	B1	B1
MIOB_DE	Y5	NC_GPU_MIOB_DE	B1	B1
MIOB_D0	Y1	NC_GPU_MIOB_D<0>	B1	B1
MIOB_D1	Y2	NC_GPU_MIOB_D<1>	B1	B1
MIOB_D2	Y3	NC_GPU_MIOB_D<2>	B1	B1
MIOB_D3	AB3	NC_GPU_MIOB_D<3>	B1	B1
MIOB_D4	AB2	NC_GPU_MIOB_D<4>	B1	B1
MIOB_D5	AB1	NC_GPU_MIOB_D<5>	B1	B1
MIOB_D6	AC4	NC_GPU_MIOB_D<6>	B1	B1
MIOB_D7	AC1	NC_GPU_MIOB_D<7>	B1	B1
MIOB_D8	AC2	NC_GPU_MIOB_D<8>	B1	B1
MIOB_D9	AC3	NC_GPU_MIOB_D<9>	B1	B1
MIOB_D10	AE3	NC_GPU_MIOB_D<10>	B1	B1
MIOB_D11	AE2	NC_GPU_MIOB_D<11>	B1	B1
MIOB_D12	U6	NC_GPU_MIOB_D<12>	B1	B1
MIOB_D13	W6	NC_GPU_MIOB_D<13>	B1	B1
MIOB_D14	Y6	NC_GPU_MIOB_D<14>	B1	B1

MIOB_HSYNC	W1	NC_GPU_MIOB_HSYNC	B1	B1
MIOB_VSYNC	W2	NC_GPU_MIOB_VSYNC	B1	B1
THERMDP	B5	GPU_TDIODE_P	B1	B1
THERMDN	B4	GPU_TDIODE_N	B1	B1

SYNC MASTER=K18 MLB

SYNC DATE=06/29/2009

NV GT216 GPIO/MIO/MISC

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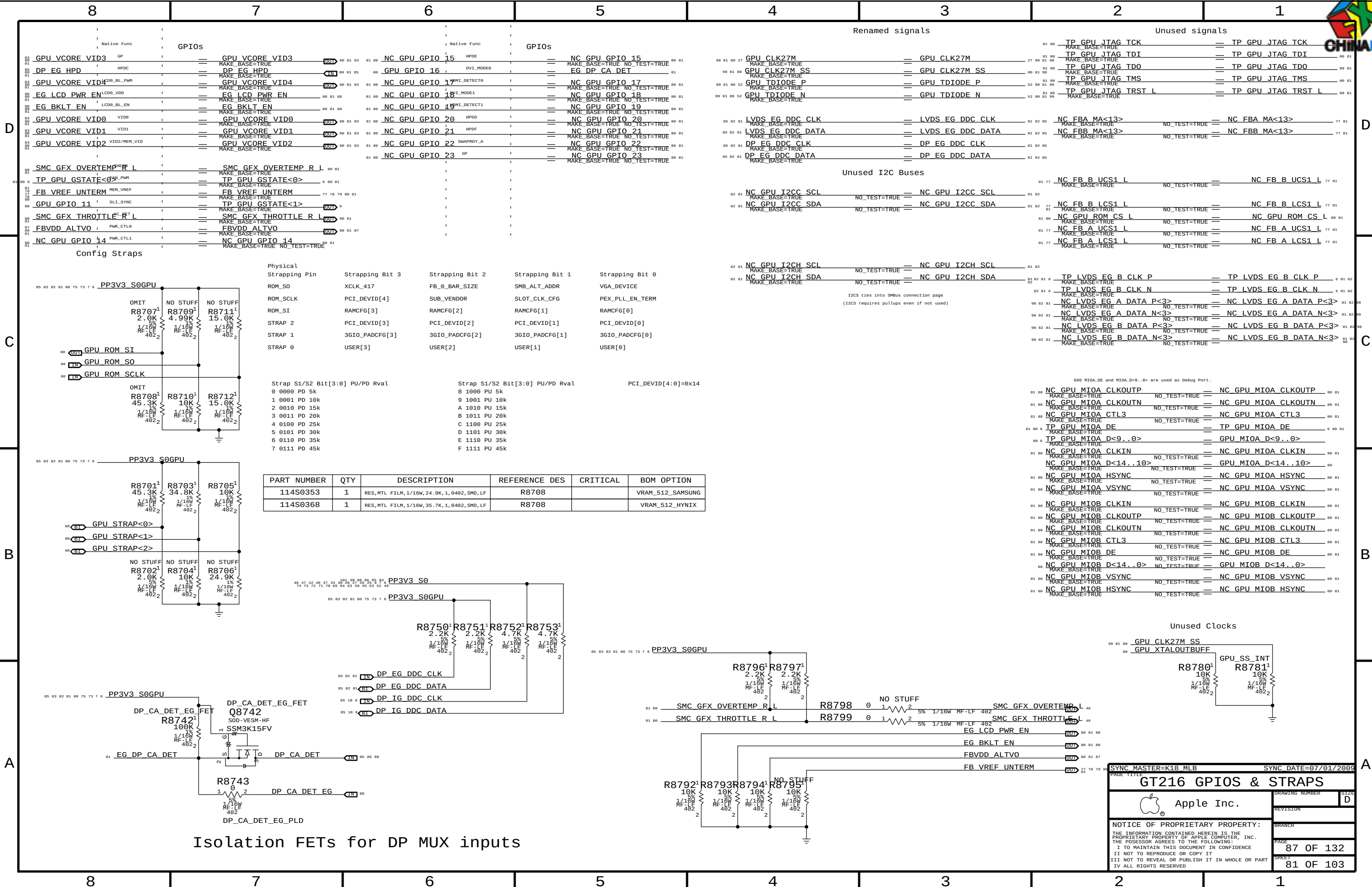
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Physical Strapping Pin	Strapping Bit 3	Strapping Bit 2	Strapping Bit 1	Strapping Bit 0
ROM_SO	XCLK_417	FB_0_BAR_SIZE	SMB_ALT_ADDR	VGA_DEVICE
ROM_SCLK	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
ROM_SI	RAMCFG[3]	RAMCFG[2]	RAMCFG[1]	RAMCFG[0]
STRAP 2	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP 1	3GIO_PADCFG[3]	3GIO_PADCFG[2]	3GIO_PADCFG[1]	3GIO_PADCFG[0]
STRAP 0	USER[3]	USER[2]	USER[1]	USER[0]

Strap S1/S2 Bit[3:0] PU/PD Rval	Strap S1/S2 Bit[3:0] PU/PD Rval	PCI_DEVID[4:0]=0x14
0 0000 PD 5k	0 1000 PU 5k	
1 0001 PD 10k	9 1001 PU 10k	
2 0010 PD 15k	A 1010 PU 15k	
3 0011 PD 20k	B 1011 PU 20k	
4 0100 PD 25k	C 1100 PU 25k	
5 0101 PD 30k	D 1101 PU 30k	
6 0110 PD 35k	E 1110 PU 35k	
7 0111 PD 45k	F 1111 PU 45k	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0353	1	RES,MTL FILM,1/16W,24.9K,1,0402,SMD,LF	R8708		VRAM_512_SAMSUNG
114S0368	1	RES,MTL FILM,1/16W,35.7K,1,0402,SMD,LF	R8708		VRAM_512_HYNIX

Isolation FETs for DP MUX inputs

SYNC MASTER=K18 MLB

SYNC DATE=07/01/2009

GT216 GPIOs & STRAPS

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Power aliases required by this page:
- =PP1V8_GPU_IFPX
- =PP3V3_GPU_IFPCD_IOVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Sum of peak currents: 240mA

73 7 6 PP1V8_GPUIFPX

L8800
FERR-220-0HM-2.5A

C8800 4.7uF
C8801 0.1uF
C8803 0.1uF

PP1V8_GPU_IFPAB_IOVDD F
MIN LINE WIDTH=0.4 mm
MIN NECK WIDTH=0.2 mm
VOLTAGE=1.8V

OMIT

U8000
NV-GT216
BGA
(5 OF 9)

IFPA_TXC AM11 LVDS EG A CLK P
IFPA_TXC* AM12 LVDS EG A CLK N
IFPA_TXD8 AM8 LVDS EG A DATA P<0>
IFPA_TXD0* AL8 LVDS EG A DATA N<0>
IFPA_TXD1 AM10 LVDS EG A DATA P<1>
IFPA_TXD1* AM9 LVDS EG A DATA N<1>
IFPA_TXD2 AK10 LVDS EG A DATA P<2>
IFPA_TXD2* AL10 LVDS EG A DATA N<2>
IFPA_TXD3 AK11 NC LVDS EG A DATA P<3>
IFPA_TXD3* AL11 NC LVDS EG A DATA N<3>
IFPB_TXC AP13 TP LVDS EG B CLK P
IFPB_TXC* AN13 TP LVDS EG B CLK N
IFPB_TXD4 AN8 LVDS EG B DATA P<0>
IFPB_TXD4* AP8 LVDS EG B DATA N<0>
IFPB_TXD5 AP10 LVDS EG B DATA P<1>
IFPB_TXD5* AN10 LVDS EG B DATA N<1>
IFPB_TXD6 AR11 LVDS EG B DATA P<2>
IFPB_TXD6* AR10 LVDS EG B DATA N<2>
IFPB_TXD7 AN11 NC LVDS EG B DATA P<3>
IFPB_TXD7* AP11 NC LVDS EG B DATA N<3>

NO STUFF
R8861 1K
NO STUFF
R8860 1K

IFPC_AUX_I2CW_SCL AP2 DP EG AUX CH P
IFPC_AUX_I2CW_SDA* AN3 DP EG AUX CH N
IFPC_L0 AM7 DP EG ML P<0>
IFPC_L0* AM6 DP EG ML N<0>
IFPC_L1 AL5 DP EG ML P<1>
IFPC_L1* AM5 DP EG ML N<1>
IFPC_L2 AM3 DP EG ML P<2>
IFPC_L2* AM4 DP EG ML N<2>
IFPC_L3 AP1 DP EG ML P<3>
IFPC_L3* AR2 DP EG ML N<3>

IFPD_AUX_I2CX_SCL AR1 NC
IFPD_AUX_I2CX_SDA* AN1 NC
IFPD_L0 AR2 NC
IFPD_L0* AR1 NC
IFPD_L1 AR3 NC
IFPD_L1* AR4 NC
IFPD_L2 AR5 NC
IFPD_L2* AR6 NC
IFPD_L3 AR7 NC
IFPD_L3* AR8 NC

IFPE_AUX_I2CY_SCL AE1 NC
IFPE_AUX_I2CY_SDA* AN1 NC
IFPE_L0 AR9 NC
IFPE_L0* AR8 NC
IFPE_L1 AR10 NC
IFPE_L1* AR11 NC
IFPE_L2 AE2 NC
IFPE_L2* AE3 NC
IFPE_L3 AE4 NC
IFPE_L3* AE5 NC

IFPF_AUX_I2CZ_SCL AE3 NC
IFPF_AUX_I2CZ_SDA* AN3 NC
IFPF_L0 AL3 NC
IFPF_L0* AL2 NC
IFPF_L1 AL4 NC
IFPF_L1* AL5 NC
IFPF_L2 AL6 NC
IFPF_L2* AL7 NC
IFPF_L3 AL8 NC
IFPF_L3* AL9 NC

DACA_RED AM16 NC
DACA_GREEN AM17 NC
DACA_BLUE AL14 NC
DACA_HSYNC AM18 NC
DACA_VSYNC AL15 NC

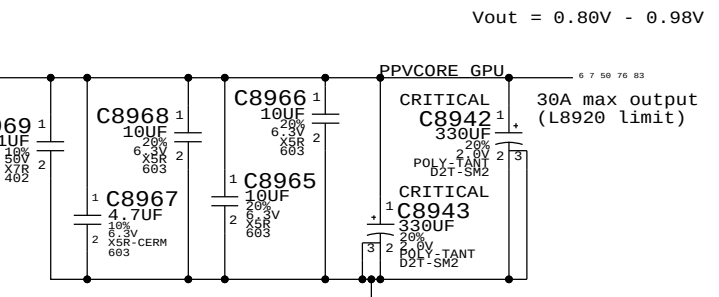
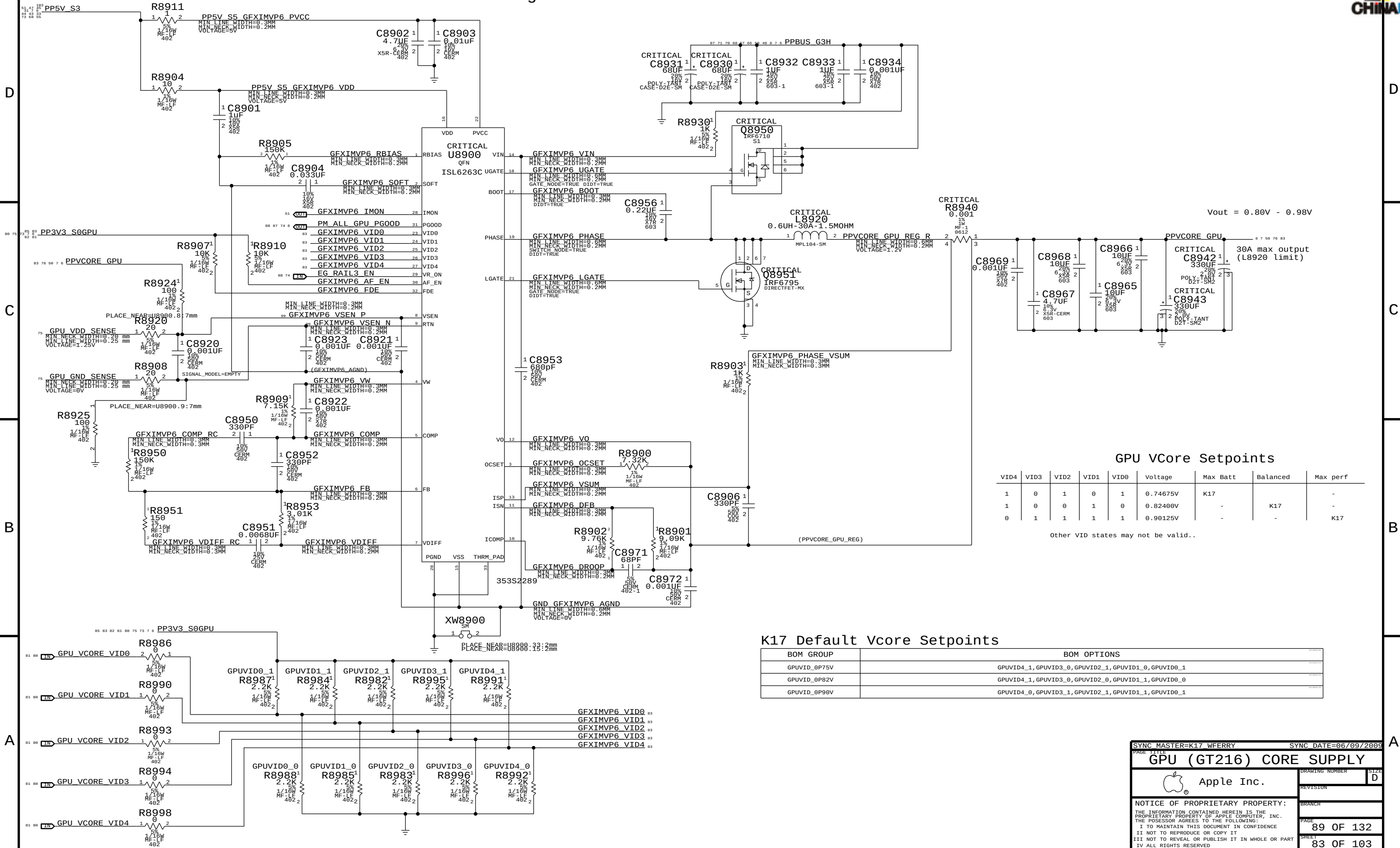
DACB_RED AK1 NC
DACB_GREEN AL16 NC
DACB_BLUE AL17 NC
CEC AR9 NC

DACB_HSYNC AK1 NC
DACB_VSYNC AK2 NC

SYNC MASTER=K18 MLB		SYNC DATE=06/29/2009	
PAGE TITLE NV GT216 VIDEO INTERFACES			
Apple Inc.		DRAWING NUMBER	SIZE D
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GPU VCore Regulator



GPU VCore Setpoints

VID4	VID3	VID2	VID1	VID0	Voltage	Max Batt	Balanced	Max perf
1	0	1	0	1	0.74675V	K17		-
1	0	0	1	0	0.82400V	-	K17	-
0	1	1	1	1	0.90125V	-	-	K17

Other VID states may not be valid..

K17 Default Vcore Setpoints

BOM GROUP	BOM OPTIONS
GPUVID_0P75V	GPUVID4_1, GPUVID3_0, GPUVID2_1, GPUVID1_0, GPUVID0_1
GPUVID_0P82V	GPUVID4_1, GPUVID3_0, GPUVID2_0, GPUVID1_1, GPUVID0_0
GPUVID_0P90V	GPUVID4_0, GPUVID3_1, GPUVID2_1, GPUVID1_1, GPUVID0_1

SYNC MASTER=K17 WFERRY

SYNC DATE=06/09/2009

GPU (GT216) CORE SUPPLY

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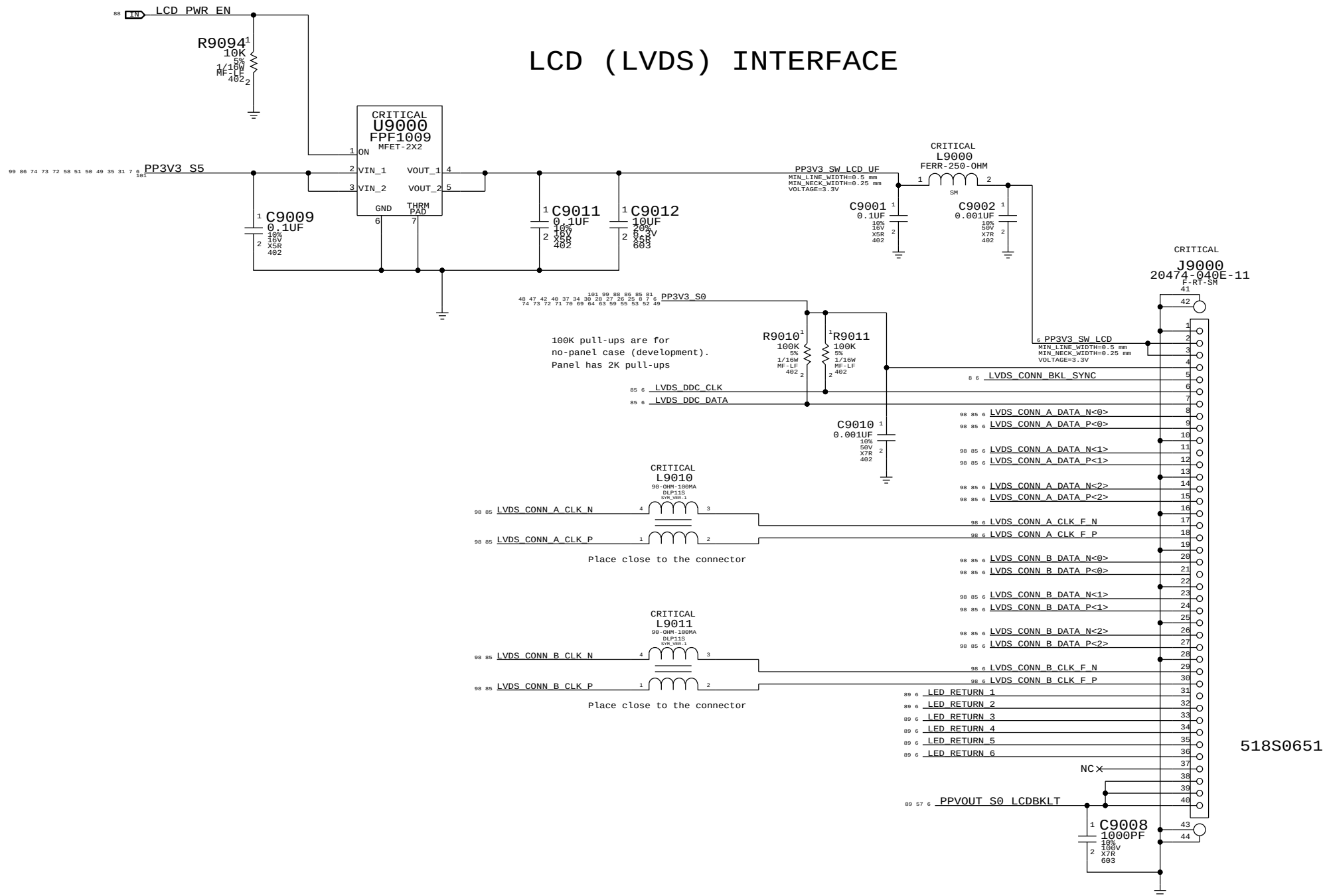
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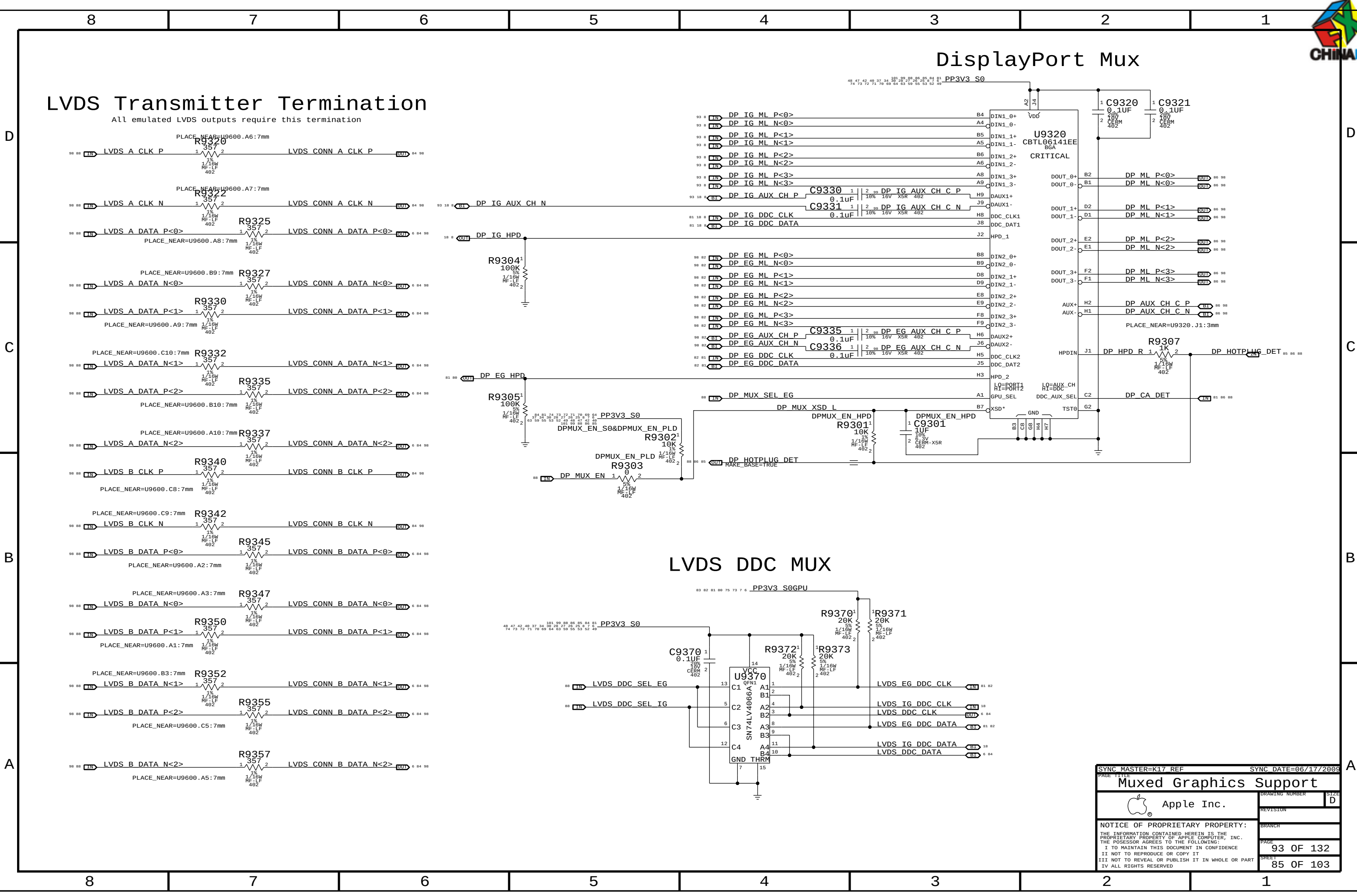
SHEET

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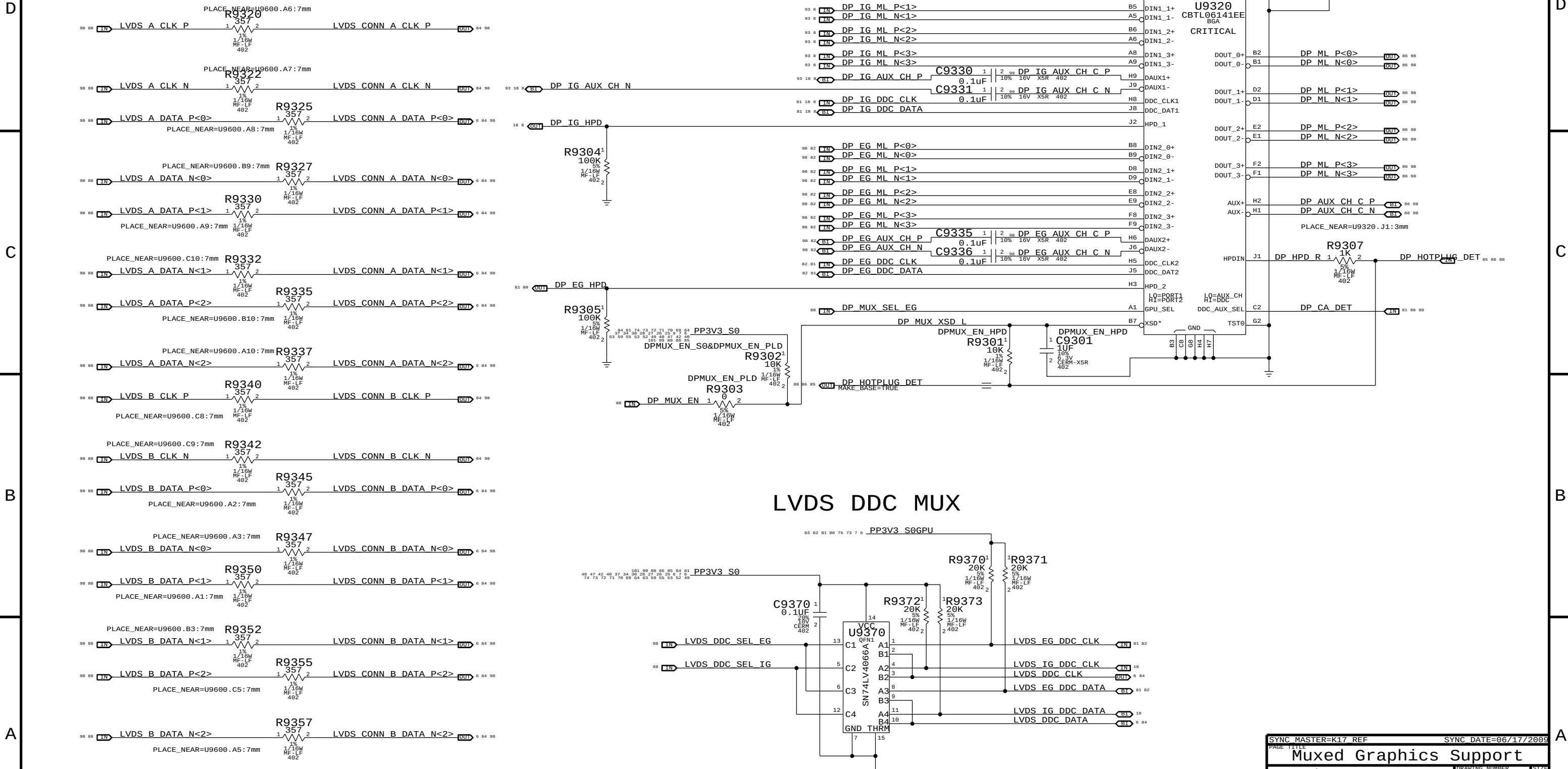
LCD (LVDS) INTERFACE



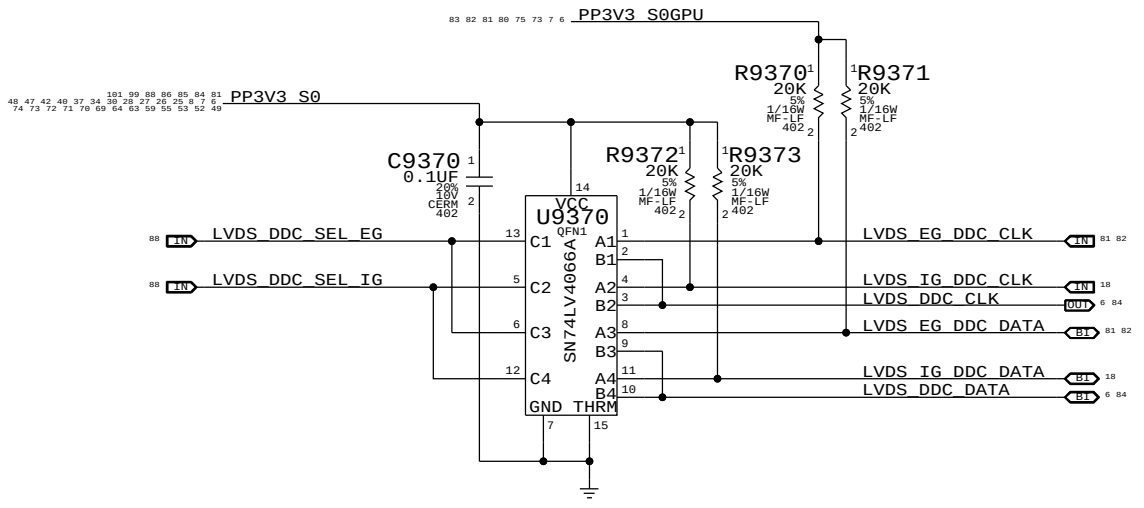


LVDS Transmitter Termination

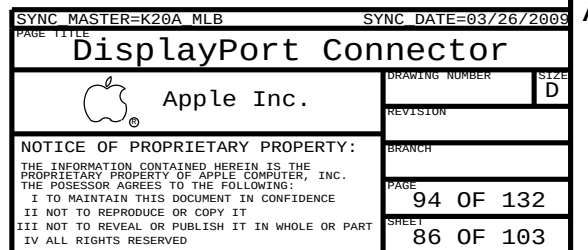
All emulated LVDS outputs require this termination



LVDS DDC MUX



SYNC MASTER=K17 REF		SYNC DATE=06/17/2009	
PAGE TITLE			
Muxed Graphics Support			
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BA



GMUX CPLD

LVDS Receiver Termination

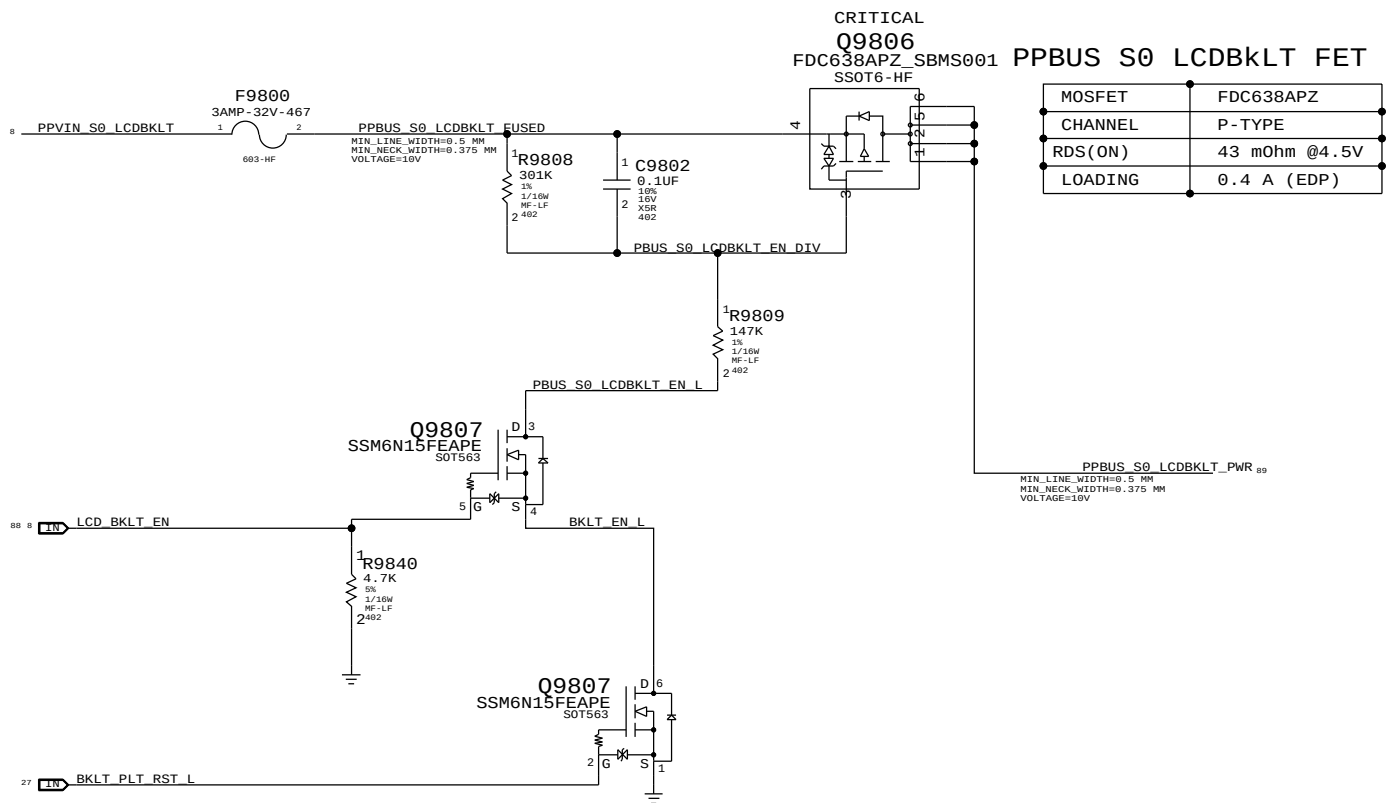
Required Pullups

Required Pulldowns

SYNC MASTER=K17 REF		SYNC DATE=06/24/2009	
PAGE TITLE			
Graphics MUX (GMUX)		DRAWING NUMBER	SIZE D
Apple Inc.		REVISION	
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SYNC MASTER=K17 VEMURI		SYNC DATE=12/16/2009	
PAGE TITLE			
LCD Backlight Driver (MC34845)			
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		REVISION	
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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	20 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

Most CPU signals with impedance requirements are 50-ohm single-ended.

Some signals require 27.4-ohm single-ended impedance.

SOURCE: Calpella SFF DG (DG-407364_v1.5), Section 2.8

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.4

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
DMI_S2N	PCIE_85D	PCIE	DMI S2N P<3:0>	9 18
DMI_S2N	PCIE_85D	PCIE	DMI S2N N<3:0>	9 18
DMI_N2S	PCIE_85D	PCIE	DMI N2S P<3:0>	9 18
DMI_N2S	PCIE_85D	PCIE	DMI N2S N<3:0>	9 18
FDI_DATA	PCIE_85D	PCIE	FDI_DATA P<7:0>	9 18
FDI_DATA	PCIE_85D	PCIE	FDI_DATA N<7:0>	9 18
	CPU_50S	CPU_AGTL	FDI_FSYNC<1..0>	9 18
	CPU_50S	CPU_AGTL	FDI_LSYNC<1..0>	9 18
	CPU_50S	CPU_AGTL	FDI_INT	9 18
CPU_PECT	CPU_50S	PCIE	CPU PECT	10 20
FSB_CPURST_L	CPU_50S	CPU_AGTL	FSB_CPURST_L	10 25
PM_SYNC	CPU_50S	CPU_AGTL	PM_SYNC	10 18
PM_MEM_PWRGD	CPU_50S	CPU_AGTL	PM_MEM_PWRGD	10 18 31
CPU_VTT_S0_PG00D	CPU_50S	CPU_AGTL	CPUVTTIS0_PG00D	10 71
XDP_XPU_PWRG00D	CPU_50S	CPU_ITP	XDP_CPUPWRGD	10 25
XDP_DBRESET_L	CPU_50S	CPU_ITP	XDP_DBRESET_L	10 25 27
XDP_PRDY_L	CPU_50S	CPU_ITP	XDP_PRDY_L	10 25
XDP_PREQ_L	CPU_50S	CPU_ITP	XDP_PREQ_L	10 25
	CPU_50S	CPU_AGTL	PM_EXT_TS_L<0>	10 47
	CPU_50S	CPU_AGTL	PM_EXT_TS_L<1>	10 47
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP0	10
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP1	10
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP2	10
CPU_CFG	CPU_50S	CPU_ITP	CPU_CFG<17..0>	8 9 25
CPU_CATERR_L	CPU_50S	CPU_AGTL	CPU_CATERR_L	10
	CPU_50S	CPU_AGTL	TP_CPU_VTT_SELECT	8 12
CPU_PROCHOT_L	CPU_50S	CPU_AGTL	CPU_PROCHOT_L	10 47 69
CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	10 20 25
PM_THRMTRIP_L	CPU_50S	CPU_8MTI	PM_THRMTRIP_L	10 20 47
FSB_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	FSB_CLK133M_CPU_P	10 20
FSB_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	FSB_CLK133M_CPU_N	10 20
FSB_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	FSB_CLK133M_ITP_P	10 25
FSB_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	FSB_CLK133M_ITP_N	10 25
PCIE_CLK100M_CPU	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_CPU_P	10 17
PCIE_CLK100M_CPU	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_CPU_N	10 17
	CPU_55S	CPU_8MTI	CPU_PSI_L	12 15 69
PM DPRSLPVR	CPU_50S	CPU_AGTL	PM DPRSLPVR	12 15 69
	CPU_27P4S	CPU_COMP	CPU_PEG_COMP	9
	CPU_27P4S	CPU_COMP	CPU_PEG_RBIA5	9
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP3	10
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP2	10
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP1	10
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP0	10
XDP_TDI	CPU_50S	CPU_ITP	XDP_TDI	25
XDP_TDO	CPU_50S	CPU_ITP	XDP_TDO	25
XDP_TMS	CPU_50S	CPU_ITP	XDP_TMS	10 25
XDP_TCK	CPU_50S	CPU_ITP	XDP_TCK	10 25
XDP_TRST_L	CPU_50S	CPU_ITP	XDP_TRST_L	10 25
XDP_BPM_L	CPU_50S	CPU_ITP	XDP_BPM_L<6..0>	10 25
XDP_BPM_L	CPU_50S	CPU_ITP	XDP_BPM_L<7>	10 25
(FSB_CPURST_L)	CPU_50S	CPU_ITP	XDP_CPURST_L	25
	CPU_55S	CPU_8MTI	CPU_VID<6..0>	8 12 15
	CPU_50S	CPU_AGTL	CPUI MVP_IMON	12 50 69
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	12 69
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	12 69
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VTTSENSE_P	12 71
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VTTSENSE_N	12 71
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	GFX_VSENSE_P	13 70
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	GFX_VSENSE_N	13 70
PM DPRSLPVR	CPU_55S	CPU_8MTI	GFX_VID<6..0>	8 13
	CPU_50S	CPU_AGTL	GFX DPRSLPVR	13 70
	CPU_50S	CPU_AGTL	GFX VR_EN	13 70
	CPU_50S	CPU_AGTL	GFXIMVP_IMON	13 70
	PCIE_85D	PCIE	PEG_R2D_P<15..0>	75
	PCIE_85D	PCIE	PEG_R2D_N<15..0>	75
PEG_R2D	PCIE_85D	PCIE	PEG_R2D_C_P<15..0>	8 75
	PCIE_85D	PCIE	PEG_R2D_C_N<15..0>	8 75
PEG_D2R	PCIE_85D	PCIE	PEG_D2R_P<15..0>	8 9 75
	PCIE_85D	PCIE	PEG_D2R_N<15..0>	8 9 75
	PCIE_85D	PCIE	PEG_D2R_C_P<15..0>	75
	PCIE_85D	PCIE	PEG_D2R_C_N<15..0>	75

SYNC MASTER=K17_WFERRY		SYNC DATE=06/09/2009	
PAGE TITLE		CPU Constraints	
		DRAWING NUMBER	SIZE D
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=3:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MILS	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

DDR3:

DQ/DM signals should be matched within 0.508mm of associated DQS pair.
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to clock matching should be within [CLK-12.7mm] and [CLK+25.4mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-12.7mm] to [CLK+0.0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to SODIMM pad is 139.7mm, from procesor ball to SODIMM pad is 114.3mm.

SOURCE: Calpella SFF Platform DG, Rev 1.5 (#407364), Section 2.2

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0>	11 28
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0>	11 28
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A CKE<3..0>	11 28
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A CS L<3..0>	11 28
MEM_A_CNT1	MEM_37S	MEM_CTRL	MEM A ODT<3..0>	11 28
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	11 28
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	11 28
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L	11 28
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L	11 28
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L	11 28
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM A DQ<7..0>	11 29
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM A DQ<15..8>	11 29
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM A DQ<23..16>	11 29
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM A DQ<31..24>	11 29
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM A DQ<39..32>	11 28 29
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM A DQ<47..40>	11 29
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM A DQ<55..48>	11 29
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM A DQ<63..56>	11 29
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM A DM<0>	11 28 29
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM A DM<1>	11 29
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM A DM<2>	11 29
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM A DM<3>	11 29
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM A DM<4>	11 29
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM A DM<5>	11 29
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM A DM<6>	11 29
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM A DM<7>	11 29
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS P<0>	11 28 29
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS N<0>	11 28 29
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS P<1>	11 29
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS N<1>	11 29
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS P<2>	11 29
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS N<2>	11 29
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS P<3>	11 29
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS N<3>	11 29
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS P<4>	11 29
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS N<4>	11 29
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS P<5>	11 29
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS N<5>	11 29
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS P<6>	11 29
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS N<6>	11 29
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS P<7>	11 29
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS N<7>	11 29
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<5..0>	11 30
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<5..0>	11 30
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B CKE<3..0>	11 30
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B CS L<3..0>	11 30
MEM_B_CNT1	MEM_37S	MEM_CTRL	MEM B ODT<3..0>	11 30
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	11 30
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	11 30
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L	11 30
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L	11 30
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L	11 30
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM B DQ<7..0>	11 29
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM B DQ<15..8>	11 29
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM B DQ<23..16>	11 29
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM B DQ<31..24>	11 29
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM B DQ<39..32>	11 29 30
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM B DQ<47..40>	11 29
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM B DQ<55..48>	11 29
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM B DQ<63..56>	11 29
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM B DM<0>	11 29 30
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM B DM<1>	11 29
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM B DM<2>	11 29
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM B DM<3>	11 29
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM B DM<4>	11 29
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM B DM<5>	11 29
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM B DM<6>	11 29
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM B DM<7>	11 29
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS P<0>	11 29 30
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS N<0>	11 29 30
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS P<1>	11 29
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS N<1>	11 29
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS P<2>	11 29
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS N<2>	11 29
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS P<3>	11 29
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS N<3>	11 29
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS P<4>	11 29
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS N<4>	11 29
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS P<5>	11 29
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS N<5>	11 29
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS P<6>	11 29
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS N<6>	11 29
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS P<7>	11 29
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS N<7>	11 29

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Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?	LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length. DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps. DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals. Max length of LVDS/DisplayPort/TMDS traces: 12 inches. SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	*	=4x_DIELECTRIC	?	SATA	TOP,BOTTOM	=3x_DIELECTRIC	?
SATA_ICOMP	*	8 MIL	?				

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.7.1.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCH_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.8

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
DP_ML	DP_85D	DISPLAYPORT	DP IG ML P<3..0>	8 85
DP_ML	DP_85D	DISPLAYPORT	DP IG ML N<3..0>	8 85
DP_AUX_CH	DP_85D	DISPLAYPORT	DP IG AUX CH P	8 18 85
DP_AUX_CH	DP_85D	DISPLAYPORT	DP IG AUX CH N	8 18 85
LVDS IG A CLK	LVDS_85D	LVDS	LVDS IG A CLK P	18 88
LVDS IG A CLK	LVDS_85D	LVDS	LVDS IG A CLK N	18 88
LVDS IG A DATA	LVDS_85D	LVDS	LVDS IG A DATA P<2..0>	18 88
LVDS IG A DATA	LVDS_85D	LVDS	LVDS IG A DATA N<2..0>	18 88
LVDS IG A DATA3	LVDS_85D	LVDS	NC LVDS IG A DATAP<3>	8 18
LVDS IG A DATA3	LVDS_85D	LVDS	NC LVDS IG A DATAN<3>	8 18
LVDS IG B CLK	LVDS_85D	LVDS	TP LVDS IG B CLKP	6 8 18
LVDS IG B CLK	LVDS_85D	LVDS	TP LVDS IG B CLKN	6 8 18
LVDS IG B DATA	LVDS_85D	LVDS	LVDS IG B DATA P<2..0>	18 88
LVDS IG B DATA	LVDS_85D	LVDS	LVDS IG B DATA N<2..0>	18 88
LVDS IG B DATA3	LVDS_85D	LVDS	NC LVDS IG B DATAP<3>	8 18
LVDS IG B DATA3	LVDS_85D	LVDS	NC LVDS IG B DATAN<3>	8 18
SATA_HDD_R2D	SATA_90D	SATA	SATA HDD R2D C P	17 42
SATA_90D	SATA	SATA	SATA HDD R2D C N	17 42
SATA_90D	SATA	SATA	SATA HDD R2D P	6 42
SATA_90D	SATA	SATA	SATA HDD R2D N	6 42
SATA_HDD_D2R	SATA_90D	SATA	SATA HDD D2R P	17 42
SATA_90D	SATA	SATA	SATA HDD D2R N	17 42
SATA_90D	SATA	SATA	SATA HDD D2R C P	42
SATA_90D	SATA	SATA	SATA HDD D2R C N	42
SATA_ODD_R2D	SATA_90D	SATA	SATA ODD R2D C P	17 42
SATA_90D	SATA	SATA	SATA ODD R2D C N	17 42
SATA_90D	SATA	SATA	SATA ODD R2D P	6 42
SATA_90D	SATA	SATA	SATA ODD R2D N	6 42
SATA_ODD_D2R	SATA_90D	SATA	SATA ODD D2R P	17 42
SATA_90D	SATA	SATA	SATA ODD D2R N	17 42
SATA_90D	SATA	SATA	SATA ODD D2R C P	42
SATA_90D	SATA	SATA	SATA ODD D2R C N	42
SATA_EXT_A_R2D	SATA_90D	SATA	TP SATA EXT_A R2D C P	8 17
SATA_90D	SATA	SATA	TP SATA EXT_A R2D C N	8 17
SATA_EXT_A_D2R	SATA_90D	SATA	TP SATA EXT_A D2R P	8 17
SATA_90D	SATA	SATA	TP SATA EXT_A D2R N	8 17
PCH_SATA_ICOMP		SATA_ICOMP	PCH SATA_ICOMP	
USB_EXT_A	USB_85D	USB	USB_EXT_A P	36 43
USB_85D	USB	USB	USB_EXT_A N	36 43
USB_EXT_B	USB_85D	USB	USB_EXT_B P	35 43
USB_85D	USB	USB	USB_EXT_B N	35 43
USB_EXT_C	USB_85D	USB	USB_EXT_C P	35 44
USB_85D	USB	USB	USB_EXT_C N	35 44
USB_HUB2_UP	USB_85D	USB	USB_HUB2_UP P	19 36
USB_85D	USB	USB	USB_HUB2_UP N	19 36
USB_MINI	USB_85D	USB	NC_USB_MINI	6
USB_85D	USB	USB	NC_USB_MINI	6
USB_HUB1_UP	USB_85D	USB	USB_HUB1_UP P	19 35
USB_85D	USB	USB	USB_HUB1_UP N	19 35
USB_CAMERA	USB_85D	USB	USB_CAMERA P	33 35
USB_85D	USB	USB	USB_CAMERA N	33 35
USB_BT	USB_85D	USB	USB_BT P	33 36
USB_85D	USB	USB	USB_BT N	33 36
USB_TP_A	USB_85D	USB	USB_TP_A P	36 54
USB_85D	USB	USB	USB_TP_A N	36 54
USB_IR	USB_85D	USB	USB_IR P	35 45
USB_85D	USB	USB	USB_IR N	35 45
USB_EXCARD	USB_85D	USB	USB_EXCARD P	8 34 36
USB_85D	USB	USB	USB_EXCARD N	8 34 36
USB_BRCRYPT	USB_85D	USB	USB_BRCRYPT P	19 103
USB_85D	USB	USB	USB_BRCRYPT N	19 103
PCH_USB_RBIAS	PCH_USB_RBIAS		PCH_USB_RBIAS	19
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_PCH P	17 26
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_PCH N	17 26
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	FSB_CLK133M_PCH P	17 26
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	FSB_CLK133M_PCH N	17 26
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT P	17 26
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT N	17 26
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA P	17 26
CLK_PCIE_90D	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA N	17 26
CPU_50S	CLK_PCIE	CLK_PCIE	PCH_CLK14P3M_REFCLK	17 26
CPU_50S	CLK_PCIE	CLK_PCIE	PCH_CLK33M_PCIEIN	17 27
GFX_CLK_DPLLSS	CLK_PCIE_90D	CLK_PCIE	GFX_CLK120M_DPLLSS P	18 17
GFX_CLK_DPLLSS	CLK_PCIE_90D	CLK_PCIE	GFX_CLK120M_DPLLSS N	18 17

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

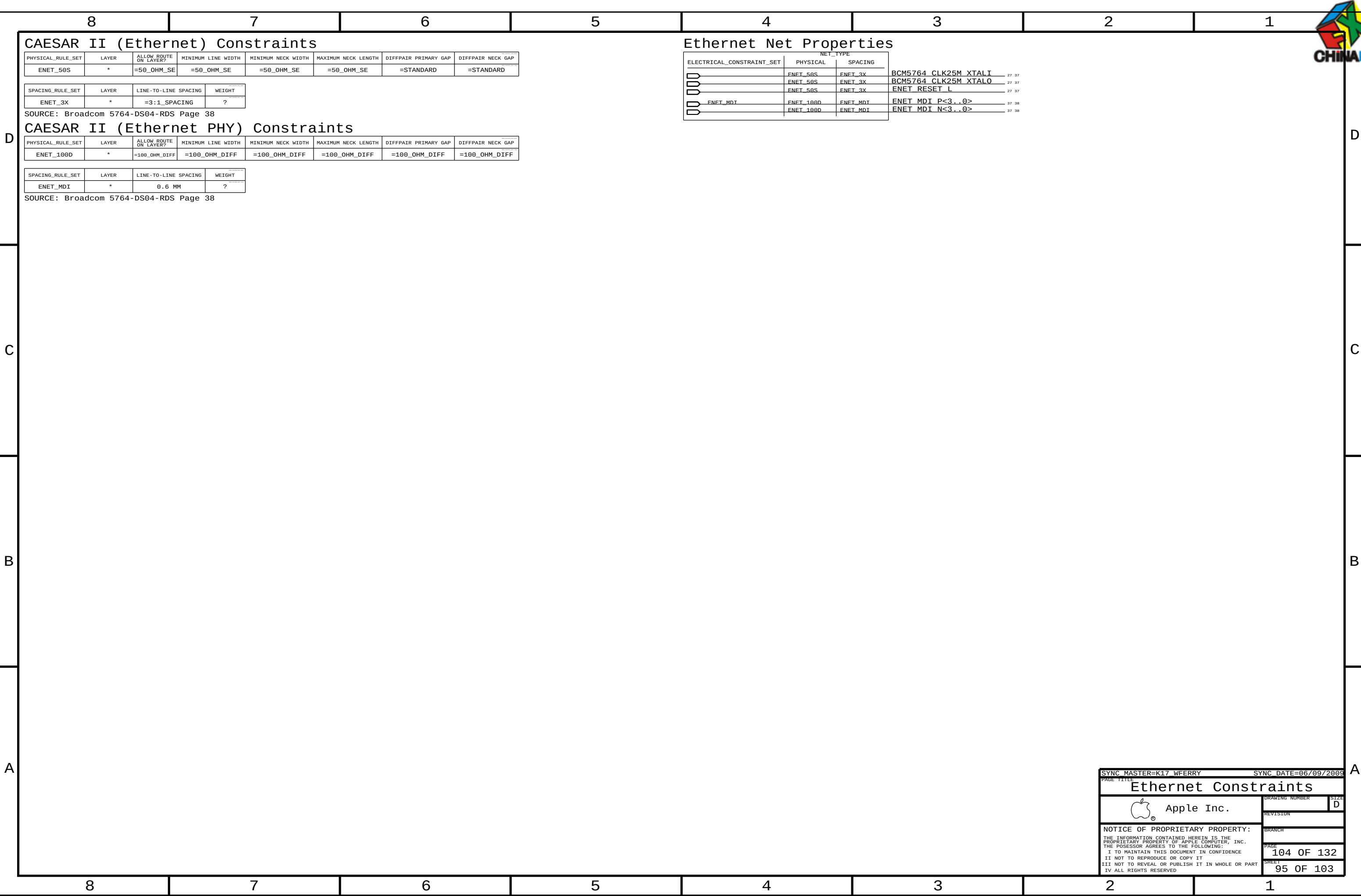
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Net Properties

NET_TYPE				
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
LPC_AD	LPC_50S	LPC	LPC AD<3..0> 6 17 46 48 88	
LPC_FRAME_L	LPC_50S	LPC	LPC FRAME L 6 17 46 48 88	
LPC_RESET_L	LPC_50S	LPC	LPCPLUS RESET L 6 27 48 88	
MCP_LPC_CLK0	CLK_LPC_50S	CLK_LPC	LPC CLK33M SMC R 19 27	
	CLK_LPC_50S	CLK_LPC	LPC CLK33M SMC 27 46	
	CLK_LPC_50S	CLK_LPC	LPC CLK33M LPCPLUS 6 27 48	
SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS PCH CLK 6 17 25 26 28 30 32 34 42 48 49	
SMBUS_PCH_DATA	SMB_50S	SMB	SMBUS PCH DATA 6 17 25 26 28 30 32 34 42 48 49	
SMBUS_PCH_0_CLK	SMB_50S	SMB	SML PCH 0 CLK 17 49	
SMBUS_PCH_0_DATA	SMB_50S	SMB	SML PCH 0 DATA 17 49	
SMBUS_PCH_1_CLK	SMB_50S	SMB	SML PCH 1 CLK 17 49	
SMBUS_PCH_1_DATA	SMB_50S	SMB	SML PCH 1 DATA 17 49	
HDA_BIT_CLK	HDA_50S	HDA	HDA BIT CLK 17 59	
	HDA_50S	HDA	HDA BIT CLK R 17	
HDA_SYNC	HDA_50S	HDA	HDA SYNC 17 59	
	HDA_50S	HDA	HDA SYNC R 17	
HDA_RST_L	HDA_50S	HDA	HDA RST R L 17	
	HDA_50S	HDA	HDA RST L 17 59	
HDA_SDIN0	HDA_50S	HDA	HDA SDIN0 17 59	
	HDA_50S	HDA	HDA SDIN CODEC 17 59	
HDA_SDOUT	HDA_50S	HDA	HDA SDOUT 17 59	
	HDA_50S	HDA	HDA SDOUT R 17	
PM_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM CLK32K SUSCLK 18 47	
SPT_CLK	SPT_55S	SPT	SPI CLK R 17 48	
	SPT_55S	SPT	SPI CLK 48	
SPT_M0SI	SPT_55S	SPT	SPI MOSI R 17 48	
	SPT_55S	SPT	SPI MOSI 48	
SPT_MISO	SPT_55S	SPT	SPI MISO 17 48	
SPT_CS0	SPT_55S	SPT	SPI CS0 R L 17 48	
	SPT_55S	SPT	SPI CS0 L 48	
	PCIE_85D	PCIE	PCIE ENET R2D P 37	
	PCIE_85D	PCIE	PCIE ENET R2D N 37	
PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE ENET R2D C P 17 37	
	PCIE_85D	PCIE	PCIE ENET R2D C N 17 37	
PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE ENET D2R P 17 37	
	PCIE_85D	PCIE	PCIE ENET D2R N 17 37	
	PCIE_85D	PCIE	PCIE ENET D2R C P 37	
	PCIE_85D	PCIE	PCIE ENET D2R C N 37	
	PCIE_85D	PCIE	PCIE AP R2D P 6 33	
	PCIE_85D	PCIE	PCIE AP R2D N 6 33	
PCIE_AP_R2D	PCIE_85D	PCIE	PCIE AP R2D C P 17 33	
	PCIE_85D	PCIE	PCIE AP R2D C N 17 33	
PCIE_AP_D2R	PCIE_85D	PCIE	PCIE AP D2R P 6 17 33	
	PCIE_85D	PCIE	PCIE AP D2R N 6 17 33	
	PCIE_85D	PCIE	PCIE FW R2D P 39	
	PCIE_85D	PCIE	PCIE FW R2D N 39	
PCIE_FW_R2D	PCIE_85D	PCIE	PCIE FW R2D C P 17 39	
	PCIE_85D	PCIE	PCIE FW R2D C N 17 39	
PCIE_FW_D2R	PCIE_85D	PCIE	PCIE FW D2R P 17 39	
	PCIE_85D	PCIE	PCIE FW D2R N 17 39	
	PCIE_85D	PCIE	PCIE FW D2R C P 39	
	PCIE_85D	PCIE	PCIE FW D2R C N 39	
	PCIE_85D	PCIE	PCIE EXCARD R2D P 6 34	
	PCIE_85D	PCIE	PCIE EXCARD R2D N 6 34	
PCIE_EXCARD_R2D	PCIE_85D	PCIE	PCIE EXCARD R2D C P 17 34	
	PCIE_85D	PCIE	PCIE EXCARD R2D C N 17 34	
PCIE_EXCARD_D2R	PCIE_85D	PCIE	PCIE EXCARD D2R P 6 17 34	
	PCIE_85D	PCIE	PCIE EXCARD D2R N 6 17 34	
MCP_PE0_REFCLK	CLK_PCTF_90D	CLK_PCTF	PEG CLK100M_P 17 75	
	CLK_PCTF_90D	CLK_PCTF	PEG CLK100M_N 17 75	
PCTF_CLK100M_ENET	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M ENET P 17 37	
	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M ENET N 17 37	
MCP_PE1_REFCLK	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M AP P 17 33	
	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M AP N 17 33	
MCP_PE2_REFCLK	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M FW P 17 39	
	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M FW N 17 39	
MCP_PE3_REFCLK	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M EXCARD P 17 34	
	CLK_PCTF_90D	CLK_PCTF	PCIE CLK100M EXCARD N 17 34	
H280	CPU_27P4S	CPU_COMP	PCH VSS NCTF<1> 6 28	
H281	CPU_27P4S	CPU_COMP	PCH VSS NCTF<2> 6 28	
H282	CPU_27P4S	CPU_COMP	PCH VSS NCTF<5> 6 28	
H283	CPU_27P4S	CPU_COMP	TP PCH VSS NCTF<7> 28	
H284	CPU_27P4S	CPU_COMP	PCH VSS NCTF<9> 6 28 94	
H285	CPU_27P4S	CPU_COMP	PCH VSS NCTF<9> 6 28 94	
H286	CPU_27P4S	CPU_COMP	PCH VSS NCTF<11> 6 28	
H287	CPU_27P4S	CPU_COMP	PCH VSS NCTF<12> 6 28	
H288	CPU_27P4S	CPU_COMP	PCH VSS NCTF<15> 6 28	
H289	CPU_27P4S	CPU_COMP	PCH VSS NCTF<17> 6 28	
H290	CPU_27P4S	CPU_COMP	PCH VSS NCTF<19> 6 28	
H291	CPU_27P4S	CPU_COMP	PCH VSS NCTF<21> 6 28	
H292	CPU_27P4S	CPU_COMP	PCH VSS NCTF<22> 28	
H293	CPU_27P4S	CPU_COMP	PCH VSS NCTF<25> 6 28	
H294	CPU_27P4S	CPU_COMP	PCH VSS NCTF<27> 6 28	
H295	CPU_27P4S	CPU_COMP	PCH VSS NCTF<29> 6 28	
H296	PCIE_AP_D2R	PCIE_85D	PCIE	CONN PCIE AP D2R P
H297	PCIE_AP_D2R	PCIE_85D	PCIE	CONN PCIE AP D2R N
H298	PCIE_AP_R2D	PCIE_85D	PCIE	CONN PCIE AP R2D P
H299	PCIE_AP_R2D	PCIE_85D	PCIE	CONN PCIE AP R2D N

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Ethernet Constraints

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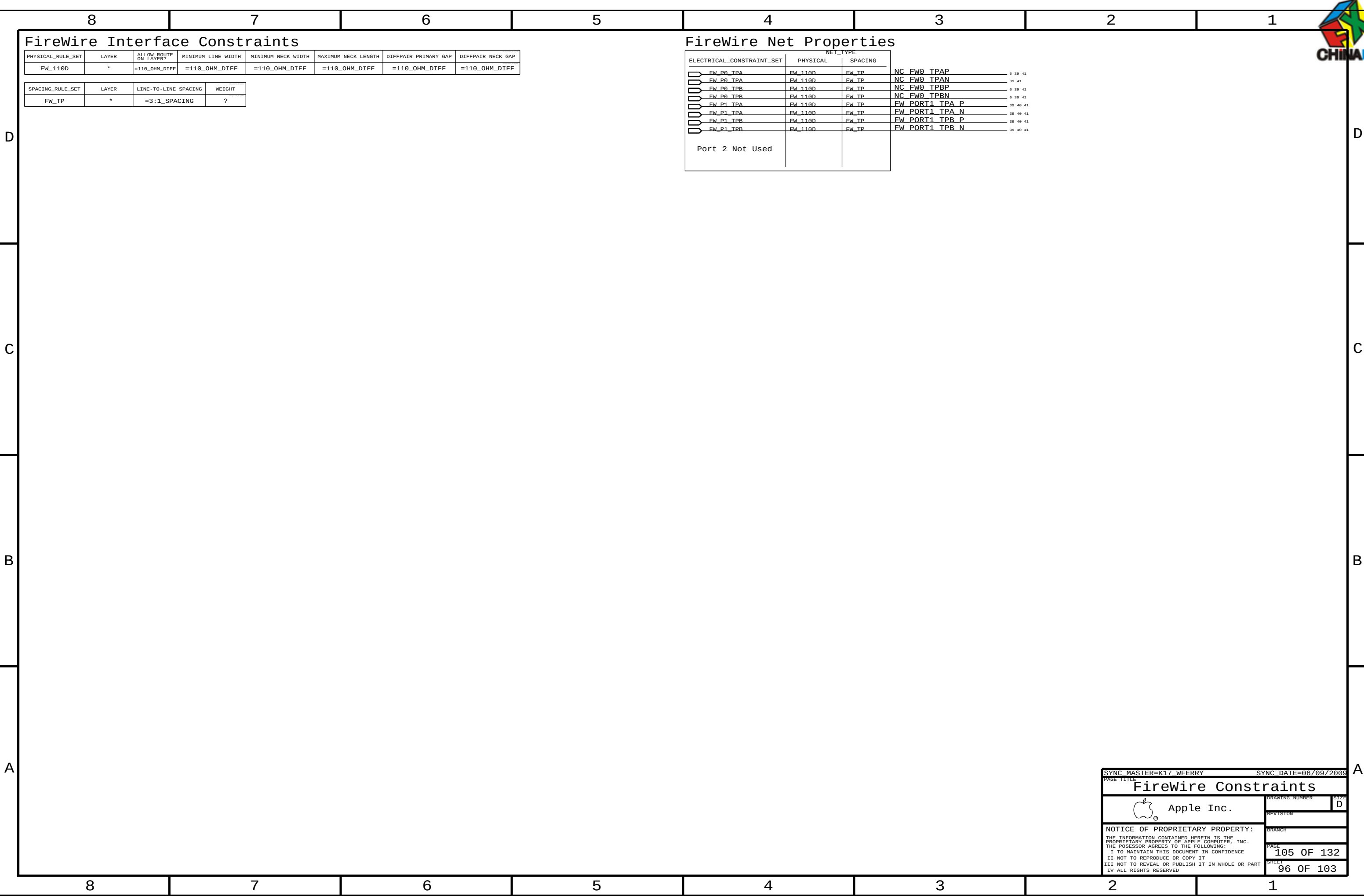
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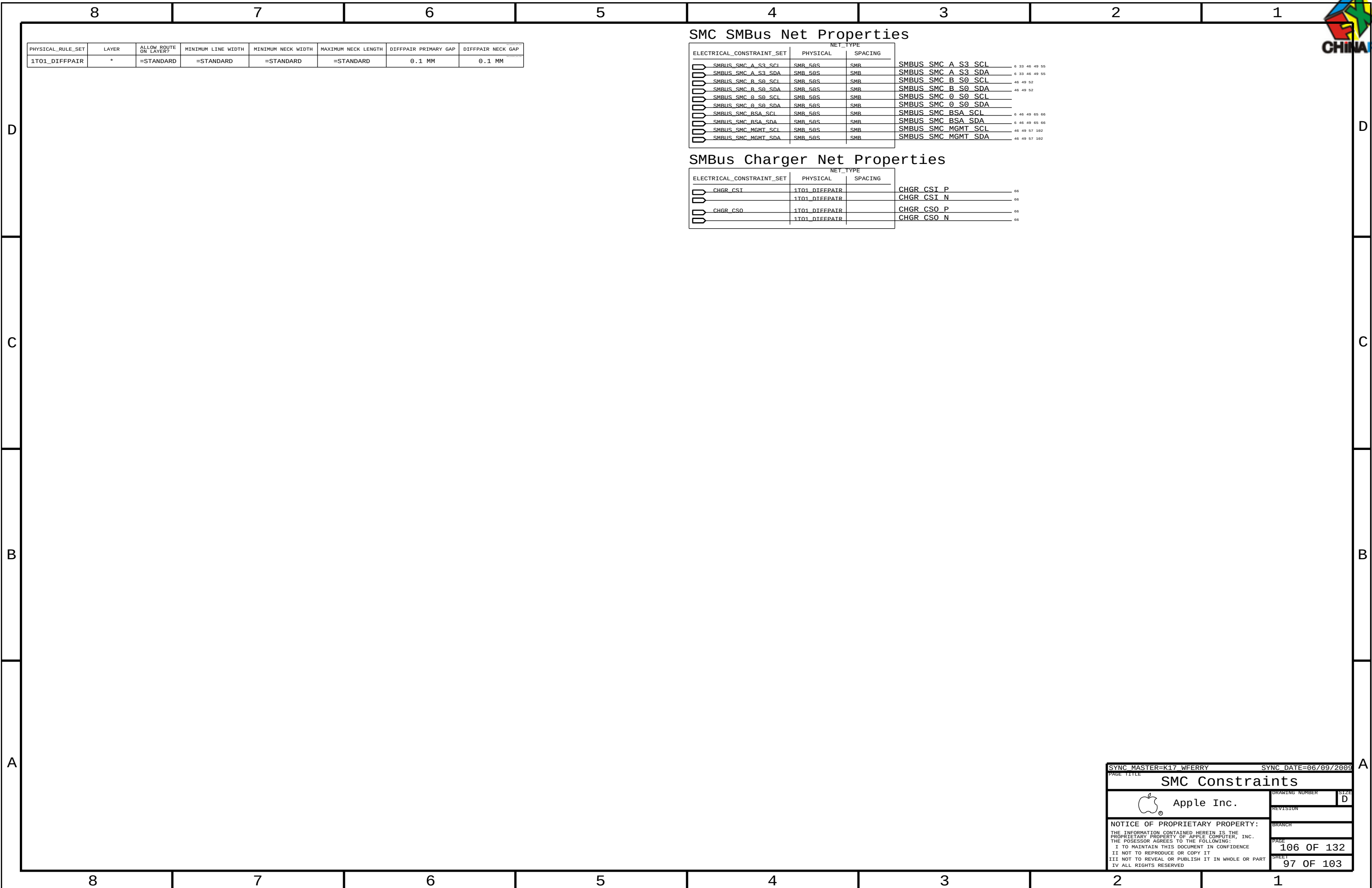
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PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SENSE_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_40S_OVERRIDE	*	VERRIDE	VERRIDE	0.09 MM_OVERRIDE	100 MIL_OVERRIDE	VERRIDE	VERRIDE
MEM_72D_OVERRIDE	*	VERRIDE	VERRIDE	0.09 MM_OVERRIDE	100 MIL_OVERRIDE	VERRIDE	VERRIDE
PCIE_85D_OVERRIDE	*	VERRIDE	VERRIDE	0.09 MM_OVERRIDE	100 MIL_OVERRIDE	VERRIDE	VERRIDE
USB_85D_OVERRIDE	TOP_OVERRIDE	VERRIDE	VERRIDE	0.1 MM_OVERRIDE	500 MIL_OVERRIDE	VERRIDE	VERRIDE
CPU_27P4S_OVERRIDE	BOTTOM_OVERRIDE	VERRIDE	VERRIDE	0.23 MM_OVERRIDE	100 MIL_OVERRIDE	VERRIDE	VERRIDE

Graphics ,SATA Constraint Relaxations

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
LVDS_85D	BGA	LVDS_85D
DP_85D	BGA	100_DIFF_BGA
SATA_90D	BGA	100_DIFF_BGA
CLK_PCIE_90D	BGA	100_DIFF_BGA

A Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_72D	BOTTOM			0.127 MM	6.35 MM		
MEM_85D	TOP			0.1 MM	6.35 MM		

K17 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	ENET_100D	ENETCONN	ENETCONN_P<3..0>
	ENET_100D	ENETCONN	ENETCONN_N<3..0>
	SATA_90D	SATA	SATA_ODD_R2D_UF_P
	SATA_90D	SATA	SATA_ODD_R2D_UF_N
	SATA_90D	SATA	SATA_ODD_D2R_UF_P
	SATA_90D	SATA	SATA_ODD_D2R_UF_N
	SATA_90D	SATA	SATA_HDD_D2R_UF_P
	SATA_90D	SATA	SATA_HDD_D2R_UF_N
	SATA_90D	SATA	SATA_HDD_R2D_UF_P
	SATA_90D	SATA	SATA_HDD_R2D_UF_N
	SATA_90D	SATA	SATA_HDD_D2R_RDRV_IN_P
	SATA_90D	SATA	SATA_HDD_D2R_RDRV_IN_N
	SATA_90D	SATA	SATA_HDD_D2R_RDRV_OUT_P
	SATA_90D	SATA	SATA_HDD_D2R_RDRV_OUT_N
	SATA_90D	SATA	SATA_HDD_R2D_RDRV_IN_P
	SATA_90D	SATA	SATA_HDD_R2D_RDRV_IN_N
	SATA_90D	SATA	SATA_HDD_R2D_RDRV_OUT_P
	SATA_90D	SATA	SATA_HDD_R2D_RDRV_OUT_N
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_CPUTHMSNS_D2_P
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_CPUTHMSNS_D2_N
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_CPU_THERMD_P
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_CPU_THERMD_N
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_GPUTHMSNS_D_P
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_GPUTHMSNS_D_N
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_GPU_TDIODE_P
	SENSE_DIEFPATR	THERM_1T01_55S	THERM_GPU_TDIODE_N
	SENSE_DIEFPATR	SENSE_1T01_55S	THERM_ISNS_3V3_S3_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	THERM_ISNS_3V3_S3_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_AIRPORT_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_AIRPORT_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_DDRISNS_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_DDRISNS_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_GPUISENS_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_GPUISENS_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_AIRPORT_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_AIRPORT_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_DDRISNS_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_DDRISNS_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_P1V8GPU_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_P1V8GPU_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_CPU_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_CPU_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_P1V8GPU_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_P1V8GPU_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_HDD_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_HDD_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_HDD_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_HDD_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_ODD_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_ODD_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_ODD_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_ODD_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_LCDBKLT_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_LCDBKLT_N
	SENSE_DIEFPATR	SENSE_1T01_55S	NC_ISNS_P1V05S0PCH_P
	SENSE_DIEFPATR	SENSE_1T01_55S	NC_ISNS_P1V05S0PCH_N
	SENSE_DIEFPATR	SENSE_1T01_55S	CPUVTTIS0_CS_P
	SENSE_DIEFPATR	SENSE_1T01_55S	CPUVTTIS0_CS_N
	SENSE_DIEFPATR	SENSE_1T01_55S	CPUVTTIS0_CS_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	CPUVTTIS0_CS_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	NC_ISNS_P3V3S0MPCH_P
	SENSE_DIEFPATR	SENSE_1T01_55S	NC_ISNS_P3V3S0MPCH_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_P3V3S0MPCH_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_P3V3S0MPCH_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	GFXTMVP_CS_P
	SENSE_DIEFPATR	SENSE_1T01_55S	GFXTMVP_CS_N
	SENSE_DIEFPATR	SENSE_1T01_55S	GFXTMVP_CS_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	GFXTMVP_CS_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP5V_S0_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP5V_S0_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	NC_ISNS_PVTTIS0PCH_P
	SENSE_DIEFPATR	SENSE_1T01_55S	NC_ISNS_PVTTIS0PCH_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PVTTIS0PCH_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PVTTIS0PCH_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP1V05_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP1V05_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP3V3_S3_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP3V3_S3_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP3V3_S5_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP3V3_S5_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP5V_S0_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP5V_S0_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP5V_S3_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_PP5V_S3_N
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_1V5_S3_P
	SENSE_DIEFPATR	SENSE_1T01_55S	ISNS_1V5_S3_N
	SENSE_DIEFPATR	SENSE_1T01_55S	THERM_ISNS_1V5_S3_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	THERM_ISNS_1V5_S3_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_AIRPORT_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	SENSE_ISNS_AIRPORT_R_N
	SENSE_DIEFPATR	SENSE_1T01_55S	GFXTMVP6_VSEN_P
	SENSE_DIEFPATR	SENSE_1T01_55S	GFXTMVP6_VSEN_N
	SENSE_DIEFPATR	SENSE_1T01_55S	GFX_ISNS_R_P
	SENSE_DIEFPATR	SENSE_1T01_55S	GFX_ISNS_R_N

K17 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	PCIE_CLK100M_AP	CLK_PCIE_90D	CLK_PCIE_PCIE_CLK100M_AP_CONN_P
		CLK_PCIE_90D	CLK_PCIE_PCIE_CLK100M_AP_CONN_N
		1T01_DIEFPATR	CHGR_CSI_R_P
		1T01_DIEFPATR	CHGR_CSI_R_N
		1T01_DIEFPATR	CHGR_CS0_R_P
		1T01_DIEFPATR	CHGR_CS0_R_N
	(USB_EXT_A)	USB_85D	USB2_EXT_A_MUXED_P
	(USB_EXT_A)	USB_85D	USB2_EXT_A_MUXED_N
	(USB_EXT_A)	USB_85D	USB2_LT1_P
	(USB_EXT_A)	USB_85D	USB2_LT1_N
	(USB_EXT_D)	USB_85D	USB_BRCRYPT_CONN_P
	(USB_EXT_D)	USB_85D	USB_BRCRYPT_CONN_N
	(USB_CAMERA)	USB_85D	USB_CAMERA_CONN_P
	(USB_CAMERA)	USB_85D	USB_CAMERA_CONN_N
		USB_85D	CONN_USB2_BT_P
		USB_85D	CONN_USB2_BT_N
		USB_85D	USB_LT2_P
		USB_85D	USB_LT2_N
		USB_85D	USB2_EXCARD_CONN_P
		USB_85D	USB2_EXCARD_CONN_N
	DP_85D	DTSPDISPLAYPORT	DP_IG_AUX_CH_C_P
	DP_85D	DTSPDISPLAYPORT	DP_IG_AUX_CH_C_N
	USB_85D	USB	USB_BRCRYPT_R_P
	USB_85D	USB	USB_BRCRYPT_R_N
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_EXCARD_CONN_N
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_EXCARD_CONN_P
	SPK_OUT	DIEFPATR	AUDIO_SPKRAMP_FL_OUT_P
		DIEFPATR	AUDIO_SPKRAMP_FL_OUT_N
	SPK_OUT	DIEFPATR	AUDIO_SPKRAMP_BL_OUT_P
		DIEFPATR	AUDIO_SPKRAMP_BL_OUT_N
	SPK_OUT	DIEFPATR	AUDIO_SPKRAMP_FR_OUT_P
		DIEFPATR	AUDIO_SPKRAMP_FR_OUT_N
	SPK_OUT	DIEFPATR	AUDIO_SPKRAMP_BR_OUT_P
		DIEFPATR	AUDIO_SPKRAMP_BR_OUT_N
	SPK_OUT	DIEFPATR	AUDIO_SPKRAMP_LFE_OUT_P
		DIEFPATR	AUDIO_SPKRAMP_LFE_OUT_N
		DIEFPATR	AUDIO_SPKRAMP_FL_IN_C_P
		DIEFPATR	AUDIO_SPKRAMP_FL_IN_C_N
		DIEFPATR	AUDIO_SPKRAMP_BL_IN_C_P
		DIEFPATR	AUDIO_SPKRAMP_BL_IN_C_N
		DIEFPATR	AUDIO_SPKRAMP_FR_IN_C_P
		DIEFPATR	AUDIO_SPKRAMP_FR_IN_C_N
		DIEFPATR	AUDIO_SPKRAMP_BR_IN_C_P
		DIEFPATR	AUDIO_SPKRAMP_BR_IN_C_N
		DIEFPATR	AUDIO_SPKRAMP_LFE_IN_C_P
		DIEFPATR	AUDIO_SPKRAMP_LFE_IN_C_N
		DIEFPATR	AUDIO_SSM2315FL_IN_P
		DIEFPATR	AUDIO_SSM2315FL_IN_N
		DIEFPATR	AUDIO_SSM2315FR_IN_P
		DIEFPATR	AUDIO_SSM2315FR_IN_N
		DIEFPATR	AUDIO_SSM2315BR_IN_P
		DIEFPATR	AUDIO_SSM2315BR_IN_N
		DIEFPATR	AUDIO_INT_MIC_F_P
		DIEFPATR	AUDIO_INT_MIC_F_N
		DIEFPATR	AUDIO_INT_MIC_P
		DIEFPATR	AUDIO_INT_MIC_N
		DIEFPATR	AUDIO_SSM2315BL_IN_P
		DIEFPATR	AUDIO_SSM2315BL_IN_N
		DIEFPATR	AUDIO_SSM2315LFE_IN_P
		DIEFPATR	AUDIO_SSM2315LFE_IN_N
	USB_85D	USB	USB_TPAD_R_P
	USB_85D	USB	USB_TPAD_R_N
	USB_85D	USB	USB_LT3_P
	USB_85D	USB	USB_LT3_N
		SB_POWER	PP3V3_S5
		SB_POWER	PP3V3_S0
		SB_POWER	PP1V5_S0
		GND	GND

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K17 Board-Specific Spacing & Physical Constraints

BOARD LAYERS			BOARD AREAS			BOARD UNITS (ALL OF MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM			NO_TYPE, BGA			MM	15.5.1
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.095 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP, BOTTOM	Y	0.185 MM	0.095 MM			
37_OHM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.250 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.165 MM	0.095 MM			
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	ISL9, ISL10	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	ISL2, ISL11	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4	Y	0.110 MM	0.110 MM		0.180 MM	0.180 MM
85_OHM_DIFF	ISL9, ISL10	Y	0.110 MM	0.110 MM		0.180 MM	0.180 MM
85_OHM_DIFF	ISL2, ISL11	Y	0.125 MM	0.125 MM		0.190 MM	0.190 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.125 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4	Y	0.102 MM	0.102 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL9, ISL10	Y	0.102 MM	0.102 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL2, ISL11	Y	0.115 MM	0.115 MM		0.230 MM	0.230 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.115 MM		0.230 MM	0.230 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL4	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL9, ISL10	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL2, ISL11	Y	0.080 MM	0.080 MM		0.220 MM	0.220 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.080 MM	0.080 MM		0.220 MM	0.220 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
110_OHM_DIFF	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.330 MM	
110_OHM_DIFF	ISL2, ISL11	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM
MEM_CLK	*	BGA	BGA_P2MM
CLK_PCIE	*	BGA	BGA_P2MM
CLK_SLOW	*	BGA	BGA_P2MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.126 MM	?
3X_DIELECTRIC	*	0.189 MM	?
4X_DIELECTRIC	*	0.252 MM	?

NOTE: From T18 MLB, changed to reflect K17 mlb stackup.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers.

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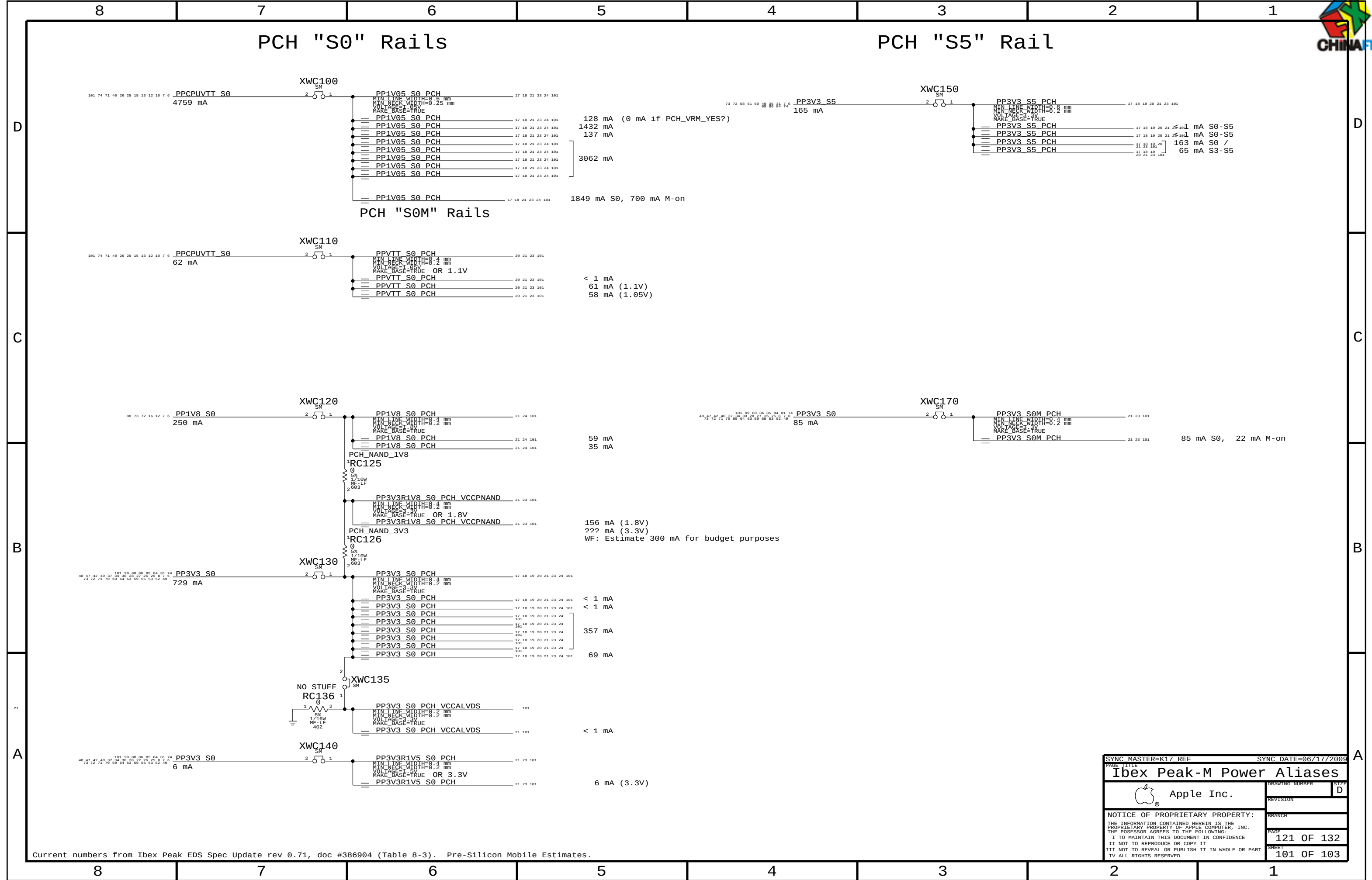
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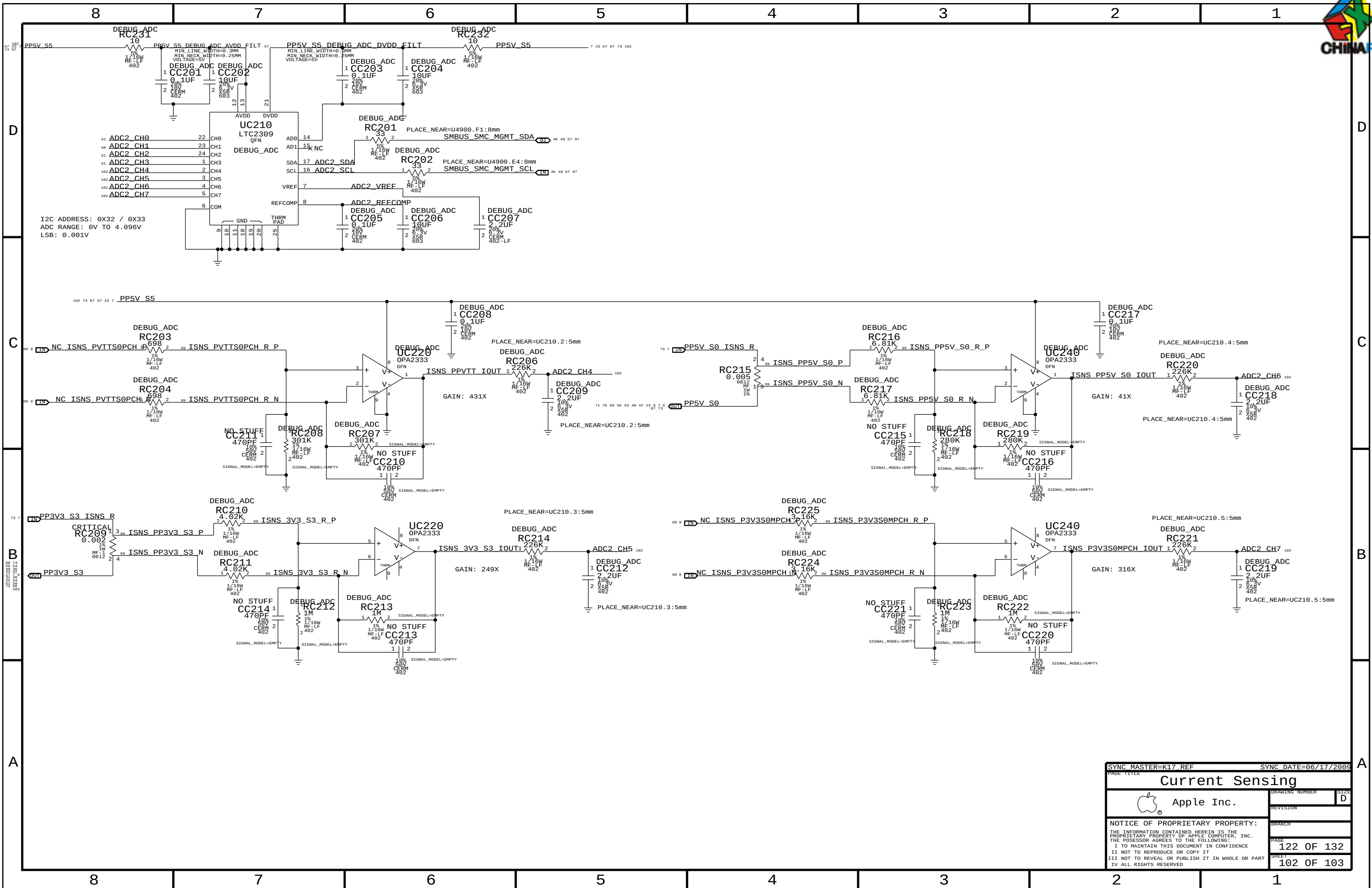
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Current numbers from Ibex Peak EDS Spec Update rev 0.71, doc #386904 (Table 8-3). Pre-Silicon Mobile Estimates.

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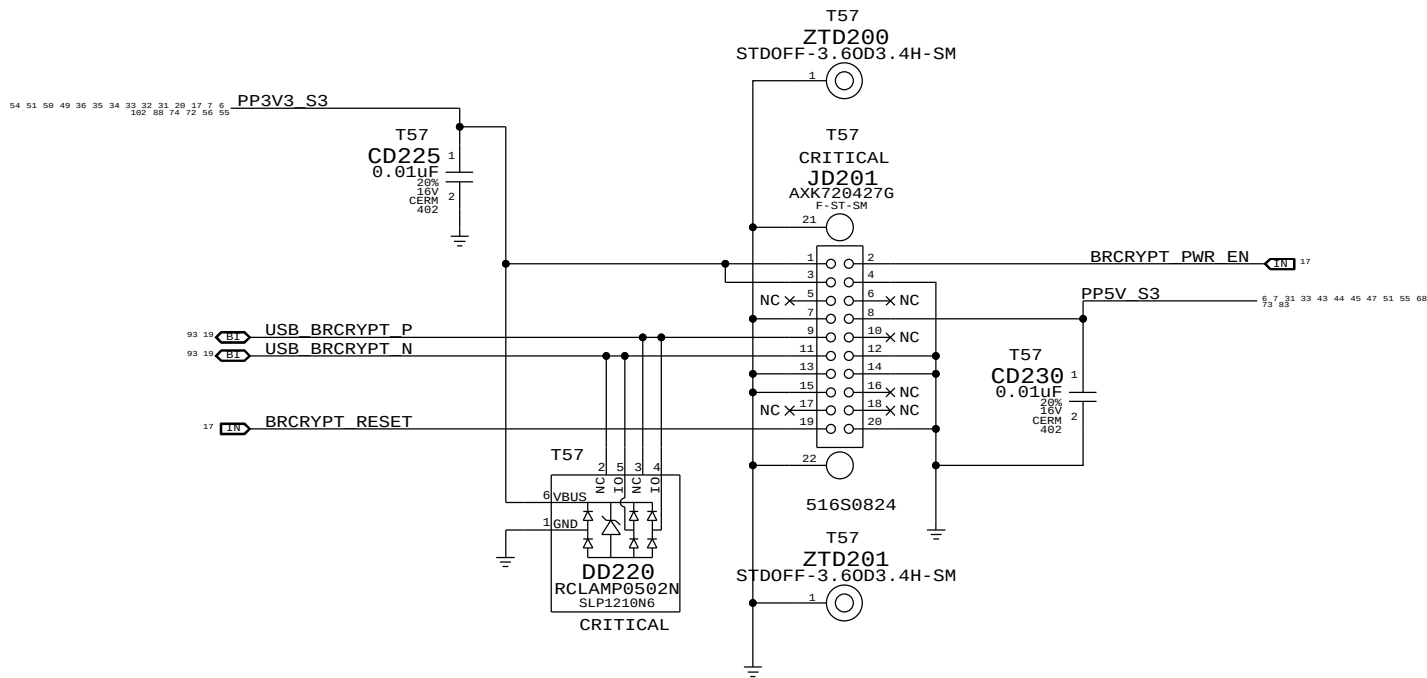
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